Lemur: Scalable Post-Quantum Synchronized Multi-Signatures

Anonymous Author(s)

Abstract

Synchronized multi-signatures allow for *non-interactive* aggregation of signatures generated for the same time step. This primitive is particularly well-suited for high-throughput blockchain protocols like Ethereum, where many distributed signers must validate the same block within a synchronized slot. In this work, we present Lemur, a post-quantum synchronized multi-signature from (module) lattices that improves upon the state-of-the-art in efficiency, scalability and flexibility. Lemur follows the blueprint of Squirrel/Chipmunk (CCS 2022/2023) but introduces a fundamentally different redesign of the foundations of the overall framework. Our revisit of the framework is also motivated by the fact that our evaluation of Chipmunk's parameter sets based on the state-of-the-art lattice security estimation methods yields only around 40-bit security (instead of the claimed 112-bit level).

First, we revisit the underlying building block of key-homomorphic one-time signature (KOTS) and introduce a novel security reduction based on a new lattice problem: the *Dual Hint-MLWE* assumption, which may be of independent interest. We then provide a formal reduction from the standard Module-LWE problem to Dual Hint-MLWE, which overall enables us to base the security of Lemur on standard Module-LWE and Module-SIS assumptions. By shifting from a statistical security argument to a computational one, our KOTS design enjoys much better compactness and scalability.

Second, we optimize the underlying homomorphic vector commitment (HVC) by transitioning from Ring-SIS to the Module-SIS setting and extending the commitment domain from vectors to matrices. This generalization allows for a more compact HVC.

After rectifying the parameters of Chipmunk for a fair comparison, our results show that Lemur's KOTS size achieves up to an order of magnitude improvement over Chipmunk's KOTS. In particular, aggregating 1 million one-time signatures requires only about 10 KB. For the total multi-signature size, Lemur demonstrates up to 2× improvement over Chipmunk. To showcase our design, we provide a full-fledged Rust implementation. Our benchmarks demonstrate an aggregate signature size of 394 KB for 2^{20} signers, a stateful signing time of roughly 4.1 ms (for a Merkle tree height of 20), and a batch verification time of 30.1 ms for an aggregated signature of 1024 signers.

CCS Concepts

• **Security and privacy → Digital signatures**.

Conference acronym 'XX, Woodstock, NY

ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

Keywords

Post-Quantum, Lattice, Multi-Signatures, Key-Homomorphic One-Time Signature, Homomorphic Vector Commitment

ACM Reference Format:
Anonymous Author(s). 2026. Lemur: Scalable Post-Quantum Synchronized Multi-Signatures. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 24 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Multi-signatures allow many signers to authenticate the same message by producing a single, compact aggregated signature [10, 15]. This primitive is a key building block in distributed systems, especially in proof-of-stake blockchains such as Ethereum and Dfinity, where large validator sets must repeatedly reach agreement on the validity of proposed blocks. By compressing many signatures into one short object, multi-signatures substantially reduce communication, storage, and verification costs in the consensus layer.

For such applications, *non-interactive* aggregation is crucial. In a public blockchain, validators may be online at different times, may not know the identities of others, and often communicate only indirectly through the network. Interactive protocols are therefore a poor fit: they require additional rounds of communication, introduce coordination bottlenecks, and increase latency in already time-sensitive consensus pipelines. By contrast, *non-interactive* aggregation allows any untrusted aggregator to collect independently generated signatures and combine them publicly, without requiring any further coordination among the signers themselves.

This functionality is already central to real-world blockchain deployments. Today, major proof-of-stake systems such as Ethereum and Dfinity achieves non-interactive aggregation using BLS signature scheme [3]. While highly compact, the security of BLS rests on discrete-logarithm assumptions in pairing-friendly groups and is therefore vulnerable to quantum attacks. As a result, replacing BLS with a post-quantum alternative that preserves public, non-interactive aggregation has become a pressing challenge for blockchain consensus.

Designing such a replacement is difficult. One possible direction is hash-based cryptography. However, hash-based signatures do not naturally support aggregation, since hash functions lack the algebraic structure that helps with making multi-signatures compact. Existing proposals [7] therefore achieve aggregation only by invoking succinct arguments, such as SNARK-like proof systems, to prove knowledge of many valid signatures. While conceptually appealing, this approach introduces heavy proof machinery and substantial computational overhead, which makes it difficult to deploy in latency-sensitive consensus settings.

A promising alternative to hash-based approach is to build non-interactive aggregation from lattices. In this direction, Squirrel [9] and its successor Chipmunk [8] constructed lattice-based non-interactive multi-signatures in the *synchronized* setting, where a common message is signed at a particular time step. This setting

N (signers)	Chipmunk (KB)		Lemur (KB)	
	One-Time Sig	Multi-Sig	One-Time Sig	Multi-Sig
2^{10}	26	237	5.6	201
2^{15}	>90	>418	7.2	284
2^{20}	>110	>831	8.6	394

Table 1: Aggregated signature sizes at 128-bit security (τ=20). Chipmunk: theoretical, from their scripts. Lemur: measured in implementation for N=2^{10}; Rice-encoded on the corresponding cells of Table 5 for $N \in \{2^{15}, 2^{20}\}$.

Operation	$N = 2^{10}$	$N = 2^{15}$	$N = 2^{20}$
Signing (BDS08)	4.1 ms	4.1 ms	4.1 ms
Signing (Tree-in-memory)	2 ms	2 ms	2 ms
Aggregation	567 ms	$\approx$ 23 s†	$\approx$ 12 min†
Batch verification	30.1 ms	812 ms§	25.3 s§
Agg. signature size	201.2 KB	283.5 KB‡	394.4 KB‡

†Linear extrapolations from $N = 8192$ timing. ‡Predicted Rice-encoded sizes on the $\tau = 20$ rows of Table 5; the artifact only has the $N = 2^{10}$ cell. §Benchmarked with a zero-public-key fixture.

Table 2: Representative Rust implementation performance using (τ=20, N=2^{10}) parameter setting; 24 CPU threads.

is particularly natural for proof-of-stake blockchains: validators are already synchronized by the protocol, sign the same candidate block in the same time step, and are expected to sign at most one message per time step. Thus, synchronized multi-signatures capture the needs of blockchain consensus well.

However, upon closer inspection, we find that Chipmunk [8] does not achieve its claimed concrete security level. In particular, its parameter selection relies on a rough heuristic evaluation of Ring-SIS hardness. When evaluating using the state-of-the-art Dilithium security estimation framework, the proposed parameters correspond to substantially lower concrete security (approximately 40-bit instead of the claimed 112-bit). The implications of this finding are significant: rectifying Chipmunk's parameters to reach a standard 128-bit security level increases their aggregate signature size by a factor of at least 2×, significantly undermining its reported efficiency [8]. Moreover, Chipmunk's reliance on Ring-SIS assumption limits its parameter flexibility compared to Module-SIS, further complicating the trade-off between security and signature compactness.

Taken together, the current landscape shows that post-quantum non-interactive aggregation is no longer out of reach, but existing approaches still leave substantial room for improvement. The central challenge is therefore to design a post-quantum synchronized multi-signature while achieving better concrete efficiency, scalability, and flexibility for blockchain-scale systems.

1.1 Our Contributions

In this work, we present Lemur, a concretely efficient, non-interactive lattice-based multi-signature scheme in the synchronized setting. Lemur builds upon the non-interactive framework in Chipmunk [8] but fundamentally redesigns the foundations of its underlying building blocks to overcome bandwidth and scalability bottlenecks. Specifically, by shifting away from the statistical security arguments that forced previous schemes to adopt large parameters, Lemur achieves significantly smaller aggregated signature sizes for large sets of signers while maintaining security against rogue-key attacks in the random oracle model.

Our improvements stem from two major technical contributions regarding the underlying cryptographic primitives. First, we revisit the key-homomorphic one-time signature (KOTS) used in Chipmunk (arising from [13]). Unlike the KOTS in Chipmunk, which relies on a statistical security argument, our construction introduces and relies upon a computational assumption: the Dual Hint-MLWE assumption. To establish confidence in this new assumption, we provide a formal, rigorous reduction from the standard Module-LWE problem to our Dual Hint-MLWE assumption. By transitioning to a computational argument, we are able to significantly improve the state-of-the-art key-homomorphic one-time signature in concrete sizes while relying on a standard post-quantum problem. Concretely, we can aggregate 1 million KOTS outputs into just 10 KB while Chipmunk requires more than 110 KB (see Table 1).

In addition to being key-homomorphic, our KOTS design enjoys a number of attractive features such as yielding *unique* signatures. As shown in [5], a unique one-time signature can be extended to a *many-time* verifiable random function (VRF) using a Merkle tree. Furthermore, the signing procedure is completely linear, which makes masking against side-channel attacks easy when needed.

Second, we extend the underlying homomorphic vector commitment (HVC) scheme used to compress the KOTS public keys. We transition the foundation of the HVC from the Ring-SIS setting to the Module-SIS setting. Furthermore, we extend the commitment domain from vectors to matrices. This generalization allows for more compact commitments and openings under aggregation, reducing the bandwidth required to support large signer sets without affecting correctness or aggregation success probability.

Combined, these theoretical optimizations allow Lemur to outperform the state-of-the-art [8, 9]. Our experimental results demonstrate that Lemur achieves significantly smaller aggregate signatures while maintaining scalability in settings involving thousands and even millions of signers. We refer to Tables 1 and 5 for a size comparison and Table 2 for the concrete Lemur runtimes. Since Chipmunk's implemented parameters are at a substantially lower security level (around 40 bits) than ours (128 bits), we are unable to provide a meaningful runtime comparison against Chipmunk.

Implementation and Benchmarks. To validate Lemur's deployability, we provide two interoperable open-source reference implementations: an optimized **Rust** crate and a readable **Python** prototype, both covering the full pipeline (KOTS, HVC, aggregation, and bit-packed encoding) and cross-validated against shared byte-level test vectors. Our default signer uses the BDS08 Merkle traversal algorithm [4], replacing a multi-GB materialized tree with a roughly 134 KB authentication-path cache against a 32-byte master seed; this brings stateful signing to roughly 4 ms on commodity hardware. At $\lambda = 128$ and $\tau = 20$, batch verification of an aggregated signature runs in 30 ms for $N = 1024$, and the aggregated signature itself is 201.2 KB at $N = 2^{10}$ (see Table 5 for larger N). The artifact is

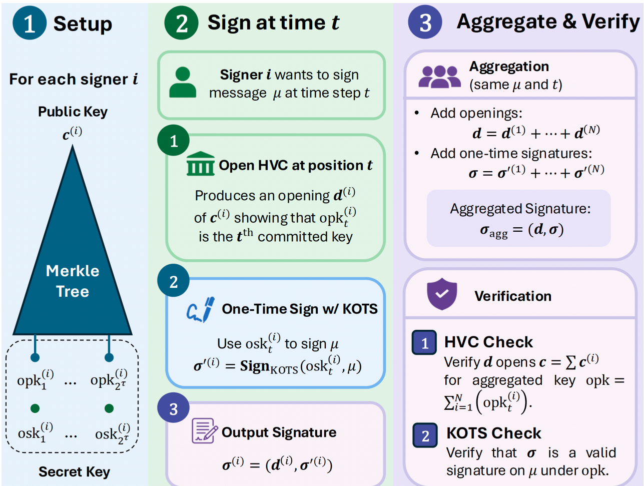

Figure 1: High Level Idea of Lemur / Chipmunk / Squirrel.

released at https://anonymous.4open.science/r/lemur-anon-5A9D/; further implementation notes appear in Section 7.2.

Open question. Our work leaves an important open question for future work. As discussed above, our KOTS achieves highly compact signatures, requiring below 10 KB for 1 million signers. However, the opening of HVC, needed for many-time signing, remains relatively large (refer to Table 5). An interesting question here is if we can find an alternative way to extend from one-time signing to many-time signing via other means, for example a solution that exploits the higher-level application layer.

1.2 Technical Overview

At a high level, Lemur follows the blueprint of Chipmunk [8]. We first recall the common high-level framework, illustrated in Figure 1, and then highlight the technical points where Lemur departs from and improves upon it.

Chipmunk Recap. In Chipmunk, signatures are aggregated only for the same message and time step, and each signer supports at most 2^τ signing opportunities before key refresh. As shown in Figure 1, during Setup (left), each signer generates 2^τ one-time key pairs and commits to the vector of one-time public keys using a homomorphic vector commitment (HVC), yielding a compact public key. To sign a message μ at time t (middle), the signer opens the commitment at position t and produces a one-time signature under the corresponding secret key, forming an individual signature consisting of an opening and a one-time signature. Verification and aggregation (right) exploit the homomorphic properties of both primitives: openings and one-time signatures are combined via sums, producing a single aggregate signature that verifies for the same message and time step under the aggregated public key. Due to security requirements, the actual aggregation is done via a *weighted-sum* (with weights derived from a random oracle), but we omit the weights here for simplicity.

Key-Homomorphic One-Time Signatures. Chipmunk's KOTS closely follows the line of work of Lyubashevsky-Micciancio [13] and Boneh-Kim [2], and is conceptually very simple. However, earlier unforgeability proofs rely on statistical arguments. As a result, the parameters must be chosen so that the secret key retains sufficiently large entropy even after revealing the public key and one signature. Our Lemur design avoids this bottleneck by replacing the statistical step with a computational argument via our new computational assumption: Dual Hint-MLWE.

At a high level, Lemur's KOTS works as follows. Each signer samples their secret key as a short (i.e., small-norm) matrix $\mathbf{S}$ from a discrete Gaussian distribution and publish $\mathbf{T} = \mathbf{SA} \bmod q'$ as the public key, where q' a system modulus parameter. To sign a message μ, the signer first maps it to a short matrix $\mathbf{H}$ via a random oracle, and outputs the signature $\mathbf{Z} = \mathbf{HS}$. Verification then consists of two checks: a norm bound on $\mathbf{Z}$, and the algebraic condition $\mathbf{ZA} = \mathbf{HT} \bmod q'$. This preserves homomorphism under aggregation.

The main challenge is security. Revealing a linear image $\mathbf{HS}$ of the secret key appears, at first glance, to leak substantial information about $\mathbf{S}$. Lemur relies on the *Dual Hint-MLWE* assumption to control this leakage. Informally, this assumption states that even given the hint $(\mathbf{H}, \mathbf{HS})$, the MLWE instance $(\mathbf{A}, \mathbf{SA})$ is computationally indistinguishable from a rerandomized instance $(\mathbf{A}, (\mathbf{S}+\mathbf{U})\mathbf{A})$, where each column of $\mathbf{U}$ is sampled from the kernel of $\mathbf{H} \bmod q'$. We formally prove the unforgeability of Lemur's KOTS under the Dual Hint-MLWE and MSIS assumption in Lemma 4.1.

Security of Dual Hint-MLWE. Kim et al. [11] introduced the Hint-MLWE assumption, which states that an MLWE sample $(\mathbf{A}, \mathbf{As})$ remains indistinguishable from uniform even when given a bounded number of noisy linear hints of the form $\mathbf{z}_i = c_i \cdot \mathbf{s} + \mathbf{y}_i$. Unfortunately, this assumption is not directly applicable to Lemur's KOTS unforgeability proof for two reasons. First, Lemur's KOTS reveals *noise-free* linear images of the secret key, rather than perturbed hints. Second, the secret key appears on opposite sides of the public key and the signature computation: the public key is of the form $\mathbf{T} = \mathbf{SA}$, while the signature reveals $\mathbf{Z} = \mathbf{HS}$. This structural mismatch necessitates a new assumption.

We therefore introduce the *Dual Hint-MLWE* problem. In this setting, the adversary is asked to distinguish between two distributions: the real sample $(\mathbf{A}, \mathbf{SA}, \mathbf{H}, \mathbf{HS})$, and the "uniform" sample $(\mathbf{A}, (\mathbf{S}+\mathbf{U})\mathbf{A}, \mathbf{H}, \mathbf{HS})$ where $\mathbf{U}$ is a mask term uniformly chosen to stay in the kernel of the hint $(\mathbf{HU} = \mathbf{0})$. We prove that the Dual Hint-MLWE is at least as hard as the standard MLWE by a sequence of hybrids in Theorem 3.1. A key technical challenge in the proof is sampling from discrete Gaussians over *non-full-rank* lattice, a setting not directly covered by existing literature. We establish a suitable sampleability lemma and provide a rigorous analysis, which may be of independent interest.

2 Preliminaries

Notations. For $n \in \mathbb{N}^+$, we identify $\mathbb{Z}_n$ with the representative set $\mathbb{Z} \cap (-n/2, n/2]$. Let $[n] = \{1, 2, \ldots, n\}$ and $[\![n]\!] = \{0, 1, \ldots, n-1\}$. For a bit string $s = (s_1, \ldots, s_\tau) \in \{0,1\}^\tau$, let s_i be the i-th bit and $s_{<i} := (s_1, \ldots, s_{i-1})$ be the first $i-1$ bits. For $i \in [\![2^\tau]\!]$, let $\mathrm{bin}_\tau(i) \in \{0,1\}^\tau$ denote the big-endian binary decomposition of i. We use bold lower-case letters for column vectors (e.g., $\mathbf{a}$) and bold upper-case for matrices (e.g., $\mathbf{A}$). For any n-dimensional vector $\mathbf{a} = (a_1, \ldots, a_n)$, let a_i be the i-th entry and $\mathbf{a}_{<i} := (a_1, \ldots, a_{i-1})$ be the prefix. We

write $\mathbf{a}^\top$ for the transpose (treated as a row vector). For an $m \times n$ matrix $\mathbf{A}$, the entry in row i and column j is $a_{i,j}$ for $i \in [m]$, $j \in [n]$. Let $\mathbf{I}_n$ denote the $n \times n$ identity matrix. For two vectors of the same length $\mathbf{a} = (a_1, \ldots, a_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$, define $\mathsf{zip}(\mathbf{a}, \mathbf{b}) = ((a_1, b_1), \ldots, (a_n, b_n))$. Define $\mathrm{vec}(\mathbf{M})$ as the vector obtained by stacking the columns of the matrix $\mathbf{M}$ one underneath the other (i.e., concatenating its columns in order). The operator inv^{-1} denotes the inverse mapping of vec operation, which reconstructs a matrix from its vectorized form. We use $[\mathbf{A}; \mathbf{B}]$ to denote the vertical stack of two matrices $\mathbf{A}$ and $\mathbf{B}$ and $[\mathbf{A} \mid \mathbf{B}]$ to denote the horizontal stack of $\mathbf{A}$ and $\mathbf{B}$.

Let d be a power of 2 and q be an integer modulus. Denote $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ and $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^d + 1)$. Multiplication between elements of $\mathcal{R}$ and $\mathcal{R}_q$ is performed modulo q by default. For a polynominal $f = \sum_{i=0}^{d-1} f_i X^i \in \mathcal{R}$, the $\ell^p (p \geq 1)$ and ℓ^∞ norms are defined as:

$$\|f\|_p := \sqrt[p]{\sum_{i=0}^{d-1} |f_i|^p}, \qquad \|f\|_\infty := \max_{i \in [\![d]\!]} |f_i|. \tag{1}$$

For any vector $\mathbf{f} = (f_1, \ldots, f_n) \in \mathcal{R}^n$, we define

$$\|\mathbf{f}\|_p := \sqrt[p]{\sum_{i=1}^{n} \|f_i\|_p^p}, \qquad \|\mathbf{f}\|_\infty := \max_{i \in [n]} \|f_i\|_\infty. \tag{2}$$

For a matrix $\mathbf{A} \in \mathcal{R}^{m \times n}$, define the max norm as: $\|\mathbf{A}\|_{\max} = \max_{i \in [m], j \in [n]} \|a_{i,j}\|_\infty$. The spectral norm $\|\mathbf{A}\|_2$ of a matrix $\mathbf{A} \in \mathcal{R}^{m \times n}$ is defined as the largest singular value of the matrix $\sigma(\mathbf{A}) \in \mathbb{Z}^{md \times nd}$, where $\sigma(\cdot)$ replaces each polynomial in $\mathcal{R}$ with its $d \times d$ negacyclic convolution matrix. For consistency, throughout the paper, we write $\|f\|_{\max}$ and $\|\mathbf{f}\|_{\max}$ to mean $\|f\|_\infty$ in (1) and $\|\mathbf{f}\|_\infty$ in (2), respectively. Define the set of ternary polynomials (coefficients in $\{-1, 0, 1\}$) with exactly α non-zero coefficients as $\mathcal{T}_\alpha = \{f \in \mathcal{R} : \|f\|_{\max} = 1 \land \|f\|_1 = \alpha\}$.

2.1 Probability Distribution

For a finite set S, we let $x \leftarrow S$ denote sampling x uniformly at random from S. More generally, for a distribution $\mathcal{D}$, $x \leftarrow D$ denotes sampling x according to D. Let D_1 and D_2 be two distributions over a countable set S. Their statistical distance is defined as $\Delta(D_1, D_2) := \frac{1}{2} \sum_{x \in S} |D_1(x) - D_2(x)|$. Let r, n be positive integers with $1 \leq r \leq n$, and let $\mathbf{B} \in \mathbb{R}^{n \times r}$ of $\mathbb{R}$-rank r. We recall that $\Lambda := \mathbf{B} \cdot \mathbb{Z}^r$ is called a rank r lattice in $\mathbb{R}^n$ with column $\mathbb{Z}$-basis $\mathbf{B}$. If $r = n$, we say that Λ is a full-rank lattice.

Definition 2.1 (Smoothing Parameter). *[Adapted from [16], Def. 3.1] For any lattice $\Lambda \subset \mathbb{R}^n$ the smoothing parameter $\eta_\varepsilon(\Lambda)$ is the smallest $s > 0$ such that $\rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \varepsilon$, where $\Lambda^* := \{\mathbf{x} \in \mathrm{Span}_{\mathbb{R}}(\Lambda) : \forall \mathbf{y} \in \Lambda, \langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z}\}$ is the dual lattice of Λ.*

Gaussian Distributions. Let $\Sigma \in \mathbb{R}^{n \times n}$ be positive definite and $\mathbf{c} \in \mathbb{R}^n$. Define the (unnormalized) elliptical Gaussian function

$$\rho_{\mathbf{c},\sqrt{\Sigma}} : \mathbb{R}^n \to (0, 1], \qquad \rho_{\mathbf{c},\sqrt{\Sigma}}(\mathbf{x}) := \exp\Bigl(-\pi\, (\mathbf{x} - \mathbf{c})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{c})\Bigr).$$

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and let $\mathbf{v} \in \mathbb{R}^n$. The discrete Gaussian distribution over the coset $\mathbf{v} + \Lambda$ is defined by

$$\mathcal{D}_{\mathbf{v}+\Lambda,\mathbf{c},\sqrt{\Sigma}}(\mathbf{x}) := \frac{\rho_{\mathbf{c},\sqrt{\Sigma}}(\mathbf{x})}{\rho_{\mathbf{c},\sqrt{\Sigma}}(\mathbf{v} + \Lambda)},$$

where $\mathbf{x} \in \mathbf{v} + \Lambda$ and $\rho_{\mathbf{c},\sqrt{\Sigma}}(\mathbf{v} + \Lambda) := \sum_{\mathbf{y} \in \mathbf{v}+\Lambda} \rho_{\mathbf{c},\sqrt{\Sigma}}(\mathbf{y})$. Note that coset sampling from $\mathcal{D}_{\mathbf{v}+\Lambda,\mathbf{0},\sqrt{\Sigma}}$ is equivalent to sampling from $\mathcal{D}_{\Lambda,-\mathbf{v},\sqrt{\Sigma}}$ and then adding the shift $\mathbf{v}$. When $\mathbf{c} = \mathbf{0}$, we omit $\mathbf{c}$ from subscripts. When $\Sigma = \alpha^2 \mathbf{I}_n$, we have $\sqrt{\Sigma} = \alpha \mathbf{I}_n$, and we may write the distribution as $\mathcal{D}_{\mathbf{v}+\Lambda,\mathbf{c},\alpha}$ (or $\mathcal{D}_{\Lambda,\mathbf{c},\alpha}$ when $\mathbf{v} = \mathbf{0}$). The scalar $\alpha > 0$ is called the width parameter. We now recall several standard results used in the security analysis of lattice-based schemes.

Lemma 2.1 ([17], Theorem 3.1). *Let $\Sigma_1, \Sigma_2 > \mathbf{0}$ be positive definite matrices, with $\Sigma = \Sigma_1 + \Sigma_2 > \mathbf{0}$. Let Λ_1, Λ_2 be lattices such that $\sqrt{\Sigma_1} \geq \eta_\varepsilon(\Lambda_1)$ for some $\varepsilon \in (0, 1/2]$. Let $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^n$. Consider the following experiment:*

(1) Sample $\mathbf{x}_2 \leftarrow \mathcal{D}_{\Lambda_2+\mathbf{c}_2,\sqrt{\Sigma_2}}$.
(2) Sample $\mathbf{e} \leftarrow \mathcal{D}_{\Lambda_1+\mathbf{c}_1-\mathbf{x}_2,\sqrt{\Sigma_1}}$ and set $\mathbf{x}_1 = \mathbf{x}_2 + \mathbf{e}$.

The marginal distribution of $\mathbf{x}_1$ is within statistical distance 8ε of $\mathcal{D}_{\Lambda_1+\mathbf{c}_1,\sqrt{\Sigma}}$.

Lemma 2.2 ([17], Lemma 2.4). *For any lattice $\Lambda \subset \mathbb{R}^n$, $\varepsilon \in (0, 1)$, $\Sigma > \mathbf{0}$ such that $\sqrt{\Sigma} \geq \eta_\varepsilon(\Lambda)$, and any $\mathbf{v} \in \mathbb{R}^n$:*

$$\rho_{\sqrt{\Sigma}}(\Lambda + \mathbf{v}) \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1\right] \cdot \rho_{\sqrt{\Sigma}}(\Lambda).$$

Lemma 2.3 ([16], Lemma 4.4). *For any n-dimensional lattice Λ, vector $\mathbf{c} \in \mathbb{R}^n$, and reals $\varepsilon \in (0, 1)$, $\alpha \geq \eta_\varepsilon(\Lambda)$, it holds that $\rho_{\mathbf{c}}(\Lambda) \geq (1 - \varepsilon) \det(\Lambda)^{-1}$ and $\rho(\Lambda) \leq (1 + \varepsilon) \det(\Lambda)^{-1}$.*

Lemma 2.4 (Adapted from [16]). *For a rank r lattice $\Lambda \subset \mathbb{R}^n$ with $1 \leq r \leq n$ and $\varepsilon > 0$, $\varepsilon \in \mathbb{R}^+$,*

$$\eta_\varepsilon(\Lambda) \leq \sqrt{\frac{\ln\bigl(2r(1 + 1/\varepsilon)\bigr)}{\pi}} \cdot \lambda_r(\Lambda) \leq \sqrt{\frac{\ln\bigl(2r(1 + 1/\varepsilon)\bigr)}{\pi}} \cdot \max_{i \in [r]} \|\mathbf{b}_i\|_2,$$

where $\lambda_r(\Lambda)$ is the smallest radius of an n-dimensional ball that contains r linearly independent vectors in Λ, and $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_r]$ is a basis of the lattice Λ.

Lemma 2.5 ([6], Lemma 4). *There is a probabilistic polynomial time algorithm that, given a basis $\mathbf{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ of a full-rank n-dimensional lattice Λ, a positive definite symmetric matrix Σ and $\mathbf{c} \in \mathbb{R}^n$ returns a sample from $\mathcal{D}_{\Lambda,\Sigma,\mathbf{c}}$, assuming that*

$$\sqrt{\ln(2n + 4)/\pi} \cdot \max_i \|\Sigma^{-1/2} \mathbf{b}_i\|_2 \leq 1.$$

The above lemma works for *full-rank* lattices, while our work needs an analogous result for general rank-r lattices in $\mathbb{R}^n$ with $r \leq n$. We show in the lemma below the extension to the general rank-r case. Due the space limit, we defer the proof to Appendix D.

Lemma 2.6 (Reduction to full-rank sampler). *Let r and n be positive integers with $1 \leq r \leq n$. Suppose there is an algorithm $\mathcal{A}$ that, given as input a full-rank column $\mathbb{Z}$-basis $\mathbf{B}' = (\mathbf{b}_1, \ldots, \mathbf{b}_r) \in \mathbb{R}^{r \times r}$ for a full-rank lattice $\Lambda' := \mathbf{B}' \mathbb{Z}^r \subseteq \mathbb{R}^r$, a vector $\mathbf{t} \in \mathbb{R}^r$ and a parameter $s \geq \eta(r) \cdot \max_i \|\mathbf{b}_i\|_2$ (for some function $\eta : \mathbb{R} \mapsto \mathbb{R}$), outputs a sample from the discrete Gaussian $\mathcal{D}_{\Lambda'+\mathbf{t},s}$ in time $T(r, \ell(\mathbf{B}'))$, where $\ell(\mathbf{B}')$ denotes the bit-length of $\mathbf{B}'$. Then there is an algorithm $\mathcal{B}$ that, given a column $\mathbb{Z}$-basis $\mathbf{B}_0 = (\mathbf{b}_{0,1}, \ldots, \mathbf{b}_{0,r}) \in \mathbb{R}^{n \times r}$ for a rank r lattice $\Lambda_0 := \mathbf{B}_0 \cdot \mathbb{Z}^r \subseteq \mathbb{R}^n$, a vector $\mathbf{c}_0 \in \mathbb{R}^n$ and a positive definite symmetric matrix $\Sigma_0 \in \mathbb{R}^{n \times n}$, returns a sample from $\mathcal{D}_{\Lambda_0+\mathbf{c}_0,\sqrt{\Sigma_0}}$, assuming that*

$$\eta(r) \cdot \max_{i \in [r]} \|\Sigma_0^{-1/2} \mathbf{b}_{0,i}\|_2 \leq 1, \tag{3}$$

in time $\mathrm{poly}(\ell(\mathbf{B}_0)) + T\big(r, \ell(\mathbf{B}_0)\big)$*. In particular, if* $\mathcal{A}$ *runs in polynomial time, then so does* $\mathcal{B}$*.*

2.2 Linear Maps

Let $\ell, k \in \mathbb{N}^+$ and $\mathbf{H} \in \mathcal{R}^{\ell \times k}$. We define the $\mathcal{R}$-module and $\mathcal{R}_q$-module homomorphisms as:

$$\varphi_{\mathbf{H}} : \mathcal{R}^k \to \mathcal{R}^\ell, \quad \varphi_{\mathbf{H}}(\mathbf{s}) = \mathbf{Hs},$$

$$\varphi_{\mathbf{H},q} : \mathcal{R}_q^k \to \mathcal{R}_q^\ell, \quad \varphi_{\mathbf{H},q}(\mathbf{s}) = \mathbf{Hs} \bmod q.$$

Their respective kernels are denoted by:

$$\ker(\varphi_{\mathbf{H}}) = \{\mathbf{s} \in \mathcal{R}^k : \mathbf{Hs} = \mathbf{0}\},$$

$$\ker(\varphi_{\mathbf{H},q}) = \{\mathbf{s} \in \mathcal{R}_q^k : \mathbf{Hs} = \mathbf{0} \bmod q\}.$$

Let $\pi : \ker(\varphi_{\mathbf{H}}) \to \ker(\varphi_{\mathbf{H},q})$ be the natural reduction map defined by $\pi(\mathbf{x}) = \mathbf{x} \bmod q$.

Lemma 2.7 (Coset Resampling). *Let* $\mathbf{H} \in \mathcal{R}^{\ell \times k}$*. For* $\alpha > 0$*, the following two distributions over* $\mathcal{R}^k \times \mathcal{R}^\ell$ *are identical:*

(1) $D_1 = \{(\mathbf{s}, \mathbf{z}) : \mathbf{s} \leftarrow \mathcal{D}_{\mathcal{R}^k,\alpha}, \mathbf{z} = \mathbf{Hs}\}$.
(2) $D_2 = \{(\tilde{\mathbf{s}}, \mathbf{z}) : \mathbf{s} \leftarrow \mathcal{D}_{\mathcal{R}^k,\alpha}, \mathbf{z} = \mathbf{Hs}, \tilde{\mathbf{s}} \leftarrow \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s},\alpha}\}$.

Proof. Fix $(\hat{\mathbf{s}}, \hat{\mathbf{z}}) \in \mathcal{R}^k \times \mathcal{R}^\ell$. Let $C(\hat{\mathbf{z}}) = \{\mathbf{x} \in \mathcal{R}^k : \mathbf{Hx} = \hat{\mathbf{z}}\}$. Note that if $\hat{\mathbf{s}} \in C(\hat{\mathbf{z}})$, then $C(\hat{\mathbf{z}}) = \ker(\varphi_{\mathbf{H}}) + \hat{\mathbf{s}}$. Under D_1, $\Pr[(\hat{\mathbf{s}}, \hat{\mathbf{z}})]$ is simply $\mathcal{D}_{\mathcal{R}^k,\alpha}(\hat{\mathbf{s}})$ if $\mathbf{H}\hat{\mathbf{s}} = \hat{\mathbf{z}}$, and 0 otherwise. Under D_2, the probability is:

$$\Pr[\mathbf{z} = \hat{\mathbf{z}}] \cdot \Pr[\tilde{\mathbf{s}} = \hat{\mathbf{s}} \mid \mathbf{z} = \hat{\mathbf{z}}] = \left(\frac{\rho_\alpha(C(\hat{\mathbf{z}}))}{\rho_\alpha(\mathcal{R}^k)}\right) \cdot \left(\frac{\rho_\alpha(\hat{\mathbf{s}})}{\rho_\alpha(C(\hat{\mathbf{z}}))}\right) = \frac{\rho_\alpha(\hat{\mathbf{s}})}{\rho_\alpha(\mathcal{R}^k)},$$

which is exactly $\mathcal{D}_{\mathcal{R}^k,\alpha}(\hat{\mathbf{s}})$ if $\hat{\mathbf{s}} \in C(\hat{\mathbf{z}})$, and 0 otherwise. □

Lemma 2.8. *Let* $\mathbf{H} \in \mathcal{R}^{\ell \times k}$*. If there exists* $\mathbf{U} \in \mathcal{R}^{k \times \ell}$ *such that* $\mathbf{HU} = \mathbf{I}_\ell$*, then* $\pi : \ker(\varphi_{\mathbf{H}}) \to \ker(\varphi_{\mathbf{H},q})$ *defined by* $\pi(\mathbf{x}) = \mathbf{x} \bmod q$ *is surjective.*

Proof. Let $\mathbf{t}_q \in \ker(\varphi_{\mathbf{H},q})$ and let $\mathbf{t} \in \mathcal{R}^k$ be any lift such that $\mathbf{t} \bmod q = \mathbf{t}_q$. Since $\mathbf{Ht} \equiv \mathbf{0} \pmod q$, we have $\mathbf{Ht} = q\mathbf{r}$ for some $\mathbf{r} \in \mathcal{R}^\ell$. Set $\mathbf{s} = \mathbf{t} - q\mathbf{Ur}$. Then $\mathbf{Hs} = \mathbf{Ht} - q(\mathbf{HU})\mathbf{r} = q\mathbf{r} - q\mathbf{r} = \mathbf{0}$. Thus $\mathbf{s} \in \ker(\varphi_{\mathbf{H}})$, and since $\mathbf{s} \equiv \mathbf{t} \equiv \mathbf{t}_q \pmod q$, the map π is surjective. □

Lemma 2.9. *Let* $\ell, k \in \mathbb{N}^+$ *such that* $k > \ell$*. Let* $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}'] \in \mathcal{R}^{\ell \times k}$*. Then,* $\mathbf{B} = [-\mathbf{H}'; \mathbf{I}_{k-\ell}] \in \mathcal{R}^{k \times (k-\ell)}$ *is a basis for* $\ker(\varphi_{\mathbf{H}})$*, and* $\mathbf{B}_q = \mathbf{B} \bmod q$ *is a basis for* $\ker(\varphi_{\mathbf{H},q})$*.*

Proof. We first prove that $\mathbf{B} = [-\mathbf{H}'; \mathbf{I}_{k-\ell}]$ is a basis of $\ker(\varphi_{\mathbf{H}})$. To show this, we prove that $\mathbf{B}$ spans $\ker(\varphi_{\mathbf{H},q})$ and is linearly independent. Let $\mathbf{x} \in \ker(\varphi_{\mathbf{H}}) \subseteq \mathcal{R}^k$ be arbitrary. Partition $\mathbf{x}$ into $\mathbf{x}_1 \in \mathcal{R}^\ell$ and $\mathbf{x}_2 \in \mathcal{R}^{k-\ell}$. Then $\mathbf{Hx} = \mathbf{0}$ implies $\mathbf{I}_\ell \mathbf{x}_1 + \mathbf{H}'\mathbf{x}_2 = \mathbf{0}$. We can therefore write $\mathbf{x} = \mathbf{Bx}_2$. We now observe that each column of $\mathbf{B}$ is linearly independent. If $\mathbf{Bc} = \mathbf{0}$, then the bottom $k - \ell$ rows give $\mathbf{I}_{k-\ell}\mathbf{c} = \mathbf{0}$, implying $\mathbf{c} = \mathbf{0}$. Thus, $\mathbf{B}$ is a basis for $\ker(\varphi_{\mathbf{H}})$.

Second, we show that $\mathbf{B}_q$ is a basis for $\ker(\varphi_{\mathbf{H},q})$. Define $\mathbf{U} = \begin{pmatrix} \mathbf{I}_\ell \\ \mathbf{0} \end{pmatrix} \in \mathcal{R}^{k \times \ell}$. Since $\mathbf{HU} = \mathbf{I}_\ell$, Lemma 2.8 implies that the reduction map $\pi : \ker(\varphi_{\mathbf{H}}) \to \ker(\varphi_{\mathbf{H},q})$ defined by $\pi(\mathbf{x}) = \mathbf{x} \bmod q$ is surjective. To show $\mathbf{B}_q$ is a basis, we show it spans $\ker(\varphi_{\mathbf{H},q})$ and is linearly independent: Let $\mathbf{u}_q \in \ker(\varphi_{\mathbf{H},q})$. By the surjectivity of π, there exists $\mathbf{u} \in \ker(\varphi_{\mathbf{H}})$ such that $\mathbf{u} \bmod q = \mathbf{u}_q$. Since $\mathbf{B}$ spans $\ker(\varphi_{\mathbf{H}})$, we have $\mathbf{u} = \mathbf{Br}$ for some $\mathbf{r} \in \mathcal{R}^{k-\ell}$. Then $\mathbf{u}_q = \mathbf{Br} \bmod q = \mathbf{B}_q(\mathbf{r} \bmod q)$, so $\mathbf{B}_q$ spans the modular kernel. Suppose $\mathbf{B}_q\mathbf{c}_q = \mathbf{0} \bmod q$ for $\mathbf{c}_q \in \mathcal{R}_q^{k-\ell}$. The last $k - \ell$ rows of this system give $\mathbf{I}_{k-\ell}\mathbf{c}_q = \mathbf{0} \bmod q$, so $\mathbf{c}_q = \mathbf{0} \in \mathcal{R}_q^{k-\ell}$. Since $\mathbf{B}_q$ is a linearly independent spanning set, it is a basis for $\ker(\varphi_{\mathbf{H},q})$, which is consequently a free module of rank $k - \ell$. □

Symbol	Description
λ	security parameter
d	a power of 2 to set $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$
q, q'	moduli for HVC and KOTS, respectively
τ	depth of the labelled full binary tree
N	number of signers in the multi-signature
η	HVC employs $(2\eta + 1)$-ary decomposition
κ, κ'	$\kappa := \lceil \log_{2\eta+1} q \rceil$ and $\kappa' := \lceil \log_{2\eta+1} q' \rceil$
ω	dimension of an HVC commitment $c \in \mathcal{R}_q^\omega$
ρ, ν	message domain of the HVC is $(\mathcal{R}_{q'}^{\rho \times \nu})^{2^\tau}$
$\mathbf{B}, \mathbf{A}_0, \mathbf{A}_1$	HVC public parameter: $\mathbf{B} \in \mathcal{R}_q^{\omega \times \rho\nu\kappa'}$ and $\mathbf{A}_0, \mathbf{A}_1 \in \mathcal{R}_q^{\omega \times \omega\kappa}$
β_{agg}	norm bound used by HVC strong verification
β_{enc}	norm bound for encoded HVC openings
W, α_w	weight set used in HVC / KOTS with $W = \mathcal{T}_{\alpha_w}$
α_H	hash output in KOTS: $\mathsf{H}(\mu) \in \mathcal{T}_{\alpha_H}^{\ell \times (k-\ell)}$
m, n	KOTS public parameter: $\mathbf{A} \in \mathcal{R}_{q'}^{m \times n}$
k	KOTS secret $\mathbf{S} \in \mathcal{R}^{k \times m}$ and public key $\mathbf{T} \in \mathcal{R}_q^{k \times n}$
ℓ	KOTS signature: $\mathbf{Z} \in \mathcal{R}^{\ell \times m}$
α	width parameter of KOTS secret
β_z	norm bound for individual signature
β_σ	norm bound for weighted sum of individual signatures
ϵ_{cor}	individual correctness error for KOTS
ϵ_{hom}	probabilistic homomorphism error (KOTS / HVC)
γ	maximum number of aggregation attempts

Table 3: Symbols used in this paper.

2.3 Lattice Assumptions

Definition 2.2 ($\mathsf{MLWE}_{q,m,n,k,\chi}$). *Let* χ *be a distribution over* $\mathcal{R}$*. Sample* $\mathbf{A}_2 \leftarrow \mathcal{R}_q^{(m-n) \times n}$*,* $\mathbf{S}_1 \leftarrow \chi^{k \times k}$*,* $\mathbf{S}_2 \leftarrow \chi^{k \times (m-k)}$*. Define* $\mathbf{A} := \begin{bmatrix} \mathbf{I}_n \\ \mathbf{A}_2 \end{bmatrix} \in \mathcal{R}_q^{m \times n}$ *and* $\mathbf{S} = [\mathbf{S}_1 \mid \mathbf{S}_2] \in \mathcal{R}^{k \times m}$*. The* $\mathsf{MLWE}_{q,m,n,k,\chi}$ *problem asks an adversary* $\mathcal{A}$ *to distinguish between the pairs* $(\mathbf{A}, \mathbf{SA})$ *and* $(\mathbf{A}, \mathbf{U})$*, where* $\mathbf{U} \leftarrow \mathcal{R}_q^{k \times n}$*. The advantage of* $\mathcal{A}$ *is defined as*

$$\mathrm{Adv}_{q,m,n,k,\chi}^{\mathsf{MLWE}}(\mathcal{A}) := \big|\Pr[\mathcal{A}(\mathbf{A}, \mathbf{SA}) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{U}) = 1]\big|.$$

Definition 2.3 ($\mathsf{MSIS}_{q,m,n,\beta}^\infty$). *Let* m, n, β, q *be positive integers such that* $0 < \beta < q$*. Then, for a given matrix* $\mathbf{A} \leftarrow \mathcal{R}_q^{m \times n}$*, the goal of the* $\mathsf{MSIS}_{q,m,n,\beta}^\infty$ *problem is to find* $\mathbf{x} \in \mathcal{R}^n$ *such that* $\mathbf{Ax} = \mathbf{0}$ *and* $0 < \|\mathbf{x}\|_{\max} \leq \beta$*. We say that a PPT adversary* $\mathcal{A}$ *has advantage* ϵ *in solving* $\mathsf{MSIS}_{q,m,n,\beta}^\infty$ *if*

$$\Pr\Big[0 < \|\mathbf{x}\|_{\max} \leq \beta \ \wedge\ \mathbf{Ax} = \mathbf{0} \ \Big|\ \mathbf{A} \leftarrow \mathcal{R}_q^{m \times n};\ \mathcal{A}(\mathbf{A}) = \mathbf{x}\Big] \geq \epsilon.$$

3 Dual Hint-MLWE Problem

We start with the problem definition.

Definition 3.1 (Dual Hint-MLWE). *Fix* $q', d, m, n, k, \ell \in \mathbb{N}^+$ *and distributions* χ, C *over* $\mathcal{R}$*. Let* $\mathbf{A} = [\mathbf{I}_n; \mathbf{A}_2] \in \mathcal{R}_{q'}^{m \times n}$ *for* $\mathbf{A}_2 \leftarrow \mathcal{R}_{q'}^{(m-n) \times n}$*,* $\mathbf{S} \leftarrow \chi^{k \times m}$*, and* $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}'] \in \mathcal{R}^{\ell \times k}$ *for* $\mathbf{H}' \leftarrow$

$C^{\ell\times(k-\ell)}$. *The Dual Hint-MLWE problem is to distinguish between* $D_0 := (\mathbf{A}, \mathbf{SA}, \mathbf{H}, \mathbf{HS})$ *and* $D_1 := (\mathbf{A}, (\mathbf{S}+\mathbf{U})\mathbf{A}, \mathbf{H}, \mathbf{HS})$, *where* $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m]$ *with* $\mathbf{u}_i \leftarrow \ker(\varphi_{\mathbf{H},q'})$ *sampled independently. The advantage of a PPT adversary* $\mathcal{A}$ *is defined as:*

$$\mathsf{Adv}^{\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\chi,C}}(\mathcal{A}) := |\Pr[\mathcal{A}(D_0)=1] - \Pr[\mathcal{A}(D_1)=1]|\,.$$

If $\chi = \mathcal{D}_{\mathcal{R},\alpha}$, *we denote the problem by* $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,C}$.

We prove the hardness of the Dual Hint-MLWE problem under the MLWE assumption when $\chi = \mathcal{D}_{\mathcal{R},\alpha}$ and $C = \mathcal{T}_{\alpha_H}$, i.e. the problem $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$.

Theorem 3.1 (Hardness of Dual Hint–MLWE). *Let* q', d, m, n, k, ℓ, $\alpha_H \in \mathbb{N}^+$ *with* $k > \ell$, *and let* $\alpha_0, \alpha, \varepsilon_2 \in \mathbb{R}_{>0}$. *Let* $\mathbf{H}' \leftarrow (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$ *and define* $\mathbf{H} := [\mathbf{I}_\ell \mid \mathbf{H}']$, $\mathbf{B} := \binom{-\mathbf{H}'}{\mathbf{I}_{k-\ell}} \in \mathcal{R}^{k\times(k-\ell)}$, *and* $\Sigma_1 := \alpha^2\mathbf{I}_k - \alpha_0^2\mathbf{B}\mathbf{B}^\top$. *If* $\alpha^2 \geq \alpha_0^2\,\|\mathbf{B}\|_2^2 + (1+\ell\alpha_H)\frac{\ln(2d(k-\ell)(1+1/\varepsilon_2))}{\pi}$, *then there exists a reduction* $\mathcal{B}$ *such that*

$$\mathsf{Adv}^{\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}}(\mathcal{A}) \leq \mathsf{Adv}^{\mathsf{MLWE}_{q',m,n,k-\ell,\alpha_0}}(\mathcal{B}) + 8m\varepsilon_2.$$

Proof. Let $\mathbf{A}_2 \leftarrow \mathcal{R}_{q'}^{(m-n)\times n}$, $\mathbf{A} := [\mathbf{I}_n; \mathbf{A}_2] \in \mathcal{R}_{q'}^{m\times n}$, $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_m] \leftarrow \mathcal{D}_{\mathcal{R}^{k\times m},\alpha}$, and $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}']$ with $\mathbf{H}' \leftarrow (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$. Consider the following sequence of hybrids:

- G_0: Output $(\mathbf{A}, \mathbf{SA}, \mathbf{H}, \mathbf{HS})$.
- G_1: For each $i \in [m]$, sample $\tilde{\mathbf{s}}_i \leftarrow \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\alpha}$. Let $\tilde{\mathbf{S}} = [\tilde{\mathbf{s}}_1, \dots, \tilde{\mathbf{s}}_m]$ and output $(\mathbf{A}, \tilde{\mathbf{S}}\mathbf{A}, \mathbf{H}, \mathbf{HS})$.
- G_2: Sample $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m]$ with $\mathbf{u}_i \leftarrow \ker(\varphi_{\mathbf{H},q'})$ and output $(\mathbf{A}, (\mathbf{S}+\mathbf{U})\mathbf{A}, \mathbf{H}, \mathbf{HS})$.

Clearly, G_0 and G_2 correspond to the Definition 3.1 distributions. We now establish the statistical equivalence of G_0 and G_1.

Lemma 3.1. *The output distributions of* G_0 *and* G_1 *are identical.*

Proof. For each column $i \in [m]$, $\mathbf{s}_i \leftarrow \mathcal{D}_{\mathcal{R}^k,\alpha}$. By Lemma 2.7, the joint distributions $(\mathbf{s}_i, \mathbf{H}\mathbf{s}_i)$ and $(\tilde{\mathbf{s}}_i, \mathbf{H}\mathbf{s}_i)$ are identical, where $\tilde{\mathbf{s}}_i \leftarrow \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\alpha}$. Since columns are independent and $\mathbf{H}$ is sampled identically in both hybrids, $(\mathbf{S}, \mathbf{H}, \mathbf{HS})$ in G_0 is distributed identically to $(\tilde{\mathbf{S}}, \mathbf{H}, \mathbf{HS})$ in G_1. Finally, since $\mathbf{A}$ is public parameter independent of the secrets and sampled identically in both games, the map $(\mathbf{S}, \mathbf{HS}, \mathbf{A}) \mapsto (\mathbf{SA}, \mathbf{HS}, \mathbf{A})$ preserves the identity in distribution. Thus, $\mathsf{G}_0 \equiv \mathsf{G}_1$. □

Let $(\mathbf{A}, \mathbf{T}') \in \mathcal{R}_{q'}^{m\times n} \times \mathcal{R}_{q'}^{(k-\ell)\times n}$ be given $\mathsf{MLWE}_{q',m,n,k-\ell,\alpha_0}$ instance. Our reduction $\mathcal{B}$ works as follows.

(1) Sample $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_m] \leftarrow \mathcal{D}_{\mathcal{R}^{k\times m},\alpha}$.
(2) For $i \in [m]$, sample $\mathbf{r}_i \leftarrow \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\sqrt{\Sigma_1}}$, relying on the $\mathcal{R}$-basis $\mathbf{B}$ for $\ker(\varphi_{\mathbf{H}})$ by Lemma 3.5. Let $\mathbf{R} := [\mathbf{r}_1, \dots, \mathbf{r}_m]$.
(3) Compute $\mathbf{T} := \mathbf{B}\,\mathbf{T}' + \mathbf{R}\,\mathbf{A} \in \mathcal{R}_{q'}^{k\times n}$.
(4) Send $(\mathbf{A}, \mathbf{T}, \mathbf{H}, \mathbf{HS})$ to $\mathcal{A}$ and output its response.

Lemmas 3.3 and 3.4 imply that $\mathcal{B}$'s output is $8m\varepsilon_2$-close to G_1 if $\mathbf{T}' = \mathbf{S}'\mathbf{A}$ for $\mathbf{S}' \leftarrow \mathcal{D}_{\mathcal{R}^{(k-\ell)\times m},\alpha_0}$ and distributed as G_2 if $\mathbf{T}'$ is uniform. Consequently, any Dual Hint-MLWE distinguishing advantage reduces to MLWE with at most $8m\varepsilon_2$ loss. □

Lemma 3.2. *The matrix* $\Sigma_1 = \alpha^2\mathbf{I}_k - \alpha_0^2\mathbf{B}\mathbf{B}^\top$ *is symmetric positive definite, and* $\sqrt{\Sigma_1} \succeq \eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}}))$.

Proof. Symmetry is immediate: $\Sigma_1^\top = (\alpha^2\mathbf{I})^\top - \alpha_0^2(\mathbf{B}\mathbf{B}^\top)^\top = \Sigma_1$. To show positive definiteness, observe that:

$$\lambda_{\min}(\Sigma_1) = \alpha^2 - \alpha_0^2\lambda_{\max}(\mathbf{B}\mathbf{B}^\top) = \alpha^2 - \alpha_0^2\|\mathbf{B}\|_2^2.$$

By assumption, $\alpha^2 - \alpha_0^2\|\mathbf{B}\|_2^2 \geq (1+\ell\alpha_H)\frac{\ln(2d(k-\ell)(1+1/\varepsilon_2))}{\pi} > 0$, making Σ_1 strictly positive definite. Next, we show that $\sqrt{\Sigma_1} \succeq \eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}}))$. Recall that $\ker(\varphi_{\mathbf{H}}) \subseteq \mathcal{R}^k$. By Lemma 2.9, the matrix $\mathbf{B} = \begin{pmatrix} -\mathbf{H}' \\ \mathbf{I}_{k-\ell} \end{pmatrix} \in \mathcal{R}^{k\times(k-\ell)}$ is an $\mathcal{R}$-basis of $\ker(\varphi_{\mathbf{H}})$. Let $n = (k-\ell)d$ be the rank of $\ker(\varphi_{\mathbf{H}})$. By Lemma 2.4,

$$\eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}})) \leq \sqrt{\frac{\ln(2n(1+1/\varepsilon_2))}{\pi}} \cdot \max_i \|\mathbf{b}_i\|_2 \tag{4}$$

$$= \sqrt{\frac{\ln(2n(1+1/\varepsilon_2))}{\pi}} \cdot \sqrt{1+\ell\alpha_H} \tag{5}$$

where $\mathbf{B} = \{\mathbf{b}_i\}$. Hence $\lambda_{\min}(\Sigma_1) \geq \eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}}))^2$ implying $\sqrt{\Sigma_1} \succeq \eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}}))$, which completes the proof. □

Lemma 3.3. *If* $\mathbf{T}' = \mathbf{S}'\mathbf{A}$ *with* $\mathbf{S}' \leftarrow \mathcal{D}_{\mathcal{R}^{(k-\ell)\times m},\alpha_0}$, *then the output of* $\mathcal{B}$ *is within statistical distance* $8m\varepsilon_2$ *of the distribution of* G_1.

Proof. We only need to show that $\mathbf{T} = (\mathbf{B}\mathbf{S}' + \mathbf{R})\mathbf{A}$ output by the reduction $\mathcal{B}$ is within $8m\varepsilon_2$ of $\tilde{\mathbf{S}}\mathbf{A}$ output in G_1.

Let $\Sigma_2 = \alpha_0^2\mathbf{B}\mathbf{B}^\top$ and $\Sigma = \Sigma_1 + \Sigma_2 = \alpha^2\mathbf{I}$. For each column $i \in [m]$, let $\mathbf{x}_{2,i} = \mathbf{B}\mathbf{s}'_i \in \ker(\varphi_{\mathbf{H}})$, where $\mathbf{s}'_i$ denotes the i-th column of $\mathbf{S}'$. Since $\mathbf{s}'_i \leftarrow \mathcal{D}_{\mathcal{R}^{k-\ell},\alpha_0}$, we have $\mathbf{x}_{2,i} \sim \mathcal{D}_{\ker(\varphi_{\mathbf{H}}),\sqrt{\Sigma_2}}$. By Lemma 3.2, $\sqrt{\Sigma_1} \succeq \eta_{\varepsilon_2}(\ker(\varphi_{\mathbf{H}}))$. Applying Lemma 2.1 with $\Lambda_1 = \Lambda_2 = \ker(\varphi_{\mathbf{H}})$, $\mathbf{c}_1 = \mathbf{s}_i$, $\mathbf{c}_2 = \mathbf{0}$, $\mathbf{x}_2 = \mathbf{x}_{2,i}$ and $\mathbf{e} = \mathbf{r}_i$, and using the fact that $\Lambda_1 + \mathbf{c}_1 - \mathbf{x}_2 = \Lambda_1 + \mathbf{c}_1$ since $\mathbf{x}_2 \in \Lambda_1$, the sum $\mathbf{r}_i + \mathbf{x}_{2,i}$ where $\mathbf{r}_i \leftarrow \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\sqrt{\Sigma_1}}$ is within statistical distance $8\varepsilon_2$ of:

$$\mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\sqrt{\Sigma_1+\Sigma_2}} = \mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\alpha}.$$

This matches the definition of $\tilde{\mathbf{s}}_i$ in G_1. Summing the statistical distance over the m columns using the independence of the columns of $\mathbf{S}'$ and $\mathbf{R}$, the total statistical distance is at most $8m\varepsilon_2$. □

Lemma 3.4. *If* $\mathbf{T}' \leftarrow \mathcal{R}_{q'}^{(k-\ell)\times n}$, *then the output of* $\mathcal{B}$ *is distributed exactly as* G_2.

Proof. The reduction $\mathcal{B}$ outputs $\mathbf{T} = \mathbf{B}\mathbf{T}' + \mathbf{R}\mathbf{A} \pmod{q'}$. Let $\mathbf{t}_j \in \mathcal{R}_{q'}^k$ denote the j-th column of $\mathbf{T}$ for $j \in [n]$. By definition of matrix multiplication, the j-th column is $\mathbf{t}_j = \mathbf{B}\mathbf{t}'_j + \sum_{i=1}^m a_{i,j}\mathbf{r}_i \pmod{q'}$, where $\mathbf{t}'_j \in \mathcal{R}_{q'}^{k-\ell}$ is the j-th column of $\mathbf{T}'$ and $a_{i,j}$ are entries of $\mathbf{A}$. Recall that each $\mathbf{r}_i$ was sampled from $\mathcal{D}_{\ker(\varphi_{\mathbf{H}})+\mathbf{s}_i,\sqrt{\Sigma_1}}$. Thus, we can write $\mathbf{r}_i = \mathbf{s}_i + \mathbf{w}_i$ for some $\mathbf{w}_i \in \ker(\varphi_{\mathbf{H}})$. Substituting this into the expression for $\mathbf{t}_j$:

$$\mathbf{t}_j = \mathbf{B}\mathbf{t}'_j + \sum_{i=1}^m a_{i,j}(\mathbf{s}_i + \mathbf{w}_i) = \underbrace{\sum_{i=1}^m a_{i,j}\mathbf{s}_i}_{(\mathbf{SA})_j} + \underbrace{\left(\mathbf{B}\mathbf{t}'_j + \sum_{i=1}^m a_{i,j}\mathbf{w}_i\right)}_{\mathbf{u}_j} \pmod{q'}.$$

By Lemma 2.9, $\mathbf{B}_{q'} = \mathbf{B} \bmod q'$ is a $\mathcal{R}_{q'}$-basis for $\ker(\varphi_{\mathbf{H},q'})$. Since $\mathbf{t}'_j$ is sampled uniformly from $\mathcal{R}_{q'}^{k-\ell}$, the term $\mathbf{B}\mathbf{t}'_j$ is distributed uniformly over $\ker(\varphi_{\mathbf{H},q'})$. For each column j, the term $\mathbf{w}'_j = \sum_{i=1}^m a_{i,j}\mathbf{w}_i \bmod q'$ is an element of $\ker(\varphi_{\mathbf{H},q'})$ because each $\mathbf{w}_i \bmod q' \in \ker(\varphi_{\mathbf{H},q'})$. Thus, $\mathbf{u}_j = \mathbf{B}\mathbf{t}'_j + \mathbf{w}'_j$ is uniformly distributed over $\ker(\varphi_{\mathbf{H},q'})$. Furthermore, since the columns $\mathbf{t}'_j$ are independent for each j, the columns $\mathbf{u}_j$ are also independent. We conclude $\mathbf{T} \equiv \mathbf{SA} + \mathbf{U} \pmod{q'}$, where $\mathbf{U} \sim [\ker(\varphi_{\mathbf{H},q'})]^m$. Recall that $\mathbf{A} = \begin{bmatrix}\mathbf{I}_n \\ \mathbf{A}_2\end{bmatrix}$. Consider the $\mathcal{R}_{q'}$-linear map $f : [\ker(\varphi_{\mathbf{H},q'})]^m \to [\ker(\varphi_{\mathbf{H},q'})]^n$, that $\mathbf{U}' \mapsto \mathbf{U}'\mathbf{A}$. For any $\mathbf{V} \in [\ker(\varphi_{\mathbf{H},q'})]^n$, choosing $\mathbf{U}' = [\mathbf{V} \mid \mathbf{0}]$ yields $f(\mathbf{U}') = \mathbf{V}$, hence f is surjective. Therefore, if $\mathbf{U}' \leftarrow [\ker(\varphi_{\mathbf{H},q'})]^m$ is uniform, then $\mathbf{U}'\mathbf{A} \leftarrow [\ker(\varphi_{\mathbf{H},q'})]^n$ is uniform. Thus, $\mathbf{T} \equiv \mathbf{SA} + \mathbf{U}'\mathbf{A} \pmod{q'}$ is distributed exactly as in G_2. □

Lemma 3.5 (Sampleability). *Let $q', k, \ell, \alpha_H \in \mathbb{N}^+$ such that $k > \ell$ and let $\alpha, \alpha_0 \in \mathbb{R}$. Let $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}']$ with $\mathbf{H}' \in (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$, and let $\mathbf{B} = \binom{-\mathbf{H}'}{\mathbf{I}_{k-\ell}} \in \mathcal{R}^{k\times(k-\ell)}$ be an $\mathcal{R}$-basis of $\Lambda = \ker(\varphi_{\mathbf{H}})$. If $\alpha^2 \geq \alpha_0^2\,\|\mathbf{B}\|_2^2 + (1+\ell\alpha_H)\cdot\frac{\ln(2(k-\ell)d+4)}{\pi}$, then there exists a PPT algorithm to sample $\mathbf{r}_i \leftarrow \mathcal{D}_{\Lambda+\mathbf{s}_i,\sqrt{\Sigma_1}}$ for $\mathbf{s}_i \in \mathcal{R}^k$.*

Proof. The rank of the lattice Λ is $n = (k-\ell)d$. Let $L = \frac{\ln(2n+4)}{\pi}$. We now verify the sampleability condition. By the Rayleigh quotient, $L\max_j \|\Sigma_1^{-1/2}\mathbf{b}_j\|_2^2 = L\max_j\left(\mathbf{b}_j^\top\Sigma_1^{-1}\mathbf{b}_j\right) \leq L\frac{\max_j\|\mathbf{b}_j\|_2^2}{\lambda_{\min}(\Sigma_1)} \leq 1$, which implies efficient sampleability by Lemma 2.6. □

4 Key-Homomorphic One-Time Signature

In this section, we first provide the syntax and definition of Key-Homomorphic One-Time Signature (KOTS) and its properties. We then present our construction and analyze its properties.

4.1 Definition

Definition 4.1 (Key-Homomorphic One-Time Signature, [8]). *Let $\mathcal{M}$ be a message space and $\mathcal{S}_{\mathsf{sk}}$ be a secret key space. Let $\mathcal{S}_{\mathsf{pk}}$ and $\mathcal{S}_{\mathsf{sig}}$ be $\mathcal{R}$-modules denoting the public key and signature spaces. A key-homomorphic one-time signature (KOTS) scheme Π_{KOTS} over $\mathcal{R}$ consists of six PPT algorithms:*

- $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$*: On input a security parameter λ, the setup algorithm outputs a set of public parameters* pp.
- $(\mathsf{osk}, \mathsf{opk}) \leftarrow \mathsf{KGen}(\mathsf{pp})$*: On input a set of public parameters* pp*, the key generation algorithm outputs a one-time signing key* $\mathsf{osk} \in \mathcal{S}_{\mathsf{sk}}$ *and the corresponding public key* $\mathsf{opk} \in \mathcal{S}_{\mathsf{pk}}$.
- $\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{osk}, \mu)$*: On input a set of public parameters* pp*, a signing key* $\mathsf{osk} \in \mathcal{S}_{\mathsf{sk}}$*, and a message $\mu \in \mathcal{M}$, the signing algorithm outputs a signature $\sigma \in \mathcal{S}_{\mathsf{sig}}$.*
- $b \leftarrow \mathsf{iVrfy}/\mathsf{sVrfy}/\mathsf{wVrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma)$*: On input a set of public parameters* pp*, a verification key* $\mathsf{opk} \in \mathcal{S}_{\mathsf{pk}}$*, a message $\mu \in \mathcal{M}$, and a signature $\sigma \in \mathcal{S}_{\mathsf{sig}}$, the individual/strong/weak verification algorithm outputs a bit b indicating acceptance or rejection.*

Definition 4.2 (Individual Correctness). *Let $\epsilon_{\mathsf{cor}} \in (0,1)$. We say that Π_{KOTS} is ϵ_{cor}-individually correct, if for all $\lambda \in \mathbb{N}$ and $\mu \in \mathcal{M}$, it holds that $\Pr\left[\mathsf{iVrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma) = 1\right] \geq 1 - \epsilon_{\mathsf{cor}}$, where $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $(\mathsf{osk}, \mathsf{opk}) \leftarrow \mathsf{KGen}(\mathsf{pp})$, and $\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{osk}, \mu)$.*

Definition 4.3 (Probabilistic Homomorphism). *Let $N \in \mathbb{N}^+$, $\epsilon_{\mathsf{hom}} \in (0,1)$, and $W \subseteq \mathcal{R}$. We say that Π_{KOTS} is $(N, W, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, if for all $\lambda \in \mathbb{N}$, $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $\mu \in \mathcal{M}$, and $(\mathsf{opk}^i, \sigma^i)_{i=1}^N \in \mathcal{S}_{\mathsf{pk}}\times\mathcal{S}_{\mathsf{sig}}$ with $\mathsf{iVrfy}(\mathsf{pp}, \mathsf{opk}^i, \mu, \sigma^i) = 1$, it holds that $\Pr_{w^i\leftarrow W}\left[\mathsf{sVrfy}\left(\mathsf{pp}, \sum_{i=1}^N w^i\mathsf{opk}^i, \mu, \sum_{i=1}^N w^i\sigma^i\right) = 1\right] \geq 1 - \epsilon_{\mathsf{hom}}$.*

Game $\mathsf{EUF\text{-}RK}^{W'}_{\Pi_{\mathsf{KOTS}},\mathcal{A}}(\lambda)$

(1) $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$
(2) $\forall i \in [N]$: $(\mathsf{osk}^i, \mathsf{opk}^i) \leftarrow \mathsf{KGen}(\mathsf{pp})$ and set $Q_i \leftarrow \emptyset$
(3) $(i^*, \mu^*, \sigma^*, w^*) \leftarrow \mathcal{A}^{\mathsf{Sign}(\cdot,\cdot)}(\mathsf{pp}, \mathsf{opk}^1, \ldots, \mathsf{opk}^N)$
(4) Output 1 iff $i^* \in [N] \wedge \mu^* \notin Q_{i^*} \wedge w^* \in W' \wedge \mathsf{wVrfy}\left(\mathsf{pp},\ w^*\cdot\mathsf{opk}^{i^*},\ \mu^*,\ \sigma^*\right) = 1$

Oracle $\mathsf{Sign}(i, \mu)$

(1) If $i \notin [N]$ or $|Q_i| = 1$ then return $\perp$
(2) Set $Q_i \leftarrow Q_i \cup \{\mu\}$
(3) Return $\mathsf{Sign}(\mathsf{pp}, \mathsf{osk}^i, \mu)$

Figure 2: EUF-RK Game for a KOTS Scheme.

Definition 4.4 (Robust Homomorphism). *Let Π_{KOTS} be a KOTS scheme. We say that Π_{KOTS} is robustly homomorphic, if for all $\lambda \in \mathbb{N}$, $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $\mu \in \mathcal{M}$, and any $\mathsf{opk}^b \in \mathcal{S}_{\mathsf{pk}}$ and $\sigma^b \in \mathcal{S}_{\mathsf{sig}}$ ($b \in \{0,1\}$) satisfying $\mathsf{sVrfy}(\mathsf{pp}, \mathsf{opk}^b, \mu, \sigma^b) = 1$, it holds that $\mathsf{wVrfy}(\mathsf{pp}, \mathsf{opk}^0 - \mathsf{opk}^1, \mu, \sigma^0 - \sigma^1) = 1$.*

Definition 4.5 (Multi-User Existential Unforgeability under Rerandomized Keys). *Let $W' := \{w^1 - w^2 : w^1, w^2 \in W \wedge w^1 \neq w^2\}$. Consider the game in Figure 2. We say that Π_{KOTS} is W'-existentially unforgeable under rerandomized keys (EUF-RK), if for all λ and PPT adversaries $\mathcal{A}$, it holds that $\mathsf{Adv}^{\mathsf{EUF\text{-}RK}}_{\Pi_{\mathsf{KOTS}}}(\mathcal{A}) \leq \mathsf{negl}(\lambda)$.*

4.2 KOTS Construction

We now present our KOTS construction in Figure 3. Some of our proofs for the KOTS require $\ell = 1$, which is already always the case in our parameter setting. However, we keep the description general by using the ℓ parameter and specify when it's restricted to $\ell = 1$.

Definition 4.6. *Let $q', d, m, n, k, \ell, \alpha_H, \beta_z, \beta_\sigma \in \mathbb{N}^+$ where d is a power of 2 and $k > \ell$. Let $\alpha \in \mathbb{R}$. Let $\mathsf{H} : \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$ be a collision-resistant hash function. We define $\Pi_{\mathsf{KOTS}} = (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{iVrfy}, \mathsf{sVrfy}, \mathsf{wVrfy})$ as the key-homomorphic one-time scheme over $\mathcal{R}$ as in Figure 3. Its secret key space is $\mathcal{S}_{\mathsf{sk}} \subseteq \mathcal{R}^{k\times m}$, public key space is $\mathcal{S}_{\mathsf{pk}} \subseteq \mathcal{R}_{q'}^{k\times n}$, and its signature space is $\mathcal{S}_{\mathsf{sig}} \subseteq \mathcal{R}^{\ell\times m}$.*

Theorem 4.1. *Let λ, t_1, t_2, p, q', d, m, n, k, ℓ, β_z, β_σ, α_H, α_w, $N \in \mathbb{N}^+$, where t_1, t_2, d are powers of 2 with $d > t_1 > 1$ and $d > t_2 > 1$, $p \equiv 2t_1 + 1 \pmod{4t_1}$ is prime, $q' \equiv 2t_2 + 1 \pmod{4t_2}$ is prime, $(\frac{1+\varepsilon_3}{1-\varepsilon_3})^{2m}p^{-dm/t_2} < 2^{-\lambda}$, and $k \geq 4\ell$. Let $\alpha, \epsilon_{\mathsf{cor}}, \epsilon_{\mathsf{hom}}, \varepsilon_1, \varepsilon_3 \in \mathbb{R}$, such that $\alpha \geq \sqrt{2}\eta_{\varepsilon_1}(\mathbb{Z})$, $\alpha \geq p\sqrt{2\alpha_H\ln(2(r-1)d(1+1/\varepsilon_3))/\pi}$, $\beta_z \geq 6\alpha\sqrt{1+(k-\ell)\alpha_H}$, and $\epsilon_{\mathsf{cor}} \geq \ell md\cdot\left(2^{-162}+2(k-\ell)\alpha_H\varepsilon_1\right)$, $\beta_\sigma \geq \beta_z\cdot\sqrt{2\alpha_w N\cdot\ln\left(\frac{2\ell md}{\epsilon_{\mathsf{hom}}}\right)}$. Let $\mathsf{H} : \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$ be a random oracle with $\left|\mathcal{T}_{\alpha_H}\right|^{r-1} \geq 2^{2\lambda}$. Let $W' = \{w_0 - w_1 \mid w_0, w_1 \in \mathcal{T}_{\alpha_w} \wedge w_0 \neq w_1\}$. If the $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$ and $\mathsf{MSIS}^\infty_{q',m,n,2\beta_\sigma+2\alpha_w\beta_z}$ problems are hard, and any $w^* \in W'$ is a unit in $\mathcal{R}_{q'}$, then the construction Π_{KOTS} in Figure 3 is an ϵ_{cor}-individually correct, $(N, \mathcal{T}_{\alpha_w}, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, robustly homomorphic KOTS that is W'-multi-user existentially unforgeable under rerandomized keys.*

$\underline{\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)}$
(1) $\mathbf{A}_2 \leftarrow \mathcal{R}_q^{(m-n)\times n}$
(2) **return** $\mathsf{pp} = \mathbf{A} = [\mathbf{I}_n; \mathbf{A}_2]$

$\underline{(\mathsf{osk}, \mathsf{opk}) \leftarrow \mathsf{KGen}(\mathsf{pp})}$
(1) $\mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{k\times m},\alpha}$
(2) $\mathbf{T} := \mathbf{S}\mathbf{A} \pmod{q'} \in \mathcal{R}_{q'}^{k\times n}$
(3) **return** $(\mathsf{osk} = \mathbf{S}, \mathsf{opk} = \mathbf{T})$

$\underline{\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{osk}, \mu)}$
(1) $\mathbf{H}' = \mathsf{H}(\mu)$
(2) $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}'] \in \mathcal{R}^{\ell\times k}$
(3) $\mathbf{Z} := \mathbf{H}\mathbf{S} \in \mathcal{R}^{\ell\times m}$
(4) **return** $\sigma = \mathbf{Z}$

$\underline{b \leftarrow \mathsf{iVrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma)}$
(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma, \beta_z)$

$\underline{b \leftarrow \mathsf{sVrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma)}$
(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma, \beta_\sigma)$

$\underline{b \leftarrow \mathsf{wVrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma)}$
(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma, 2\beta_\sigma)$

$\underline{b \leftarrow \mathsf{Vrfy}(\mathsf{pp}, \mathsf{opk}, \mu, \sigma, \beta)}$
(1) Parse $\mathsf{pp} = \mathbf{A}$, $\mathsf{opk} = \mathbf{T}$ and $\sigma = \mathbf{Z}$
(2) $\mathbf{H}' = \mathsf{H}(\mu)$, and $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}']$
(3) **if** $\|\mathbf{Z}\|_{\max} > \beta$ **or** $\mathbf{Z}\mathbf{A} \neq \mathbf{H}\mathbf{T} \pmod{q'}$, **return** 0
(4) **return** 1

Figure 3: Our KOTS Construction Π_{KOTS}.

Proof. We prove the theorem via Lemmas D.1, 4.1, D.2 and D.3. □

As the main technically challenging step is the unforgeability proof, we provide its full proof here and refer to Appendix D for other proofs.

Lemma 4.1. *Let $\lambda, p, t_1, t_2, q', d, m, n, k, \ell, r, \beta_\sigma, \alpha_H, \alpha_w, N \in \mathbb{N}^+$ and $\varepsilon_3 \in \mathbb{R}$ such that $\ell = 1$, $k \geq 4\ell$, $r = k - 2\ell$, $d > t_1 > 1$ and $d > t_2 > 1$ are powers of 2, $p \equiv 2t_1 + 1 \pmod{4t_1}$ is prime, $(\frac{1+\varepsilon_3}{1-\varepsilon_3})^{2m} p^{-dm/t_1} < 2^{-\lambda}$, and $q' \equiv 2t_2 + 1 \pmod{4t_2}$ is prime. Let $\alpha \in \mathbb{R}$ such that $\alpha \geq p\sqrt{2\alpha_H \ln(2(r-1)d(1+1/\varepsilon_3))/\pi}$. Let $\mathsf{H}\colon \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell\times(k-\ell)}$ be a random oracle with $\left|\mathcal{T}_{\alpha_H}\right|^{r-1} \geq 2^{2\lambda}$. If the $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$ and $\mathsf{MSIS}^\infty_{q',m,n,2\beta_\sigma+2\alpha_w\beta_z}$ problems are hard, and any $w^* \in W'$ is a unit in $\mathcal{R}_{q'}$, then Π_{KOTS} is W'-EUF-RK secure in the ROM. Specifically, for any PPT adversary $\mathcal{A}$ making Q_H hash queries, there exist PPT algorithms $\mathcal{B}$ and $\mathcal{D}$ such that:*

$$Adv^{\mathsf{EUF\text{-}RK}}_{\Pi_{\mathsf{KOTS}}}(\mathcal{A}) \leq N(Q+1)^2\Big(Adv^{\mathsf{DualHintMLWE}}(\mathcal{B}) + Adv^{\mathsf{MSIS}^\infty}(\mathcal{D})\Big) + \frac{Q_H^2}{|\mathcal{T}_{\alpha_H}|^{k-2\ell-1}} + negl(\lambda),$$

where $Q := Q_H + N$ is the total number of distinct messages queried.

Proof. We proceed via a sequence of games $\mathsf{G}_0, \mathsf{G}_1, \mathsf{G}_2, \mathsf{G}_3$. Let $p_i := \Pr[\mathsf{G}_i = 1]$ denote the probability that the adversary wins in Game i.

Game G_0. This is the original EUF-RK game. The challenger generates N key pairs, answers hash queries, and provides at most one signature per user. By definition:$p_0 = \mathsf{Adv}^{\mathsf{EUF\text{-}RK}}_{\Pi_{\mathsf{KOTS}}}(\mathcal{A})$.

Game G_1. This game is identical to G_0, except the challenger tracks all Q_H queries made to the random oracle H. For any hash matrix $\mathbf{H}_i$, partition it as $[\mathbf{I}_\ell \mid \mathbf{H}_i^{(1)} \mid \mathbf{H}_i^{(2)}]$, where the last block has $r = k - 2\ell$ columns. For $i = \{1, 2\}$, let $\mathbf{H}_i^{(2)} = [\overline{h}_i^{(1)}, \ldots, \overline{h}_i^{(r)}]$ (it has one row since $\ell = 1$) and define $\overline{g}_i^{(j)} := \overline{h}_i^{(j)} / \overline{h}_i^{(1)}$ for $j \in \{2, \ldots, r\}$. The challenger aborts if there exists any pair of distinct queries resulting in hashes $\mathbf{H}_1, \mathbf{H}_2$ such that $\overline{g}_2^{(j)} - \overline{g}_1^{(j)} = 0$ for all $j \in \{2, \ldots, r\}$.

Claim 4.1. $|p_0 - p_1| \leq \frac{Q_H^2}{|\mathcal{T}_{\alpha_H}|^{r-1}}$.

Proof. Since H is modeled as a random oracle, each query yields an independent, uniformly random matrix in $\mathcal{T}^r_{\alpha_H}$ for the last block of r columns. For any fixed pair of queries, the relation $\overline{g}_2^{(j)} - \overline{g}_1^{(j)} = 0$ holds for a specific j with probability at most $1/|\mathcal{T}_{\alpha_H}|$. The probability it holds for columns $j \in \{2, \ldots, r\}$ simultaneously is at most $1/|\mathcal{T}_{\alpha_H}|^{r-1}$. Applying a union bound over all $\leq Q_H^2$ pairs of queries gives the claimed bound. Conditioned on not aborting, the game is identical to G_0. □

Game G_2. The challenger guesses the target user $i_0 \leftarrow [N]$, the hash query index $j_1 \in [Q]$ corresponding to the signing query, and the index $j_2 \in [Q]$ corresponding to the forgery. It aborts if $\mathcal{A}$'s actions do not match these guesses.

Claim 4.2. $p_2 \geq \frac{1}{N(Q+1)^2} p_1$.

Proof. There are at most N choices for the target user i^*, and at most $Q + 1$ possible query positions for the signing message and the forged message. With probability $1/(N(Q+1)^2)$, the guesses match the adversary's behavior. Conditioned on a correct guess, the simulation is perfectly identical to G_1. □

Game G_3. The challenger alters how the target key $\mathbf{T}^{i_0}$ is generated. It samples $\mathbf{A}_2 \leftarrow \mathcal{R}_{q'}^{(m-n)\times n}$, $\mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}_{q'}^{k\times m},\alpha}$, sets $\mathbf{A} = [\mathbf{I}_n; \mathbf{A}_2]$, and programs the hashes $\mathbf{H}_1, \mathbf{H}_2$ at j_1, j_2. Using the intersection sampling from Lemma 4.3, it samples $\mathbf{U} \in \ker(\varphi_{[\mathbf{H}_1;\mathbf{H}_2],q'})$, sets $\mathbf{T} := (\mathbf{S} + \mathbf{U})\mathbf{A}$, and answers the signing query with $\mathbf{Z}_1 := \mathbf{H}_1\mathbf{S}$.

Claim 4.3. *There exists a PPT distinguisher $\mathcal{B}$ such that $|p_2 - p_3| \leq Adv^{\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}}(\mathcal{B}) + negl(\lambda)$.*

Proof. On input $(\mathbf{A}', \mathbf{T}', \mathbf{H}', \mathbf{Z}')$, algorithm $\mathcal{B}$ sets $(\mathbf{A}, \mathbf{T}, \mathbf{H}_1, \mathbf{Z}_1) := (\mathbf{A}', \mathbf{T}', \mathbf{H}', \mathbf{Z}')$ and simulates the exact environment described in G_2 or G_3. If $\mathcal{B}$ receives a Case 0 Dual Hint-MLWE challenge ($\mathbf{T} = \mathbf{S}\mathbf{A}$), the simulated view corresponds to G_2. If it receives Case 1 ($\mathbf{T} = (\mathbf{S} + \mathbf{U})\mathbf{A}$), the view corresponds to G_3. The distinguishing advantage is exactly $|p_2 - p_3|$. □

Claim 4.4. *Assume w^* is a unit in $\mathcal{R}_{q'}$. There exists a PPT MSIS solver $\mathcal{D}$ such that $p_3 \leq Adv^{\mathsf{MSIS}^\infty_{q',n,m,2\beta_\sigma+2\alpha_w\beta_z}}(\mathcal{D}) + negl(\lambda)$.*

Proof. Algorithm $\mathcal{D}$ receives an MSIS challenge $\mathbf{A}' \leftarrow \mathcal{R}_{q'}^{n\times m}$. It initiates the G_3 challenger by setting $\mathbf{A} = (\mathbf{A}')^\top \in \mathcal{R}_{q'}^{m\times n}$. It follows the G_3 procedure: sampling $\mathbf{S}$, programming $\mathbf{H}_1, \mathbf{H}_2$, and sampling $\mathbf{U}$ from their joint kernel. If $\mathcal{A}$ outputs a successful forgery $(i^*, \mu^*, \mathbf{Z}^*, w^*)$, $\mathcal{D}$ outputs a non-zero column of the matrix $\mathbf{X}^\top := (\mathbf{Z}^* - w^*\mathbf{H}_2\mathbf{S})^\top$.

First, $\mathcal{D}$'s simulation is perfect. The verification equation $\mathbf{Z}_1\mathbf{A} = \mathbf{H}_1\mathbf{T} \pmod{q'}$ holds since $\mathbf{H}_1\mathbf{U} = \mathbf{0} \pmod{q'}$, making the view identical to G_3. Because $\mathcal{A}$ wins, verification passes: $\mathbf{Z}^*\mathbf{A} = \mathbf{H}_2(w^*\mathbf{T}) \pmod{q'}$ and $\|\mathbf{Z}^*\|_{\max} \leq 2\beta_\sigma$. Since w^* is a scalar and $\mathbf{H}_2\mathbf{U} = \mathbf{0} \pmod{q'}$, substituting $\mathbf{T}$ yields:

$$\mathbf{X}\mathbf{A} = (\mathbf{Z}^* - w^*\mathbf{H}_2\mathbf{S})\mathbf{A} = \mathbf{H}_2(w^*\mathbf{T}) - w^*\mathbf{H}_2\mathbf{S}\mathbf{A} = \mathbf{0} \pmod{q'}.$$

Taking the transpose gives $\mathbf{A}'\mathbf{X}^\top = \mathbf{0} \pmod{q'}$. The norm bound is satisfied via the triangle inequality:

$$\|\mathbf{X}\|_{\max} \leq \|\mathbf{Z}^*\|_{\max} + \|w^*\|_1 \cdot \|\mathbf{H}_2\mathbf{S}\|_{\max} \leq 2\beta_\sigma + 2\alpha_w\beta_z.$$

The remaining failure condition is $\mathbf{X} = \mathbf{0}$. We show this happens with negligible probability. Partition $\mathbf{S} \in \mathcal{R}^{k\times m}$ into $[\mathbf{S}^{(0)}; \mathbf{S}^{(1)}; \mathbf{S}^{(2)}]$ where $\mathbf{S}^{(0)}, \mathbf{S}^{(1)} \in \mathcal{R}^{\ell\times m}$ and $\mathbf{S}^{(2)} \in \mathcal{R}^{(k-2\ell)\times m}$. Correspondingly, for $i \in \{1,2\}$, we partition the programmed hashes $\mathbf{H}_i = [\mathbf{I}_\ell \mid \mathbf{H}_i^{(1)} \mid \mathbf{H}_i^{(2)}] \in \mathcal{R}^{\ell\times k}$ with $\mathbf{H}_i^{(1)} \in \mathcal{R}^{\ell\times\ell}$ and $\mathbf{H}_i^{(2)} \in \mathcal{R}^{\ell\times(k-2\ell)}$. The adversary's view includes $\mathbf{T} = (\mathbf{S}+\mathbf{U})\mathbf{A}$ and the signature $\mathbf{Z}_1 = \mathbf{H}_1\mathbf{S}$. By Lemma 4.3, $\mathbf{U}$ is sampled such that it is uniform in the intersection of the kernels of $\mathbf{H}_1$ and $\mathbf{H}_2$. Specifically, the last $k-2\ell$ rows of $\mathbf{U}$ (corresponding to $\mathbf{S}^{(2)}$) provide enough degrees of freedom to statistically mask $\mathbf{S}^{(2)}$ given $\mathbf{T}$. Conditioned on the adversary's view, the probability that the forgery satisfies the zero-condition is:

$$\Pr_{\mathbf{S}\leftarrow\mathcal{D}_{\mathcal{R}^{k\times m},\alpha}}[\mathbf{X} = \mathbf{0} \mid \mathbf{H}_1\mathbf{S} = \mathbf{Z}_1 \wedge \mathbf{T} = (\mathbf{U}+\mathbf{S})\mathbf{A}] \tag{6}$$

$$= \Pr_{\mathbf{S}\leftarrow\mathcal{D}_{\mathcal{R}^{k\times m},\alpha}}[w^*\mathbf{H}_2\mathbf{S} = \mathbf{Z}^* \mid \mathbf{H}_1\mathbf{S} = \mathbf{Z}_1 \wedge \mathbf{T} = (\mathbf{U}+\mathbf{S})\mathbf{A}] \tag{7}$$

$$= \Pr_{\mathbf{S}\leftarrow\mathcal{D}_{\mathcal{R}^{k\times m},\alpha}}[\mathbf{H}_2\mathbf{S} = (w^*)^{-1}\mathbf{Z}^* \mid \mathbf{H}_1\mathbf{S} = \mathbf{Z}_1 \wedge \mathbf{T} = (\mathbf{U}+\mathbf{S})\mathbf{A}] \tag{8}$$

$$\le \max_{\mathbf{Y}} \Pr_{\mathbf{S}\leftarrow\mathcal{D}_{\mathcal{R}^{k\times m},\alpha}}[\mathbf{H}_2\mathbf{S} = \mathbf{Y} \mid \mathbf{H}_1\mathbf{S} = \mathbf{Z}_1 \wedge \mathbf{T} = (\mathbf{U}+\mathbf{S})\mathbf{A}] \tag{9}$$

$$\le \max_{\mathbf{Y}} \Pr_{\mathbf{S}\leftarrow\mathcal{D}_{\mathcal{R}^{k\times m},\alpha}}[\mathbf{H}_2\mathbf{S} = \mathbf{Y} \mid \mathbf{H}_1\mathbf{S} = \mathbf{Z}_1 \wedge \mathbf{V} = \mathbf{U}+\mathbf{S}] \tag{10}$$

$$\le \max_{\mathbf{Y}^{(2)}} \Pr_{\mathbf{S}^{(2)}\leftarrow\mathcal{D}_{\mathcal{R}^{(k-2\ell)\times m},\alpha}}[\mathbf{H}_2^{(2)}\mathbf{S}^{(2)} = \mathbf{Y}^{(2)} \mid \mathbf{H}_1^{(2)}\mathbf{S}^{(2)} = \mathbf{Z}_1^{(2)}] \tag{11}$$

$$\le \left(\frac{1+\epsilon}{1-\epsilon}\right)^{2m} p^{-dm/t_1} < 2^{-\lambda}. \tag{12}$$

In the above derivation, (8) holds since w^* is a unit. In (10), $\mathbf{V}$ represents the part of $\mathbf{S}$ leaked via $\mathbf{T}$. By Lemma 4.3, the last $k-2\ell$ rows of $\mathbf{U}$ are uniform and mask $\mathbf{S}^{(2)}$, leading to (11). Applying Lemma 4.2 with $r = k-2\ell$, $\overline{\mathbf{H}}_1 = \mathbf{H}_1^{(2)}$, and $\overline{\mathbf{H}}_2 = \mathbf{H}_2^{(2)}$, we get (12) for some prime p. Thus, with overwhelming probability, $\mathcal{D}$ computes a nonzero $\mathbf{X}$ satisfying $\mathbf{X}\mathbf{A} = \mathbf{0} \pmod{q'}$ and $\|\mathbf{X}\|_{\max} \le 2\beta_\sigma + 2\alpha_w\beta_z$. This constitutes a valid solution to the $\mathsf{MSIS}^\infty_{q',n,m,2\beta_\sigma+2\alpha_w\beta_z}$ problem. Thus, $p_3 \le \mathsf{Adv}^{\mathsf{MSIS}^\infty_{q',n,m,2\beta_\sigma+2\alpha_w\beta_z}}(\mathcal{D}) + \mathsf{negl}(\lambda)$. □

Putting Claim 4.1, Claim 4.2, Claim 4.3, Claim 4.4 together completes the proof. □

Lemma 4.2. *Let $r, d, m \in \mathbb{N}^+$ and $\ell = 1$ such that $r \ge 2\ell$ and d is a power of 2. Let p be prime such that $X^d + 1$ factors modulo p into t_1 irreducible polynomials of degree d/t_1 (e.g., $p \equiv 2t_1 + 1 \pmod{4t_1}$). Let $\varepsilon_3 \in (0,1)$ and $\alpha \ge p\sqrt{2\alpha_H \ln(2(r-1)d(1+1/\varepsilon_3))/\pi}$. For $i \in \{1,2\}$, let $\overline{\mathbf{H}}_i = [\overline{h}_i^{(1)} \cdots \overline{h}_i^{(r)}] \in (\mathcal{T}_{\alpha_H})^{1\times r}$. Assume $\overline{h}_i^{(j)}$ is invertible in $\mathcal{R}_p$ (which naturally holds for $p = 5$ and $t_1 = 2$ by Lemma A.2). Define $\overline{g}_i^{(j)} := \overline{h}_i^{(j)}/\overline{h}_i^{(1)}$. Assume there exists $j^* \in \{2,\dots,r\}$ such that $\delta^{(j^*)} := \overline{g}_2^{(j^*)} - \overline{g}_1^{(j^*)} \ne 0$.*

Let $\overline{\Lambda}_1 := \ker(\varphi_{\overline{\mathbf{H}}_1}) \subseteq \mathcal{R}^r$ and $\overline{\Lambda}_{12} := \{\mathbf{u} \in \overline{\Lambda}_1 : \overline{\mathbf{H}}_2\mathbf{u} = \mathbf{0} \pmod p\}$. Let $\overline{\mathbf{S}} \leftarrow \mathcal{D}_{\mathcal{R}^{r\times m},\alpha}$ and define $\overline{\mathbf{Z}}_i := \overline{\mathbf{H}}_i\overline{\mathbf{S}}$ for $i \in \{1,2\}$. Then for every fixed $\hat{\mathbf{Z}}_1$,

$$\max_{\hat{\mathbf{Z}}_2} \Pr[\overline{\mathbf{Z}}_2 = \hat{\mathbf{Z}}_2 \mid \overline{\mathbf{Z}}_1 = \hat{\mathbf{Z}}_1] \le \left(\frac{1+\varepsilon_3}{1-\varepsilon_3}\right)^{2m} p^{-dm/t_1}. \tag{13}$$

Proof. We first upper bound $\eta_{\varepsilon_3}(\overline{\Lambda}_{12})$ to show $\alpha \ge \eta_{\varepsilon_3}(\overline{\Lambda}_{12})$ and hence $\alpha \ge \eta_{\varepsilon_3}(\overline{\Lambda}_1)$. Observe that $p\cdot\overline{\Lambda}_1' \subseteq \overline{\Lambda}_{12}$, where $\overline{\Lambda}_1'$ is generated by column basis $\mathbf{B}'$ formed by placing $-\overline{h}_1^{(j)}$ in first row and $\overline{h}_1^{(1)}$ along the diagonal for $j \ge 2$. Under coefficient embedding, all columns of $\mathbf{B}'$ have weight $2\alpha_H$ and coefficients in $\{-1,1\}$, giving a Euclidean norm of $\sqrt{2\alpha_H}$. Since $\overline{\Lambda}_1'$ has $\mathbb{Z}$-rank $(r-1)d$, applying Lemma 2.4 gives $\eta_{\varepsilon_3}(\overline{\Lambda}_{12}) \le p\cdot\eta_{\varepsilon_3}(\overline{\Lambda}_1') \le p\sqrt{\frac{2\alpha_H\ln(2(r-1)d(1+1/\varepsilon_3))}{\pi}} \le \alpha$. Since $\overline{\Lambda}_{12} \subseteq \overline{\Lambda}_1$, we get $\alpha \ge \eta_{\varepsilon_3}(\overline{\Lambda}_1)$.

Fix $j \in [m]$ and write $\overline{\mathbf{s}} := \overline{\mathbf{s}}_j \leftarrow \mathcal{D}_{\mathcal{R}^r,\alpha}$, $\overline{\mathbf{z}}_1 := \overline{\mathbf{H}}_1\overline{\mathbf{s}}$, $\overline{\mathbf{z}}_2 := \overline{\mathbf{H}}_2\overline{\mathbf{s}}$. Fix $\hat{\mathbf{z}}_1 \in \mathcal{R}^\ell$ and choose any $\overline{\mathbf{c}}_1 \in \mathcal{R}^r$ with $\overline{\mathbf{H}}_1\overline{\mathbf{c}}_1 = \hat{\mathbf{z}}_1$. Then $\{\overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1\}$ is equivalent to $\{\overline{\mathbf{s}} \in \overline{\mathbf{c}}_1 + \overline{\Lambda}_1\}$. If the system ($\overline{\mathbf{H}}_1\overline{\mathbf{s}} = \hat{\mathbf{z}}_1$, $\overline{\mathbf{H}}_2\overline{\mathbf{s}} = \hat{\mathbf{z}}_2 \pmod p$) is solvable, its solution set is a coset $\overline{\mathbf{c}}_{12} + \overline{\Lambda}_{12}$. Thus,

$$\begin{aligned}\Pr[\overline{\mathbf{z}}_2 = \hat{\mathbf{z}}_2 \mid \overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1] &= \frac{\Pr[\overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1 \wedge \overline{\mathbf{z}}_2 = \hat{\mathbf{z}}_2]}{\Pr[\overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1]} && (14)\\ &\le \frac{\Pr[\overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1 \wedge \overline{\mathbf{z}}_2 = \hat{\mathbf{z}}_2 \pmod p]}{\Pr[\overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1]} && (15)\\ &= \frac{\rho_\alpha(\overline{\mathbf{c}}_{12} + \overline{\Lambda}_{12})}{\rho_\alpha(\overline{\mathbf{c}}_1 + \overline{\Lambda}_1)} && (16)\\ &\le \frac{\rho_\alpha(\overline{\Lambda}_{12})}{\rho_\alpha(\overline{\mathbf{c}}_1 + \overline{\Lambda}_1)} \quad \text{(Lemma 2.2)}. && (17)\end{aligned}$$

Since $\alpha \ge \eta_{\varepsilon_3}(\overline{\Lambda}_1)$,

$$\begin{aligned}\rho_\alpha(\overline{\mathbf{c}}_1 + \overline{\Lambda}_1) &\ge \frac{1-\varepsilon_3}{1+\varepsilon_3}\rho_\alpha(\overline{\Lambda}_1) \quad \text{(Lemma 2.2)} && (18)\\ &\ge \frac{(1-\varepsilon_3)^2}{1+\varepsilon_3}\det(\overline{\Lambda}_1)^{-1} \quad \text{(Lemma 2.3)}. && (19)\end{aligned}$$

Since $\alpha \ge \eta_{\varepsilon_3}(\overline{\Lambda}_{12})$, by Lemma 2.3,

$$\rho_\alpha(\overline{\Lambda}_{12}) \le (1+\varepsilon_3)\det(\overline{\Lambda}_{12})^{-1}. \tag{20}$$

Combining (17), (19), (20) yields

$$\Pr[\overline{\mathbf{z}}_2 = \hat{\mathbf{z}}_2 \mid \overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1] \le \left(\frac{1+\varepsilon_3}{1-\varepsilon_3}\right)^2 \cdot \frac{\det(\overline{\Lambda}_1)}{\det(\overline{\Lambda}_{12})}. \tag{21}$$

Define $\psi : \overline{\Lambda}_1 \to \mathcal{R}_p^\ell$ by $\psi(\mathbf{u}) = \overline{\mathbf{H}}_2\mathbf{u} \pmod p$. By definition of $\overline{\Lambda}_{12}$, we have $\ker(\psi) = \overline{\Lambda}_{12}$. By the first isomorphism theorem,

$$\frac{\det(\overline{\Lambda}_1)}{\det(\overline{\Lambda}_{12})} = 1/|\overline{\Lambda}_1/\overline{\Lambda}_{12}| = 1/|\mathrm{Im}(\psi)|, \tag{22}$$

hence it suffices to lower bound $|\mathrm{Im}(\psi)|$. We first observe that because $\overline{\Lambda}_1' \subseteq \overline{\Lambda}_1$, we have $|\mathrm{Im}(\psi)| \ge |\mathrm{Im}(\psi')|$, where $\psi' : \overline{\Lambda}_1' \to \mathcal{R}_p^\ell$ is the restriction of ψ to the sublattice $\overline{\Lambda}_1'$. We have:

$$\begin{aligned}|\mathrm{Im}(\psi')| &:= |\{y \in \mathcal{R}_p : \exists \mathbf{u} \in \overline{\Lambda}_1' \text{ s.t. } \overline{\mathbf{H}}_2\mathbf{u} = y\} && (23)\\ &= |\{y \in \mathcal{R}_p : \exists \mathbf{v} \in \mathcal{R}_p^{r-1} \text{ s.t. } \overline{\mathbf{H}}_2\mathbf{B}'\mathbf{v} = y\}| && (24)\\ &= |\{y \in \mathcal{R}_p : \exists \mathbf{v} \in \mathcal{R}_p^{r-1} \text{ s.t. } \overline{\mathbf{c}}^\top \cdot \mathbf{v} = y\}| && (25)\\ &= |\mathbf{c}^\top \cdot \mathcal{R}_p^{r-1}|, && (26)\end{aligned}$$

where $\mathbf{c}^\top := \overline{\mathbf{H}}_2\mathbf{B}' = (c^{(2)},\dots,c^{(r)}) \in \mathcal{R}_p^{r-1}$ with $c^{(j)} = \overline{h}_1^{(1)}\overline{h}_2^{(j)} - \overline{h}_2^{(1)}\overline{h}_1^{(j)} = \overline{h}_1^{(1)}\overline{h}_2^{(1)}\delta^{(j)}$ for $j \in \{2,\dots,r\}$. Therefore, $|\mathrm{Im}(\psi')|$ is the cardinality of the ideal $\mathcal{I}$ in $\mathcal{R}_p$ generated by $c^{(2)},\dots,c^{(r)}$. From the assumption that there exists some $j^* \in \{2,\dots,r\}$ such that $\delta^{(j^*)} \ne 0$ and the invertibility of $\overline{h}_1^{(1)}\overline{h}_2^{(1)}$ in $\mathcal{R}_p$, it follows that we know that $c^{(j^*)} \ne 0$ and hence $\mathcal{I}$ is a non-trivial ideal in $\mathcal{R}_p$. We recall that $\mathcal{R}_p$ is a principal ideal domain, and $X^d + 1$ factors mod p

into irreducible factors $\phi_i(X)$ over $\mathbb{Z}_p$ each of degree d/t_1. Hence, any non-trivial proper ideal of $\mathcal{R}_p$ is generated by $\prod_{i\in S}\phi_i(X)$ for some proper subset $S \subset [t_1]$. It follows that the smallest cardinality over all non-trivial proper ideals of $\mathcal{R}_p$ is p^{d/t_1} and we conclude that $|\mathrm{Im}(\psi)| \geq |\mathrm{Im}(\psi')| \geq p^{d/t_1}$. Then $\frac{\det(\overline{\Lambda}_1)}{\det(\overline{\Lambda}_{12})} = [\overline{\Lambda}_1 : \overline{\Lambda}_{12}]^{-1} = p^{-d/t_1}$. Plugging this into (21) gives the single-column bound

$$\max_{\hat{\mathbf{z}}_2\in\mathcal{R}^\ell} \Pr[\overline{\mathbf{z}}_2 = \hat{\mathbf{z}}_2 \mid \overline{\mathbf{z}}_1 = \hat{\mathbf{z}}_1] \leq \left(\frac{1+\varepsilon_3}{1-\varepsilon_3}\right)^2 p^{-d/t_1}. \tag{27}$$

Finally, conditioning on $\overline{\mathbf{Z}}_1 = \hat{\mathbf{Z}}_1$ fixes each per-column condition $\overline{\mathbf{z}}_{1,j} = \hat{\mathbf{z}}_{1,j}$ and preserves independence across columns, hence

$$\Pr[\overline{\mathbf{Z}}_2 = \hat{\mathbf{Z}}_2 \mid \overline{\mathbf{Z}}_1 = \hat{\mathbf{Z}}_1] \leq \left(\left(\frac{1+\varepsilon_3}{1-\varepsilon_3}\right)^2 p^{-d/t_1}\right)^m, \tag{28}$$

and taking maximum over $\hat{\mathbf{Z}}_2$ yields the claim. □

LEMMA 4.3. *Let $d \geq t_2 > 1$ be powers of 2 and $q' \equiv 2t_2 + 1 \pmod{4t_2}$ be a prime. Let $\ell = 1, k, \alpha_H \in \mathbb{N}^+$ such that $k > 2\ell$. Fix matrices $\mathbf{H}_i = [\mathbf{I}_\ell \mid \mathbf{H}_i^{(1)} \mid \mathbf{H}_i^{(2)}] \in \mathcal{R}^{\ell\times k}$ for $i \in \{1,2\}$, where $\mathbf{H}_i^{(1)} \in (\mathcal{T}_{\alpha_H})^{\ell\times\ell}$. Assume $\overline{\mathbf{X}} := \mathbf{H}_1^{(1)} - \mathbf{H}_2^{(1)} \pmod{q'}$ is invertible. Define*

$$\mathbf{B}_{q'} := \begin{bmatrix} -\mathbf{H}_1^{(1)}\overline{\mathbf{X}}^{-1}(\mathbf{H}_2^{(2)} - \mathbf{H}_1^{(2)}) - \mathbf{H}_1^{(2)} \\ \overline{\mathbf{X}}^{-1}(\mathbf{H}_2^{(2)} - \mathbf{H}_1^{(2)}) \\ \mathbf{I}_{k-2\ell} \end{bmatrix} \pmod{q'}.$$

Let $\widetilde{\mathbf{H}} = [\mathbf{H}_1; \mathbf{H}_2]$. The mapping $\phi(\mathbf{x}) = \mathbf{B}_{q'}\mathbf{x}$ is a bijection onto $\ker(\varphi_{\widetilde{\mathbf{H}},q'})$. Thus, sampling m uniform $\mathbf{x} \leftarrow \mathcal{R}_{q'}^{k-2\ell}$ yields a matrix $\mathbf{U}$ with i.i.d. columns uniform in the joint kernel of $\mathbf{H}_1$ and $\mathbf{H}_2$.

PROOF. Write any $\mathbf{u} \in \mathcal{R}_{q'}^k$ as $\mathbf{u} = (\mathbf{u}_0; \mathbf{u}_1; \mathbf{u}_2)$ with $\mathbf{u}_0, \mathbf{u}_1 \in \mathcal{R}_{q'}^\ell$ and $\mathbf{u}_2 \in \mathcal{R}_{q'}^{k-2\ell}$. The kernel constraints $\mathbf{H}_1\mathbf{u} = \mathbf{0}$ and $\mathbf{H}_2\mathbf{u} = \mathbf{0} \pmod{q'}$ expand to $\mathbf{u}_0 + \mathbf{H}_1^{(1)}\mathbf{u}_1 + \mathbf{H}_1^{(2)}\mathbf{u}_2 = \mathbf{0}$ and $\mathbf{u}_0 + \mathbf{H}_2^{(1)}\mathbf{u}_1 + \mathbf{H}_2^{(2)}\mathbf{u}_2 = \mathbf{0}$. Subtracting the second from the first eliminates $\mathbf{u}_0$, and solving for $\mathbf{u}_1$ gives $\mathbf{u}_1 = \overline{\mathbf{X}}^{-1}(\mathbf{H}_2^{(2)} - \mathbf{H}_1^{(2)})\mathbf{u}_2$. Plugging this back into first equation determines $\mathbf{u}_0 = -\mathbf{H}_1^{(1)}\overline{\mathbf{X}}^{-1}(\mathbf{H}_2^{(2)} - \mathbf{H}_1^{(2)})\mathbf{u}_2 - \mathbf{H}_1^{(2)}\mathbf{u}_2$. Because both $\mathbf{u}_0$ and $\mathbf{u}_1$ are uniquely determined by a free choice of $\mathbf{u}_2$, every $\mathbf{u} \in \ker(\varphi_{\widetilde{\mathbf{H}},q'})$ equals $\mathbf{B}_{q'}\mathbf{u}_2$. Therefore, $\phi(\mathbf{x})$ generates the exact kernel. □

5 Homomorphic Vector Commitment

In this section, we describe our optimization of the Chipmunk Homomorphic Vector Commitment (HVC) framework [8]. Specifically, we extend the supported commitment domain $\mathcal{S}_{\mathsf{dom}}$ from vectors to matrices, and transition from the Ring-SIS assumption to the more versatile Module-SIS assumption. Due to limited space and the fact that our main contribution in this work is on KOTS, we only provide the main HVC construction here in Figure 4 and refer the reader to Appendix B for more details.

At a high level, we set up some random matrices (to be used mainly as commitment keys) in Setup. The commitment function Com creates a Merkle-tree while instantiating the underlying hash function using an Ajtai hash (over module lattices in our case). The opening algorithm Open returns the authentication path for a particular leaf node indicated by t. The verification Vrfy checks that the

<u>$\mathsf{Setup}(1^\lambda)$</u>

(1) sample $\mathbf{B} \leftarrow \mathcal{R}_q^{\omega\times\rho\nu\kappa'}$
(2) sample $\mathbf{A}_0, \mathbf{A}_1 \leftarrow \mathcal{R}_q^{\omega\times\omega\kappa}$
(3) **return** $\mathsf{pp} := (\mathbf{B}, \mathbf{A}_0, \mathbf{A}_1)$

<u>$\mathsf{Com}(\mathsf{pp}, \widetilde{\mathbf{M}})$</u>

(1) $\mathbf{p}_0 := \mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\widetilde{\mathbf{M}}, \epsilon)$
(2) $c := \mathsf{proj}_q(\mathbf{p}_0) \in \mathcal{R}_q^\omega$
(3) **return** c

<u>$\mathsf{Open}(\mathsf{pp}, c, \widetilde{\mathbf{M}}, t)$</u>

(1) $\tilde{t} := \mathsf{bin}_\tau(t-1)$
(2) **for** $1 \leq j \leq \tau$ **do**
 · $\mathbf{p}_j := \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j} \,\|\, \tilde{t}_j)$
 · $\overline{\mathbf{p}}_j := \mathsf{Encode}_{\eta,q}^{\mathcal{B}}(\mathbf{p}_j)$
 · $\mathbf{s}_j := \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j} \,\|\, (\tilde{t}_j \oplus 1))$
(3) $\mathbf{u} := \mathsf{dec}_{q'}(\mathsf{vec}(\mathbf{M}_t)) \in \mathcal{R}^{\rho\nu\kappa'}$
(4) **return** $d := ((\overline{\mathbf{p}}_j)_j, (\mathbf{s}_j)_j, \mathbf{u})$

<u>$\mathsf{iVrfy}(\mathsf{pp}, c, t, d)$</u>

(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, c, t, d, \eta)$

<u>$\mathsf{sVrfy}(\mathsf{pp}, c, t, d)$</u>

(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, c, t, d, \beta_{\mathsf{agg}})$

<u>$\mathsf{wVrfy}(\mathsf{pp}, c, t, d)$</u>

(1) **return** $\mathsf{Vrfy}(\mathsf{pp}, c, t, d, 2\beta_{\mathsf{agg}})$

<u>$\mathsf{Vrfy}(\mathsf{pp}, c, t, d, \beta)$</u>

(1) $\tilde{t} := \mathsf{bin}_\tau(t-1)$; parse d as $((\overline{\mathbf{p}}_j)_j, (\mathbf{s}_j)_j, \mathbf{u})$
(2) **if** $\|\mathbf{u}\|_{\max} > \beta$ **return** $\perp$
(3) $\mathsf{hint}_\tau \leftarrow \mathbf{B}\cdot\mathbf{u} \in \mathcal{R}_q^\omega$
(4) **for** $j = \tau, \tau-1, \ldots, 1$ **do**
 · $\mathbf{p}_j \leftarrow \mathsf{Decode}_{\eta,q}^{\mathcal{B}}(\overline{\mathbf{p}}_j, \mathsf{hint}_j)$
 · **if** $\|\mathbf{p}_j\|_{\max} > \beta$ **or** $\|\mathbf{s}_j\|_{\max} > \beta$ **return** $\perp$
 · **if** $\|\mathsf{proj}_{\eta,\kappa}(\mathbf{p}_j)\|_{\max} > \frac{q\beta}{2\eta}$ **or** $\|\mathsf{proj}_{\eta,\kappa}(\mathbf{s}_j)\|_{\max} > \frac{q\beta}{2\eta}$ **return** $\perp$
 · $\mathsf{hint}_{j-1} \leftarrow \mathbf{A}_{\tilde{t}_j}\cdot\mathbf{p}_j + \mathbf{A}_{\tilde{t}_j\oplus 1}\cdot\mathbf{s}_j$
(5) **if** $c \neq \mathsf{hint}_0$ **return** $\perp$
(6) **return** $\mathsf{vec}^{-1}(\mathsf{proj}_{q'}(\mathbf{u})) \in \mathcal{R}_{q'}^{\rho\times\nu}$

Figure 4: Our HVC Construction Π_{HVC}.

authentication path correctly constructs the root and that certain components along the way are bounded in norm. The variants of Vrfy simply impose different norm bounds. The Encode/Decode algorithms are used to compress authentication path elements and the labelling function label is used for specifying how every node in a full binary tree is mathematically computed and represented. Finally, the decomposition dec is used to map high-norm ring elements into vectors of small-norm limbs compatible with the underlying Ajtai hash functions, while projection proj acts as its linear inverse to reconstruct the original elements.

6 Lemur: Synchronized Multi-Signatures

In this section, we briefly describe our synchronized multi-signature scheme, Lemur. At a high level, our construction follows the modular design patterns introduced by Squirrel [9] and Chipmunk [8]. We utilize a Homomorphic Vector Commitment (HVC) to manage a signer's history of one-time keys and a Key-Homomorphic One-Time Signature (KOTS) to enable the non-interactive aggregation of individual signatures. These blocks are composed to support synchronized signing across 2^τ time steps.

The full construction is given in Figure 5. We first set up the parameters for underlying KOTS and HVC. The KGen algorithm creates 2^τ KOTS key pairs and constructs an HVC out of the corresponding KOTS public keys. Signing at time t is done by using the t-th KOTS secret key to generate a one-time signature and attaching the corresponding opening (authentication path) of the HVC. Individual verification iVrfy calls corresponding individual verification functions of the underlying KOTS and HVC. Aggregation simply computes a weighted-sum of the one-time signatures and HVC openings. Thanks to homomorphic properties of KOTS and HVC, the resulting aggregate signature-opening pair is valid and can be

$\underline{\mathsf{Setup}(1^\lambda)}$

(1) $\mathsf{pp}_{\mathsf{KOTS}} \leftarrow \Pi_{\mathsf{KOTS}}.\mathsf{Setup}(1^\lambda)$
(2) $\mathsf{pp}_{\mathsf{HVC}} \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{Setup}(1^\lambda)$
(3) **return** $\mathsf{pp} := (\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{pp}_{\mathsf{HVC}})$

$\underline{\mathsf{Sign}(\mathsf{pp}, \mathsf{sk}, t, \mu)}$

(1) $\sigma' \leftarrow \Pi_{\mathsf{KOTS}}.\mathsf{Sign}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{osk}^{(t)}, \mu)$
(2) $d \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{Open}(\mathsf{pp}_{\mathsf{HVC}}, c, \mathsf{OPK}, t)$
(3) **return** $\sigma := (\sigma', d)$

$\underline{\mathsf{Aggregate}(\mathsf{pp}, P, t, \mu, S)}$

(1) **if** $|S| \neq N$ **or** $|P| \neq N$ **return** $\perp$
(2) **for** $(\mathsf{pk}, \sigma) \in \mathsf{zip}(P, S)$ **do**
(3) **if** $\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}, t, \mu, \sigma) = 0$ **return** $\perp$
(4) $j \leftarrow 0$
(5) **do**
- $j \leftarrow j + 1$
- $(w^{(1)}, \ldots, w^{(N)}) \leftarrow H(t, \mu, P, j)$
- $\sigma' \leftarrow \sum_{i=1}^{N} w^{(i)} \cdot \sigma'^{(i)}$
- $d \leftarrow \sum_{i=1}^{N} w^{(i)} \cdot d^{(i)}$

(6) **while** $\mathsf{aVrfy}(\mathsf{pp}, P, t, \mu, (\sigma', d, j)) = 0$ **and** $j < \gamma$
(7) **return** $\sigma_{\mathsf{agg}} := (\sigma', d, j)$

$\underline{\mathsf{KGen}(\mathsf{pp})}$

(1) **for** $i \in [2^\tau]$ **do**
(2) $(\mathsf{osk}^{(i)}, \mathsf{opk}^{(i)}) \leftarrow \Pi_{\mathsf{KOTS}}.\mathsf{KGen}(\mathsf{pp}_{\mathsf{KOTS}})$
(3) $\mathsf{OSK} := (\mathsf{osk}^{(1)}, \ldots, \mathsf{osk}^{(2^\tau)})$
(4) $\mathsf{OPK} := (\mathsf{opk}^{(1)}, \ldots, \mathsf{opk}^{(2^\tau)})$
(5) $c \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{Com}(\mathsf{pp}_{\mathsf{HVC}}, \mathsf{OPK})$
(6) **return** $(\mathsf{sk}, \mathsf{pk}) := ((\mathsf{OSK}, \mathsf{OPK}), c)$

$\underline{\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}, t, \mu, \sigma)}$

(1) Parse $\mathsf{pk} = c$ and $\sigma = (\sigma', d)$
(2) $\mathsf{opk} \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c, t, d)$
(3) **if** $t > 2^\tau$ **or** $\mathsf{opk} = \perp$ **return** 0
(4) **return** $\Pi_{\mathsf{KOTS}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}, \mu, \sigma')$

$\underline{\mathsf{aVrfy}(\mathsf{pp}, P, t, \mu, \sigma_{\mathsf{agg}})}$

(1) Parse $\sigma_{\mathsf{agg}} = (\sigma', d, j)$
(2) $(w^{(1)}, \ldots, w^{(N)}) \leftarrow H(t, \mu, P, j)$
(3) $c \leftarrow \sum_{i=1}^{N} w^{(i)} \cdot c^{(i)}$
(4) $\mathsf{opk} \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c, t, d)$
(5) **if** $|P| \neq N$ **or** $\mathsf{opk} = \perp$ **return** 0
(6) **return** $\Pi_{\mathsf{KOTS}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}, \mu, \sigma')$

Figure 5: Lemur Multi-Signature Construction.

checked using the corresponding aggregate public key (computed similarly via a weighted-sum). In Appendix C, we provide the full details of Lemur including formal definitions and proofs.

7 Implementation and Benchmarks

In this section, we provide concrete parameter setting for Lemur and evaluate its performance across various scales of signer sets N and time periods 2^τ. We compare our results with Chipmunk [8].

7.1 Parameters and Security Estimates

In this subsection, we describe how Lemur's parameters are chosen in practice. Table 4 summarizes all constraints required by Lemur construction. Table 6 reports the resulting sizes of Lemur, including public parameters, public and secret keys, and individual and aggregated signature sizes. We provide a parameter generation script[1]. The script is built on top of the Chipmunk codebase, but the underlying parameter-selection logic is fundamentally different. In particular, MSIS hardness is estimated using the state-of-the-art Dilithium security estimation framework[2], and MLWE hardness is evaluated using the lattice estimator[3] [1].

Optimization Goal. Our primary optimization goal is to minimize the size of the *aggregated signature*, subject to satisfying all constraints in Table 4. The aggregated signature consists of three components (see also Table 6 for a summary):

(1) **Path nodes.** There are $\tau\omega\kappa$ polynomials in $\mathcal{R}$ with ℓ^∞-norm bounded by β_{agg}, and an additional $\tau\omega\kappa$ polynomials with norm bounded by β_{encode}. This contributes (at most) $\tau\omega\kappa d\big(\log(2\beta_{\text{agg}}+1) + \log(2\beta_{\text{encode}}+1)\big)$ bits without any entropic encoding.
(2) **Aggregated one-time public keys.** These consist of $kn\kappa'$ polynomials in $\mathcal{R}$ with norm bounded by β_{agg}, contributing (at most) $kn\kappa' d \log(2\beta_{\text{agg}}+1)$ bits without any entropic encoding.
(3) **Aggregated KOTS signature.** This contains ℓm polynomials in $\mathcal{R}$ with norm bounded by β_σ, which is (at most) $\ell m d \log(2\beta_\sigma+1)$ bits without any entropic encoding.

[1]https://anonymous.4open.science/r/lemur-anon-5A9D/
[2]https://github.com/pq-crystals/security-estimates
[3]https://github.com/malb/lattice-estimator

#	Source	Constraint
1	Lemma B.5	$\beta_{\text{agg}} \geq \eta\sqrt{2\alpha_w N \ln L}$ $L = \frac{2d}{\epsilon_{\text{hom}}} \cdot N(2\tau\omega\kappa + kn\kappa' + 2\tau\omega)$
2	Lemma B.7	$\mathsf{MSIS}_{q,\omega,2\omega\kappa,4\beta_{\text{agg}}}$ is hard
3	Lemma B.7	$\mathsf{MSIS}_{q,\omega,kn\kappa',4\beta_{\text{agg}}}$ is hard
4	Lemma B.2	$\beta_{\text{encode}} < \frac{\beta_{\text{agg}}}{2\eta} + \frac{1}{2}$
5	Lemma D.1	$\alpha \geq \sqrt{2}\eta_{\varepsilon_1}(\mathbb{Z})$
6	Lemma D.1	$\beta_z \geq 6\alpha\sqrt{1+(k-\ell)\alpha_H}$
7	Lemma D.1	$\epsilon_{\text{cor}} \geq \ell m d \cdot \left(2^{-162} + 2(k-\ell)\alpha_H\varepsilon_1\right)$
8	Lemma D.2	$\beta_\sigma \geq \beta_z \cdot \sqrt{2\alpha_w N \cdot \ln\left(\frac{2\ell m d}{\epsilon_{\text{hom}}}\right)}$
9	Lemma 4.1	$\ell = 1, k \geq 4\ell, r = k - 2\ell$
10	Lemma 4.1	$p \equiv 2t_1 + 1 \pmod{4t_1}$
11	Lemma 4.2	$\left(\frac{1+\varepsilon_3}{1-\varepsilon_3}\right)^{2m} \cdot p^{-dm/t_1} < 2^{-\lambda}$
12	Lemma 4.2	$\alpha \geq p\sqrt{2\alpha_H \ln(2(r-1)d(1+1/\varepsilon_3))/\pi}$
13	Lemma 4.3	$q' \equiv 2t_2 + 1 \pmod{4t_2}$
14	Lemma 4.1	$\lvert\mathcal{T}_{\alpha_H}\rvert^{r-1} \geq 2^{2\lambda}$
15	Lemma 4.1	$\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$ is hard
16	Lemma 4.1	$\mathsf{MSIS}^{\infty}_{q',n,m,2\beta_\sigma+2\alpha_w\beta_z}$ is hard
17	Lemma C.2	$\gamma \geq \frac{\lambda}{\log_2\left(\frac{1}{2\epsilon_{\text{hom}}}\right)}$
18	Lemma C.3	$\lvert\mathcal{T}_{\alpha_w}\rvert = \binom{d}{\alpha_w} \cdot 2^{\alpha_w} \geq 2^\lambda$

Table 4: Parameter constraints of Lemur.

Parameter selection strategy. We divide the parameters into three categories: parameters shared by KOTS and HVC, KOTS-specific parameters, and HVC-specific parameters. Our strategy is to fix as many global parameters as possible, and then exhaustively search over the remaining degrees of freedom. We target a security level of $\lambda = 128$ bits. We begin by fixing the common parameters: the ring dimension $d \in \{128, 256\}$, the number of signers $N \in \{2^{10}, 2^{15}, 2^{17}, 2^{20}\}$, and the Merkle-tree height $\tau \in \{12, 16, 20, 24\}$. For ease of illustration, we describe the parameter-selection process for a representative setting $d = 128$, $N = 2^{20}$, and $\tau = 24$ from now on. Once d and λ are fixed, Constraint 18 (from Table 4) determines $\alpha_w = 31$. We set the probabilistic homomorphism error to $\epsilon_{\text{hom}} = 2^{-15}$, which in turn fixes the number of aggregation attempts to $\gamma = 10$ via Constraint 17.

KOTS parameters. We next determine the KOTS parameters. We set $\ell = 1$, $k = 5$, and $r = 3$ satisfying Constraint 9. Given (r, λ), Constraint 14 fixes $\alpha_H = 31$. We set $t_1 = 2$ and $p = 5$, $\varepsilon_3 = 2^{-0.5}$, after which Constraint 10 and Constraint 12 are automatically satisfied for all valid choices of m. We set $t_2 = 8$ and $q' \equiv 17 \pmod{32}$ by Constraint 13. The choice of the Gaussian width α is more delicate, as it must simultaneously satisfy Constraints 5, 15, and 12. As discussed later after HVC parameters, we select $\alpha = 61$, which meets all three constraints with margin. Once α is fixed, Constraint 6 determines individual-signature norm bound $\beta_z = 8185$. The remaining KOTS parameters (m, n, q') are chosen by enumerating feasible ranges for m and n, and selecting NTT-friendly primes q' such that both $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$ (Constraint 15) and

τ	N	Chipmunk (d = 1024) Sig.	Open	Total	Lemur (d = 256) Sig.	Open	Total
12	2^{10}	26	133	159	7	152	159
	2^{15}	> 90	204	> 294	9	212	220
	2^{17}	> 100	267	> 367	9	235	244
	2^{20}	> 110	449	> 559	10	293	303
16	2^{10}	26	172	198	7	192	199
	2^{15}	> 90	266	> 356	9	267	276
	2^{17}	> 100	347	> 447	9	297	306
	2^{20}	> 110	585	> 695	10	370	380
20	2^{10}	26	211	237	7	232	239
	2^{15}	> 90	328	> 418	9	323	331
	2^{17}	> 100	427	> 527	9	358	367
	2^{20}	> 110	721	> 831	10	447	457
24	2^{10}	26	250	276	7	272	279
	2^{15}	> 90	390	> 480	9	378	387
	2^{17}	> 100	507	> 607	9	420	429
	2^{20}	> 110	857	> 967	10	524	534

Table 5: Comparison of aggregated one-time signature, opening, and total aggregated sizes (KB) for $\lambda = 128$ and root Hermite factor $\mathsf{RHF} \leq 1.0045$. These are theoretical values without any entropic encoding. As given in Table 1, Lemur achieves about 14% shorter sizes with Rice encoding. Chipmunk data for $N \in \{2^{15}, 2^{17}, 2^{20}\}$ represent lower bounds, as feasible parameters could not be found for $d = 1024$ with $\mathsf{RHF} \leq 1.0045$. For these cases, we relax the KOTS requirements: $(N, \mathsf{RHF}_{\mathsf{KOTS}}) \in \{(2^{15}, 1.00464), (2^{17}, 1.0048), (2^{20}, 1.00501)\}$. Lemur uses the fixed $d = 256, k = 4$ parameter set and maintains $\mathsf{RHF} \leq 1.0045$ throughout.

$\mathsf{MSIS}^{\infty}_{q',n,m,2\beta_\sigma+2\alpha_w\beta_z}$ (Constraint 16) meet target security level. For each candidate, we compute β_σ using Constraint 8 and select the parameters minimizing the aggregated KOTS signature size. Constraint 7 is checked to ensure that the individual correctness error is negligible.

HVC parameters. Finally, we determine the HVC parameters. To reduce the search space, we enumerate candidate bit-lengths for q in the range $[16, 63]$ bits. For each candidate q, we enumerate small integer values of κ and ω and retain those satisfying Constraints 1, 2, 3, and 4. Among all valid candidates, we select the parameter set that minimizes the HVC opening size. Overall, this systematic parameter selection procedure yields compact aggregated signatures while satisfying all correctness and security constraints.

How to choose α. We convert all the three constraints we require for α in Table 4 into explicit numerical bounds and explain how the final value is chosen.

(1) Constraint 5: Upper bound $\eta_{\varepsilon_1}(\mathbb{Z})$ by Lemma 2.4, we obtain $\alpha \geq \sqrt{\frac{4(1+1/\varepsilon_1)}{\pi}} = 5.4$ for $\varepsilon_1 = 2^{-150}$. Substituting $\ell = 1$, $m = 20$, $d = 128$, $k = 5$, $\alpha_H = 31$, $\varepsilon_1 = 2^{-150}$ into Constraint 7 gives $\epsilon_{\text{cor}} \approx 2^{-130}$, which is negligible.
(2) Constraint 15: By Theorem 3.1, $\mathsf{DualHintMLWE}_{q',m,n,k,\ell,\alpha,\mathcal{T}_{\alpha_H}}$ is at least as hard as $\mathsf{MLWE}_{q',m,n,k-\ell,\alpha_0}$ provided that $\alpha^2 \geq \alpha_0^2 \|\mathbf{B}\|_2^2 + (1+\ell\alpha_H)\frac{\ln(2d(k-\ell)(1+1/\varepsilon_2))}{\pi}$, where $\mathbf{B} = \binom{-\mathbf{H}'}{\mathbf{I}_{k-\ell}}$. To evaluate this bound concretely, we estimate $\|\mathbf{B}\|_2$ numerically by sampling $\mathbf{H}'$ according to $\mathcal{T}_{\alpha_H}$ and computing the spectral norm of $\mathbf{B}$. Across 10^3, 10^4, 10^5, 10^6 trials, the empirical distribution of $\|\mathbf{B}\|_2$ follows a Gaussian distribution. We therefore conservatively bound $\|\mathbf{B}\|_2$ by taking the empirical mean plus 12× standard deviations, capturing the tail behavior. This yields $\|\mathbf{B}\|_2 \approx 33.36$. Setting $\alpha_0 = \sqrt{2\pi/3}$ and $\varepsilon_2 = 2^{-136}$, chosen so that the term $8m\varepsilon_2$ in Lemma 3.3 is negligible, and substituting all parameters into the above inequality gives $\alpha \geq 57.97$.
(3) Constraint 12: Substituting the above concrete parameters, we obtain $\alpha \geq 60.93$.

Taking the maximum over all three bounds, we set $\alpha = 61$. Substituting this value back into the Dual Hint-MLWE sampleability condition yields a revised bound $\alpha_0 = 1.5$, which is used in the standard MLWE estimation for the parameter generation script.

7.2 Implementation Notes

The main advantage of Lemur over non-aggregated signatures is the smaller signature size. However, its resource utilization is also remarkably modest.

The main deployment challenge is signer state: a naively materialized HVC tree at $\tau = 20$ occupies many GB of RAM, and τ is limited by available RAM (as in the Chipmunk reference implementation). Our default signer instead uses the BDS08 Merkle traversal algorithm [4], which maintains an amortized-$O(\tau)$ authentication-path cache of ≈ 134 KB against the same 32-byte master seed. This brings signing below 10 ms on commodity hardware after a one-time offline key generation, without carrying a multi-GB tree on disk or in RAM. With the entire tree in RAM, consecutive signatures can be generated in 2 ms. We consider the use of BDS08 as a significant practical contribution. Signature verification times are close to those of ML-DSA and Falcon at the same security level.

Each polynomial in the aggregated signature is bit-packed independently: the aggregated KOTS signature uses a fixed-width encoding tuned to the CLT-predicted standard deviation of $\mathbf{Z}$, and the HVC opening uses Golomb-Rice coding for components whose folded-Gaussian model yields a shorter average code than the fixed-width alternative. On the implemented (τ=20, N=2^{10}) parameter setting, the encoder brings the aggregate size from the 239 KB bound (Table 5) down to 201.2 KB; the same encoder yields ≈ 14% savings on the larger-N cells (e.g. 394.4 KB vs. 457 KB at $N = 2^{20}$).

We provide two independent, fully interoperable open-source reference implementations: an optimized **Rust** crate and a readable **Python** prototype. Both cover the full pipeline (KOTS, HVC, aggregation, encoding) and are cross-validated against shared byte-level test vectors, so the specification admits a single concrete realization. The Rust artifact combines a Montgomery-domain NTT for the HVC ring with a CRT-via-auxiliary-primes backend for the KOTS ring (whose modulus $q' \equiv 17 \pmod{32}$ is not natively NTT-friendly), a discrete-Gaussian CDT sampler whose precision and tail-cut are pinned by a Rényi-divergence analysis at $\lambda = 128$, and three interchangeable signing paths—seed-only, BDS08 cache [4], and an optional in-memory materialized tree—that produce byte-identical signatures. The implementations and more complete benchmarks are available at https://anonymous.4open.science/r/lemur-anon-5A9D/.

References

[1] Martin R. Albrecht, Rachel Player, and Sam Scott. 2015. On the concrete hardness of Learning with Errors. *J. Math. Cryptol.* 9, 3 (2015), 169–203.

[2] Dan Boneh and Sam Kim. 2020. One-Time and Interactive Aggregate Signatures from Lattices. https://api.semanticscholar.org/CorpusID:221364120

[3] Dan Boneh, Ben Lynn, and Hovav Shacham. 2001. Short Signatures from the Weil Pairing. In *ASIACRYPT 2001 (LNCS)*. Springer, 514–532.

[4] Johannes Buchmann, Erik Dahmen, and Michael Schneider. 2008. Merkle Tree Traversal Revisited. In *Post-Quantum Cryptography (PQCrypto) 2008 (LNCS, Vol. 5299)*. Springer, 63–78.

[5] Maxime Buser, Rafael Dowsley, Muhammed F. Esgin, Shabnam Kasra Kermanshahi, Veronika Kuchta, Joseph K. Liu, Raphaël C.-W. Phan, and Zhenfei Zhang. 2022. Post-Quantum Verifiable Random Function from Symmetric Primitives in PoS Blockchain. In *ESORICS (1) (LNCS)*. Springer, 25–45.

[6] Julien Devevey, Alain Passelègue, and Damien Stehlé. 2023. G+G: A Fiat-Shamir Lattice Signature Based on Convolved Gaussians. In *ASIACRYPT 2023 (LNCS)*. Springer, 37–64.

[7] Justin Drake, Dmitry Khovratovich, Mikhail A. Kudinov, and Benedikt Wagner. 2025. Hash-Based Multi-Signatures for Post-Quantum Ethereum. *IACR Commun. Cryptol.* 2, 1 (2025), 13.

[8] Nils Fleischhacker, Gottfried Herold, Mark Simkin, and Zhenfei Zhang. 2023. Chipmunk: Better Synchronized Multi-Signatures from Lattices. In *CCS 2023*. ACM, 386–400.

[9] Nils Fleischhacker, Mark Simkin, and Zhenfei Zhang. 2022. Squirrel: Efficient Synchronized Multi-Signatures from Lattices. In *CCS 2022*. ACM, 1109–1123.

[10] Kazuharu Itakura and Katsuhiro Nakamura. 1983. A public-key cryptosystem suitable for digital multisignatures. *NEC Research & Development* 71 (1983), 1–8.

[11] Duhyeong Kim, Dongwon Lee, Jinyeong Seo, and Yongsoo Song. 2023. Toward Practical Lattice-Based Proof of Knowledge from Hint-MLWE. In *CRYPTO 2023 (LNCS, Vol. 14085)*. Springer, 549–580.

[12] Vadim Lyubashevsky. 2012. Lattice Signatures without Trapdoors. In *EUROCRYPT 2012 (LNCS, Vol. 7237)*. Springer, 738–755.

[13] Vadim Lyubashevsky and Daniele Micciancio. 2008. Asymptotically Efficient Lattice-Based Digital Signatures. In *TCC 2008 (LNCS)*. Springer, 37–54.

[14] Vadim Lyubashevsky and Gregor Seiler. 2018. Short, Invertible Elements in Partially Splitting Cyclotomic Rings and Applications to Lattice-Based Zero-Knowledge Proofs. In *EUROCRYPT 2018 (LNCS, Vol. 10820)*. Springer, 204–224.

[15] Silvio Micali, Kazuo Ohta, and Leonid Reyzin. 2001. Accountable-subgroup multisignatures: extended abstract. In *CCS 2001*. ACM, 245–254.

[16] Daniele Micciancio and Oded Regev. 2004. Worst-Case to Average-Case Reductions Based on Gaussian Measures. In *FOCS 2004*. IEEE Computer Society, 372–381.

[17] Chris Peikert. 2010. An Efficient and Parallel Gaussian Sampler for Lattices. In *CRYPTO 2010 (LNCS, Vol. 6223)*. Springer, 80–97.

Open Science

In line with the ACM CCS open-science policy, the artifacts required to reproduce the core claims of this paper are made available at https://anonymous.4open.science/r/lemur-anon-5A9D/. The Lemur artifact contains three independent components:

- **Python reference implementation** (`lemur-py`): A readable prototype covering the full Lemur pipeline (KOTS, HVC, aggregation, and bit-packed encoding), together with its command-line interface, unit tests, and a test-vector dumper. Primarily intended as a "golden reference" and functional test vector generator.
- **Rust reference implementation** (`lemur-rs`): A performance-oriented implementation covering the same pipeline, together with benchmarking tools and integration tests for robustness. Cross-validated byte-for-byte against the Python implementation through a shared set of test vectors.
- **Parameter-selection script**: The scripts used to produce the concrete parameter sets reported in this paper. Sweeps candidate dimensions and other parameters against the scheme constraints, together with the MSIS / MLWE hardness estimates.

Each component ships with its own `README` and build instructions. In addition to open-source tools such as compilers and Sagemath, no additional datasets, models, or proprietary artifacts are required to evaluate the claims of this paper.

Ethical Considerations

This work presents a theoretical study of lattice-based multi-signature schemes and does not involve human subjects, user data, or experimentation on deployed systems. Consequently, it does not raise direct ethical concerns related to privacy, data protection, or user consent, and no institutional review was required.

We explicitly consider the broader societal implications of our work, in particular the balance between its potential benefits and risks. On the positive side, our contributions advance the design and analysis of post-quantum secure signature schemes, which are important for strengthening the long-term security and integrity of digital communications and distributed systems. Such advances are especially relevant in the context of emerging quantum threats.

At the same time, we acknowledge the inherent dual-use nature of cryptographic research. Cryptographic primitives, including multi-signature schemes, may be deployed both in benign applications (e.g., secure collaboration, blockchain systems) and in settings that could potentially conceal malicious activities. This dual-use characteristic is a fundamental property of cryptography and is not specific to our construction.

To mitigate potential risks, our work is limited to theoretical modeling and formal analysis. We do not provide implementations, system deployments, or tools that could be directly misused in practice. Furthermore, we adhere to established norms in the cryptographic research community, aiming to promote transparency, rigor, and responsible dissemination of results.

Overall, we assess the ethical risk of this work to be minimal, while its potential benefits to the security community are significant.

Generative AI Usage

Generative AI assistance was used to create and validate experiments and implementation artifacts. The main tools were OpenAI Codex with GPT-5.5 and Anthropic Claude Code with Opus-4.7.

A Additional Preliminaries

Lemma A.1. *For $a, b \in \mathcal{R}$, it holds that*

(1) $\|a\|_{\max} \leq \|a\|_2 \leq \|a\|_1$,
(2) $\|a+b\|_{\max} \leq \|a\|_{\max} + \|b\|_{\max}$,
(3) $\|a \cdot b\|_{\max} \leq \|a\|_1 \cdot \|b\|_{\max}$.

Lemma A.2 ([14], Corollary 1.2). *Let $d \geq t > 1$ be powers of 2 and let $p \equiv 2t+1 \pmod{4t}$ be a prime. Then, any $y \in \mathcal{R}_p$ that satisfies $0 < \|y\|_{\max} < \frac{1}{\sqrt{t}} \cdot p^{1/t}$ has an inverse in $\mathcal{R}_p$.*

Lemma A.3 ([8], Lemma 4). *Let $d, \alpha_w, N, \beta \in \mathbb{N}^+$ with d a power of 2. Then for any $x_1, \ldots, x_N \in \mathcal{R}$ with $\|x_i\|_{\max} \leq \beta$ for all i any growing factors $\zeta \geq 1$, it holds that*

$$\Pr_{w_1,\ldots,w_N \leftarrow \mathcal{T}_{\alpha_w}}\left[\left\|\sum_{i=1}^{N} w_i x_i\right\|_{\max} > \zeta\beta\right] \leq 2d \cdot \exp\left(-\frac{\zeta^2}{2\alpha_w N}\right).$$

Lemma A.4 (Extended version of Lemma 3 of [11], Theorem 3.1 of [17]). *Let $k \in \mathbb{N}^+$, and $\varepsilon \in (0, \frac{1}{2})$ such that $\alpha \geq \sqrt{2}\eta_\varepsilon(\mathbb{Z})$. If $x_1, \ldots, x_k$ are independent samples from $\mathcal{D}_{\mathbb{Z},\alpha}$, then*

$$\Delta\left(\sum_{i=1}^{k} x_i,\ \mathcal{D}_{\mathbb{Z},\sqrt{k}\alpha}\right) \leq 2(k-1)\varepsilon.$$

Lemma A.5 ([12], Lemma 4.4; [11], Lemma 1). *For $\alpha \in \mathbb{R}_{>0}$ and $z \leftarrow \mathcal{D}_{\mathbb{Z},\alpha}$, $\Pr[|z| < 6\alpha] > 1 - 2^{-162}$.*

Definition A.1 (Labeled Full Binary Tree). *For a depth $\tau \in \mathbb{N}^+$, we identify the nodes of a full binary tree with the set of bit-strings $\mathcal{V} = \bigcup_{i=0}^{\tau}\{0,1\}^i$. The root is the empty string ε. For any internal node $v \in \{0,1\}^{<\tau}$, its left and right children are $v\|0$ and $v\|1$, respectively. The set of leaves is $\mathcal{L} = \{0,1\}^\tau$. A labeled tree is defined by a function $\mathsf{label} : \mathcal{V} \to \mathcal{X}$, which assigns a value from a set $\mathcal{X}$ to each node in the tree.*

Definition A.2 (R-module). *Let $(R, +, \times)$ be a commutative ring. We say that a set M is an R-module when $(M, +)$ is an abelian group, and when there exists an operation $\cdot$ (product) such that for all $r, s \in R$ and $x, y \in M$, it holds that*

(1) $1_R \cdot x = x$,
(2) $(r + s) \cdot x = r \cdot x + s \cdot x$,
(3) $r \cdot (x + y) = r \cdot x + r \cdot y$,
(4) $(r \times s) \cdot x = r \cdot (s \cdot x)$.

Remark A.1. *For $n \in \mathbb{N}^+$ and prime q, $\mathcal{R}^n$ is an $\mathcal{R}$-module. Additionally, $\mathcal{R}_q^n$ is an $\mathcal{R}_q$-module.*

B Details of Homomorphic Vector Commitment

B.1 Definition

We formally define homomorphic vector commitments here.

Definition B.1 (Homomorphic Vector Commitment (HVC), [8]). *Let $\tau, d \in \mathbb{N}^+$ such that d is a power of 2. Let $\mathcal{R} = \mathbb{Z}[X]/(X^d+1)$. Let $\mathcal{S}_{\mathsf{dom}}, \mathcal{S}_{\mathsf{com}}, \mathcal{S}_{\mathsf{open}}$ be $\mathcal{R}$-modules. A homomorphic vector commitment scheme (HVC) for domain $\mathcal{S}_{\mathsf{dom}}$ and vectors of length 2^τ consists of six PPT algorithms $\Pi_{\mathsf{HVC}} = (\mathsf{Setup}, \mathsf{Com}, \mathsf{Open}, \mathsf{iVrfy}, \mathsf{sVrfy}, \mathsf{wVrfy})$.*

- $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$*: On input a security parameter $\lambda \in \mathbb{N}^+$, the setup algorithm outputs a set of public parameters* pp.
- $c \leftarrow \mathsf{Com}(\mathsf{pp}, \widetilde{\mathbf{M}})$*: On input a set of public parameters* pp *and a vector of matrices $\widetilde{\mathbf{M}} = \{\mathbf{M}_1, \ldots, \mathbf{M}_{2^\tau}\} \in \mathcal{S}_{\mathsf{dom}}^{2^\tau}$, the commitment algorithm outputs a commitment $c \in \mathcal{S}_{\mathsf{com}}$.*
- $d \leftarrow \mathsf{Open}(\mathsf{pp}, c, \widetilde{\mathbf{M}}, t)$*: On input a set of public parameters* pp*, a commitment c, a committed vector $\widetilde{\mathbf{M}}$, and an index $t \in [2^\tau]$, the opening algorithm outputs a decommitment $d \in \mathcal{S}_{\mathsf{open}}$*
- $\mathbf{M}_t/\bot \leftarrow \mathsf{iVrfy}/\mathsf{sVrfy}/\mathsf{wVrfy}(\mathsf{pp}, c, t, d)$*: On input a set of public parameters* pp*, a commitment c, an index t, and a decommitment d, the individual/strong/weak verification algorithm outputs either $\mathbf{M}_t \in \mathcal{S}_{\mathsf{dom}}$ or an error symbol $\bot$.*

Our HVC construction, operates over the message space $\mathcal{S}_{\mathsf{dom}} = \mathcal{R}_{q'}^{\rho\times\nu}$ for $\rho, \nu \in \mathbb{N}^+$ and a prime q'. In contrast, Chipmunk considers the domain $\mathcal{S}_{\mathsf{dom}} = \mathcal{R}_{q'}^{\ell_{\mathsf{dom}}}$. The commitment and opening spaces will be of the form $\mathcal{S}_{\mathsf{com}} = \mathcal{R}_q^{\ell_{\mathsf{com}}}$ and $\mathcal{S}_{\mathsf{open}} = \mathcal{R}^{\ell_{\mathsf{open}}}$ for a prime q as in Chipmunk. We require a HVC to satisfy the following four properties: individual correctness, probabilistic homomorphism, robust homomorphism, and position binding. These properties are formally defined below.

Definition B.2 (Individual Correctness, [8]). *We say that Π_{HVC} is individually correct, if for all $\lambda \in \mathbb{N}^+$, $\widetilde{\mathbf{M}} \in \mathcal{S}_{\mathsf{dom}}^{2^\tau}$, $t \in [2^\tau]$, $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $c \leftarrow \mathsf{Com}(\mathsf{pp}, \widetilde{\mathbf{M}})$, and $d \leftarrow \mathsf{Open}(\mathsf{pp}, c, \widetilde{\mathbf{M}}, t)$ it holds that $\mathsf{iVrfy}(\mathsf{pp}, c, t, d) = \mathbf{M}_t$.*

Definition B.3 (Probabilistic Homomorphism, [8]). *Let $N \in \mathbb{N}^+$, $\epsilon_{\mathsf{hom}} \in (0,1)$, and $W \subseteq \mathcal{R}$. We say that Π_{HVC} is $(N, W, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, if for all $\lambda \in \mathbb{N}^+$, $t \in [2^\tau]$, $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, and $(c^i \in \mathcal{S}_{\mathsf{com}}, d^i \in \mathcal{S}_{\mathsf{open}})$ where $\mathsf{iVrfy}(\mathsf{pp}, c^i, t, d^i) = \mathbf{M}_t^i \neq \bot$ for all $i \in [N]$, it holds that:*

$$\Pr_{w^i \leftarrow W}\left[\mathsf{sVrfy}\Big(\mathsf{pp}, \sum_{i=1}^{N} w^i c^i, t, \sum_{i=1}^{N} w^i d^i\Big) = \sum_{i=1}^{N} w^i \mathbf{M}_t^i\right] \geq 1 - \epsilon_{\mathsf{hom}}.$$

Definition B.4 (Robust Homomorphism, [8]). *We say that Π_{HVC} is robustly homomorphic, if for all $\lambda \in \mathbb{N}^+$, $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $t \in [2^\tau]$, any $c^b \in \mathcal{S}_{\mathsf{com}}$ and $d^b \in \mathcal{S}_{\mathsf{open}}$ ($b \in \{0,1\}$) with $\mathsf{sVrfy}(\mathsf{pp}, c^b, t, d^b) = \mathbf{M}^b \neq \bot$, it holds that $\mathsf{wVrfy}(\mathsf{pp}, c^0 - c^1, t, d^0 - d^1) = \mathbf{M}^0 - \mathbf{M}^1$.*

Definition B.5 (Position Binding, [8]). *We say that Π_{HVC} is position binding, if for all $\lambda \in \mathbb{N}^+$ and PPT algorithms $\mathcal{A}$ it holds that $\Pr[\mathbf{M}^0 \neq \mathbf{M}^1 \wedge \bot \notin \{\mathbf{M}^0, \mathbf{M}^1\}] \leq \mathsf{negl}(\lambda)$, where $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $(c, t, d^0, d^1) \leftarrow \mathcal{A}(\mathsf{pp})$, and $\mathbf{M}^b := \mathsf{wVrfy}(\mathsf{pp}, c, t, d^b)$ for $b \in \{0,1\}$.*

B.2 Projection and Decomposition

Chipmunk constructs a homomorphic vector commitment (HVC) using a Merkle tree, where homomorphism is achieved by replacing the standard hash function with the Ajtai hash. This design, however, introduces an inherent technical constraint: the Ajtai hash requires inputs to be "small", whereas its outputs are not guaranteed to satisfy this property. To address this issue, Chipmunk introduces a pair of canonical transformations: a decomposition map, which converts an element of $\mathcal{R}_q$ into a short vector over $\mathcal{R}$, and a corresponding projection map, which recombines such a short vector into the original ring element. Informally, decomposition enables "large" elements to be represented in a form suitable for Ajtai hashing, while projection ensures consistency with the underlying algebraic structure. In this subsection, we first recall the projection and decomposition maps used in Chipmunk, along with their key properties. We then extend these maps to Lemur's label space by applying them independently to each block of a tree node.

Projection and Decomposition in Chipmunk. We first provide additional intuition for why projection and decomposition are necessary in the construction of a homomorphic Merkle tree. For clarity, we focus on the core idea and omit some technical details.

In Chipmunk, the internal node of the Merkle tree is computed using an Ajtai hash. Concretely, given two child nodes $\mathbf{c}_1, \mathbf{c}_2 \in \mathcal{R}_q^{\ell_{\mathsf{dom}}}$, the parent node is computed as

$$\mathbf{p} = \mathbf{a}_1^\top \mathbf{c}_1 + \mathbf{a}_2^\top \mathbf{c}_2 \mod q,$$

where $\mathbf{a}_1, \mathbf{a}_2 \in \mathcal{R}_q^{\ell_{\mathsf{dom}}}$ are public parameters. While this operation preserves homomorphism, it creates a mismatch in representation: the resulting $\mathbf{p}$ lies in $\mathcal{R}_q$, whereas the Ajtai hash requires inputs to be vectors of "small" elements over $\mathcal{R}$. To enable iterative hashing across tree levels, Chipmunk employs a balanced $(2\eta+1)$-ary decomposition on $\mathbf{p}$, where coefficients lie in small sets $\{-\eta, \ldots, \eta\}$. The projection map serves as an inverse transformation and ensures that this transformation remains consistent with the original algebraic value, i.e.,

$$\overline{\mathsf{proj}}_q(\mathbf{p}) = \mathbf{a}_1^\top \mathbf{c}_1 + \mathbf{a}_2^\top \mathbf{c}_2 \mod q.$$

We will see in the following that the projection map enjoys useful homomorphic properties.

(1) **Projection and Decomposition over $\mathcal{R}$.** Let $\eta, \kappa \in \mathbb{N}^+$. Define the projection map

$$\overline{\mathsf{proj}}_{\eta,\kappa} : \mathcal{R}^\kappa \to \mathcal{R}, \qquad \overline{\mathsf{proj}}_{\eta,\kappa}(a_1, \ldots, a_\kappa) := \sum_{i=1}^{\kappa} a_i \cdot (2\eta+1)^{i-1}.$$

The corresponding decomposition map

$$\overline{\mathsf{dec}}_{\eta,\kappa} : \mathcal{R} \to \mathcal{R}^\kappa, \qquad a \mapsto (a_1, \ldots, a_\kappa)$$

satisfies

$$a = \sum_{i=1}^{\kappa} a_i \, (2\eta+1)^{i-1}, \text{ and } \|a_i\|_{\max} \le \eta \text{ for all } 1 \le i < \kappa.$$

(2) **Projection and Decomposition over $\mathcal{R}_q$.** Let $\eta \in \mathbb{N}^+$ and q be an odd prime. Define $\kappa := \lceil \log_{2\eta+1} q \rceil$. The projection map is given by

$$\overline{\mathsf{proj}}_q : \mathcal{R}^\kappa \to \mathcal{R}_q, \qquad \overline{\mathsf{proj}}_q(\mathbf{a}) := \overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{a}) \bmod q,$$

The corresponding decomposition map is defined as

$$\overline{\mathsf{dec}}_q : \mathcal{R}_q \to \mathcal{R}^\kappa, \qquad \overline{\mathsf{dec}}_q(a) := \overline{\mathsf{dec}}_{\eta,\kappa}(a'),$$

where $a' \in \mathcal{R}$ is the centered representative of $a \in \mathcal{R}_q$ with coefficients in $\left\{-\frac{q-1}{2}, \ldots, \frac{q-1}{2}\right\}$.

Proposition B.1. *The following facts are established in Proposition 13 of Chipmunk [8].*

(1) The maps $\overline{\mathsf{proj}}_{\eta,\kappa}$ and $\overline{\mathsf{proj}}_q$ are $\mathcal{R}$-linear, i.e., for all $\mathbf{a}_0, \mathbf{a}_1 \in \mathcal{R}^\kappa$ and $w_0, w_1 \in \mathcal{R}$,

$$\overline{\mathsf{proj}}_{\eta,\kappa}(w_0 \cdot \mathbf{a}_0 + w_1 \cdot \mathbf{a}_1) = w_0 \cdot \overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{a}_0) + w_1 \cdot \overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{a}_1),$$
$$\overline{\mathsf{proj}}_q(w_0 \cdot \mathbf{a}_0 + w_1 \cdot \mathbf{a}_1) = w_0 \cdot \overline{\mathsf{proj}}_q(\mathbf{a}_0) + w_1 \cdot \overline{\mathsf{proj}}_q(\mathbf{a}_1).$$

(2) For all $a \in \mathcal{R}_q$, it holds that $\overline{\mathsf{proj}}_q(\overline{\mathsf{dec}}_q(a)) = a$.

(3) For all $a \in \mathcal{R}$, it holds that $\overline{\mathsf{proj}}_{\eta,\kappa}(\overline{\mathsf{dec}}_{\eta,\kappa}(a)) = a$.

(4) For all $a \in \mathcal{R}_q$, it holds that $\|\overline{\mathsf{dec}}_q(a)\|_{\max} \le \eta$.

Proposition B.2 (Proposition 22 of [8]). *Let $q, \eta \in \mathbb{N}^+$ with q prime. Set $B := 2\eta + 1$ and $\kappa := \lceil \log_B q \rceil$. Define lattices*

$$\Lambda_{\mathcal{R}} := \{\mathbf{v} \in \mathcal{R}^\kappa \mid \mathsf{proj}_{\eta,\kappa}(\mathbf{v}) = 0\}.$$

Let

$$\mathcal{B} := \begin{pmatrix} -B & 0 & \cdots & 0 \\ 1 & -B & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & -B \\ 0 & 0 & \cdots & 1 \end{pmatrix} \in \mathcal{R}^{\kappa \times (\kappa-1)}.$$

Then, $\Lambda_{\mathcal{R}}$ is an $\mathcal{R}$-module lattice. Moreover, the columns of $\mathcal{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_{\kappa-1}]$ form an $\mathcal{R}$-basis of $\Lambda_{\mathcal{R}}$.

Projection and Decomposition in Lemur. We now explain why it is necessary to extend the projection and decomposition maps of Chipmunk. In Lemur, the binding property of the Ajtai hash is based on the MSIS assumption, whereas Chipmunk relies on RSIS. Concretely, given two child nodes $\mathbf{c}_1, \mathbf{c}_2 \in \mathcal{R}_q^{\ell_{\text{dom}}}$, the hash computation in Lemur takes the form

$$\mathbf{y} = \mathbf{A}_1\mathbf{c}_1 + \mathbf{A}_2\mathbf{c}_2 \in \mathcal{R}_q^\omega,$$

where $\mathbf{A}_1, \mathbf{A}_2 \in \mathcal{R}_q^{\omega \times \ell_{\text{dom}}}$ are public parameters. Thus, in contrast to Chipmunk, where the hash output is a single element of $\mathcal{R}_q$ element, the output in Lemur is a vector of $\mathcal{R}^\omega$. This distinction leads to a natural employment of the decomposition map entry-wise. Moving forward, after decomposition, Merkle tree node labels in Lemur lie in $\mathcal{R}^{\omega\kappa}$. We may view any vector $\mathbf{v} \in \mathcal{R}^{\omega\kappa}$ as a concatenation of ω blocks, i.e.,

$$\mathbf{v} = \left(\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(\omega)}\right), \qquad \mathbf{v}^{(r)} \in \mathcal{R}^\kappa \text{ for } r \in [\omega].$$

We extend the maps $\overline{\mathsf{proj}}_{\eta,\kappa}, \overline{\mathsf{proj}}_q, \overline{\mathsf{dec}}_{\eta,\kappa}, \overline{\mathsf{dec}}_q$ to this setting by applying them independently to each block. Specifically, we define

(1) **Projection and Decomposition over $\mathcal{R}$.** Let $\eta, \kappa, \omega \in \mathbb{N}^+$. Define the projection map $\mathsf{proj}_{\eta,\kappa} : \mathcal{R}^{\omega\kappa} \to \mathcal{R}^\omega$, with

$$\mathsf{proj}_{\eta,\kappa}(\mathbf{v}) := \left(\overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{v}^{(1)}), \ldots, \overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{v}^{(\omega)})\right)$$

The corresponding decomposition map $\mathsf{dec}_{\eta,\kappa} : \mathcal{R}^\omega \to \mathcal{R}^{\omega\kappa}$, with

$$\mathsf{dec}_{\eta,\kappa}(\mathbf{y}) := \left(\overline{\mathsf{dec}}_{\eta,\kappa}(y_1), \ldots, \overline{\mathsf{dec}}_{\eta,\kappa}(y_\omega)\right).$$

(2) **Projection and Decomposition over $\mathcal{R}_q$.** Let $\eta \in \mathbb{N}^+$ and q be an odd prime. Define $\kappa := \lceil \log_{2\eta+1} q \rceil$. The projection map is given by $\mathsf{proj}_q : \mathcal{R}^{\omega\kappa} \to \mathcal{R}_q^\omega$, with

$$\mathsf{proj}_q(\mathbf{v}) := \left(\overline{\mathsf{proj}}_q(\mathbf{v}^{(1)}), \ldots, \overline{\mathsf{proj}}_q(\mathbf{v}^{(\omega)})\right).$$

The corresponding decomposition map is defined as $\mathsf{dec}_q : \mathcal{R}_q^\omega \to \mathcal{R}^{\omega\kappa}$, with

$$\mathsf{dec}_q(\mathbf{y}) := \left(\overline{\mathsf{dec}}_q(y_1), \ldots, \overline{\mathsf{dec}}_q(y_\omega)\right).$$

Proposition B.3. *The following facts are obtained by independently employing Proposition B.1 block by block.*

(1) The maps $\mathsf{proj}_{\eta,\kappa}$ and proj_q are $\mathcal{R}$-linear, i.e., for all $\mathbf{a}_0, \mathbf{a}_1 \in \mathcal{R}^{\omega\kappa}$ and $w_0, w_1 \in \mathcal{R}$,

$$\mathsf{proj}_{\eta,\kappa}(w_0 \cdot \mathbf{a}_0 + w_1 \cdot \mathbf{a}_1) = w_0 \cdot \mathsf{proj}_{\eta,\kappa}(\mathbf{a}_0) + w_1 \cdot \mathsf{proj}_{\eta,\kappa}(\mathbf{a}_1),$$
$$\mathsf{proj}_q(w_0 \cdot \mathbf{a}_0 + w_1 \cdot \mathbf{a}_1) = w_0 \cdot \mathsf{proj}_q(\mathbf{a}_0) + w_1 \cdot \mathsf{proj}_q(\mathbf{a}_1).$$

(2) For all $\mathbf{a} \in \mathcal{R}_q^\omega$, it holds that $\mathsf{proj}_q(\mathsf{dec}_q(\mathbf{a})) = \mathbf{a}$.
(3) For all $\mathbf{a} \in \mathcal{R}^\omega$, it holds that $\mathsf{proj}_{\eta,\kappa}(\mathsf{dec}_{\eta,\kappa}(\mathbf{a})) = \mathbf{a}$.
(4) For all $\mathbf{a} \in \mathcal{R}_q^\omega$, it holds that $\|\mathsf{dec}_q(\mathbf{a})\|_{\max} \le \eta$.

B.3 Encode / Decode Algorithms

Projection and decomposition give us canonical short representatives for node labels, but they do not by themselves yield small *aggregatable* openings: since dec_q is not additive, aggregated openings cannot be verified from sibling nodes alone without also sending the intermediate path labels. Chipmunk avoids this blow-up using an additional encoding step that makes the missing path information recoverable from a short "hint". In this subsection, we migrate

Chipmunk's encoding techinique to our label space $\mathcal{R}^{\omega\kappa}$. The resulting algorithms $\mathsf{Encode}^{\mathcal{B}}_{\eta,q}$ and $\mathsf{Decode}^{\mathcal{B}}_{\eta,q}$ are applied block-wise and allow the verifier to reconstruct each path label from its encoding and the corresponding hint. See Figure 6 for the explicit algorithms.

$\underline{\mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{v})}$

(1) Set $\kappa \leftarrow \lceil \log_{2\eta+1}(q) \rceil$.
(2) Parse $\mathbf{v} \in \mathcal{R}^{\omega\kappa}$ as $\mathbf{v} = (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(\omega)})$, $\mathbf{v}^{(r)} \in \mathcal{R}^{\kappa}$.
(3) For each $r \in [\omega]$ do:
  (a) $\mathsf{hint}^{(r)} \leftarrow \overline{\mathsf{proj}}_q(\mathbf{v}^{(r)}) \in \mathcal{R}_q$.
  (b) Represent $\mathsf{hint}^{(r)}$ by $\mathsf{hint}'^{(r)} \in \mathcal{R}$ with coefficients in $\left[-\frac{q-1}{2}, \frac{q-1}{2}\right]$.
  (c) $\alpha_*^{(r)} \leftarrow \dfrac{\overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{v}^{(r)}) - \mathsf{hint}'^{(r)}}{q} \in \mathcal{R}$
  (d) $\boldsymbol{\delta}^{(r)} \leftarrow \mathbf{v}^{(r)} - \overline{\mathsf{dec}}_{\eta,\kappa}(\overline{\mathsf{proj}}_{\eta,\kappa}(\mathbf{v}^{(r)})) \in \mathcal{R}^{\kappa}$.
  (e) Find unique $\alpha_1^{(r)}, \ldots, \alpha_{\kappa-1}^{(r)} \in \mathcal{R}$ such that
$$\boldsymbol{\delta}^{(r)} = \alpha_1^{(r)}\mathbf{b}_1 + \cdots + \alpha_{\kappa-1}^{(r)}\mathbf{b}_{\kappa-1},$$
  where $\mathbf{b}_1, \ldots, \mathbf{b}_{\kappa-1}$ is the basis of $\Lambda_{\mathcal{R}} = \ker(\overline{\mathsf{proj}}_{\eta,\kappa}) \subset \mathcal{R}^{\kappa}$.
(4) Return the encoded vector $\overline{\mathbf{v}} \in \mathcal{R}^{\omega\kappa}$ obtained by concatenating, for each $r \in [\omega]$,
$$\overline{\mathbf{v}}^{(r)} \leftarrow (\alpha_*^{(r)}, \alpha_1^{(r)}, \ldots, \alpha_{\kappa-1}^{(r)}) \in \mathcal{R}^{\kappa}, \qquad \overline{\mathbf{v}} \leftarrow (\overline{\mathbf{v}}^{(1)}, \ldots, \overline{\mathbf{v}}^{(\omega)}).$$

$\underline{\mathsf{Decode}^{\mathcal{B}}_{\eta,q}(\overline{\mathbf{v}}, \mathsf{hint})}$

(1) Set $\kappa \leftarrow \lceil \log_{2\eta+1}(q) \rceil$.
(2) Parse $\overline{\mathbf{v}} \in \mathcal{R}^{\omega\kappa}$ as $\overline{\mathbf{v}} = (\overline{\mathbf{v}}^{(1)}, \ldots, \overline{\mathbf{v}}^{(\omega)})$, with $\overline{\mathbf{v}}^{(r)} \in \mathcal{R}^{\kappa}$. Parse $\mathsf{hint} \in \mathcal{R}_q^{\omega}$ as $\mathsf{hint} = (\mathsf{hint}^{(1)}, \ldots, \mathsf{hint}^{(\omega)})$.
(3) For each $r \in [\omega]$ do:
  (a) Represent $\mathsf{hint}^{(r)}$ by $\mathsf{hint}'^{(r)} \in \mathcal{R}$ with coefficients in $\left[-\frac{q-1}{2}, \frac{q-1}{2}\right]$.
  (b) Parse $\overline{\mathbf{v}}^{(r)}$ as $(\alpha_*^{(r)}, \alpha_1^{(r)}, \ldots, \alpha_{\kappa-1}^{(r)})$.
  (c) $h''^{(r)} \leftarrow \mathsf{hint}'^{(r)} + q \cdot \alpha_*^{(r)} \in \mathcal{R}$.
  (d) $\boldsymbol{\delta}^{(r)} \leftarrow \alpha_1^{(r)}\mathbf{b}_1 + \cdots + \alpha_{\kappa-1}^{(r)}\mathbf{b}_{\kappa-1} \in \mathcal{R}^{\kappa}$.
  (e) $\mathbf{v}^{(r)} \leftarrow \overline{\mathsf{dec}}_{\eta,\kappa}(h''^{(r)}) + \boldsymbol{\delta}^{(r)} \in \mathcal{R}^{\kappa}$.
(4) Return $\mathbf{v} \leftarrow (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(\omega)}) \in \mathcal{R}^{\omega\kappa}$.

Figure 6: Algorithms for encoding and decoding an element $\mathbf{v} \in \mathcal{R}^{\omega\kappa}$, which is viewed as $\mathbf{v} = (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(\omega)})$, $\mathbf{v}^{(r)} \in \mathcal{R}^{\kappa}$.

Lemma B.1 (Properties of $\mathsf{Encode}^{\mathcal{B}}_{\eta,q}$ and $\mathsf{Decode}^{\mathcal{B}}_{\eta,q}$, adopted from [8]). *Let $q, \eta, \omega, \kappa \in \mathbb{N}^+$ with q prime and $\kappa := \lceil \log_{2\eta+1}(q) \rceil$. Then the deterministic encoding and decoding algorithms defined in Figure 6 satisfy the following properties:*

(1) For $r \in [\omega]$, coefficients $(\alpha_*^{(r)}, \alpha_1^{(r)}, \ldots, \alpha_{\kappa-1}^{(r)})$ as required in $\mathsf{Encode}^{\mathcal{B}}_{\eta,q}$ exist, are unique and can be found in polynomial-time.
(2) For any $\mathbf{v} \in \mathcal{R}^{\omega\kappa}$, $\mathsf{hint} = \mathsf{proj}_q(\mathbf{v})$, and $\overline{\mathbf{v}} \leftarrow \mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{v})$, we have $\mathsf{Decode}^{\mathcal{B}}_{\eta,q}(\overline{\mathbf{v}}, \mathsf{hint}) = \mathbf{v}$.

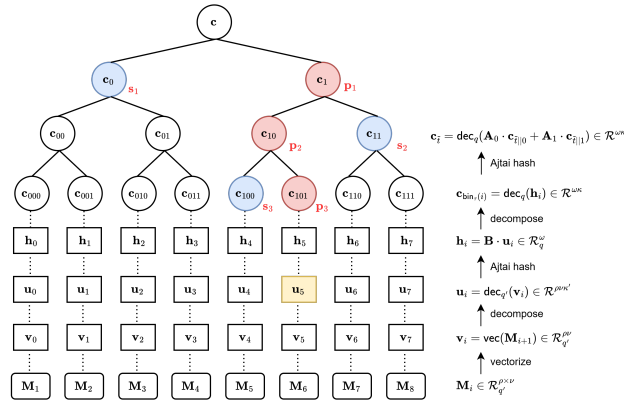

Figure 7: Visualization of an opening for the message $\mathbf{M}_6 \in \mathcal{R}_{q'}^{\rho\times\nu}$ at time slot $t = 6$ ($\tilde{t} = \mathsf{bin}_\tau(t-1) = 101$). Let $\mathbf{B} \in \mathcal{R}_q^{\omega\times\rho\nu\kappa'}$, $\mathbf{A}_0, \mathbf{A}_1 \in \mathcal{R}_q^{\omega\times\omega\kappa}$ be the public parameters of the Ajtai hash. The opening consists of three distinct components: (i) the path nodes $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ (highlighted in pink) which correspond to the ancestors of the target leaf; (ii) the auxiliary sibling nodes $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3$ (highlighted in blue) required to reconstruct the root commitment c; and (iii) the decomposed message vector $\mathbf{u}_5$ (highlighted in yellow) derived from the target matrix $\mathbf{M}_6$. Note that the path nodes $\mathbf{p}_i$ depicted here are in unencoded form; in the final Lemur HVC construction, these are encoded as $\bar{\mathbf{p}}_i$ to minimize bandwidth.

(3) For any $\overline{\mathbf{v}} \in \mathcal{R}^{\omega\kappa}$, $\mathsf{hint} \in \mathcal{R}_q^{\omega}$, and $\mathbf{v} \leftarrow \mathsf{Decode}^{\mathcal{B}}_{\eta,q}(\overline{\mathbf{v}}, \mathsf{hint})$, we have $\mathsf{proj}_q(\mathbf{v}) = \mathsf{hint}$, and $\mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{v}) = \overline{\mathbf{v}}$.
(4) For any $\mathbf{v} \in \mathcal{R}^{\omega\kappa}$ and $\overline{\mathbf{v}} \leftarrow \mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{v})$, with $\overline{\mathbf{v}} \leftarrow (\overline{\mathbf{v}}^{(1)}, \ldots, \overline{\mathbf{v}}^{(\omega)})$ and $\overline{\mathbf{v}}^{(r)} \leftarrow (\alpha_*^{(r)}, \alpha_1^{(r)}, \ldots, \alpha_{\kappa-1}^{(r)}) \in \mathcal{R}^{\kappa}$, we have
$$\|\alpha_*^{(r)}\|_{\max} \leq \frac{\|\mathsf{proj}_{\eta,\kappa}(\mathbf{v}^{(r)})\|_{\max}}{q} + \frac{1}{2},$$
$$\|\alpha_i^{(r)}\|_{\max} \leq \frac{\|\mathbf{v}^{(r)}\|_{\max}}{2\eta} + \frac{1}{2}.$$

Proof. By definition, $\mathsf{Encode}^{\mathcal{B}}_{\eta,q}$ and $\mathsf{Decode}^{\mathcal{B}}_{\eta,q}$ act independently on each block $r \in [\omega]$. For each fixed r, we can apply Lemma 23 in Chipmunk to the single-block input $\mathbf{v}^{(r)} \in \mathcal{R}^{\kappa}$ and obtain the stated lemma. □

B.4 HVC Construction

Given the building blocks introduced in the previous two subsections, we are now ready to present our concrete HVC construction. We begin by formally defining a labeled full binary tree.

Definition B.6 (Labeled Full Binary Tree). *Let $d, q, q', \rho, \nu, \omega, \eta \in \mathbb{N}^+$ with d a power of 2 and q, q' primes. Define $\kappa = \lceil \log_{2\eta+1} q \rceil$ and $\kappa' = \lceil \log_{2\eta+1} q' \rceil$. Let $\widetilde{\mathbf{M}} = (\mathbf{M}_1, \ldots, \mathbf{M}_{2^\tau}) \in (\mathcal{R}_{q'}^{\rho\times\nu})^{2^\tau}$, $\mathbf{B} \in \mathcal{R}_q^{\omega\times\rho\nu\kappa'}$, and $\mathbf{A}_0, \mathbf{A}_1 \in \mathcal{R}_q^{\omega\times\omega\kappa}$ be fixed. We define the labeling function $\mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1} : (\mathcal{R}_{q'}^{\rho\times\nu})^{2^\tau} \times \{0,1\}^{\leq\tau} \mapsto \mathcal{R}^{\omega\kappa}$ for a labeled full* *binary tree of depth τ as* $\mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\widetilde{\mathbf{M}}, v)$ *and define it as:*

$$\begin{cases} \mathrm{dec}_q\big(\mathbf{B}\cdot \mathrm{dec}_{q'}(\mathrm{vec}(\mathbf{M}_v))\big) & \text{if } |v| = \tau, \\ \mathrm{dec}_q\Big(\mathbf{A}_0\cdot \mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\widetilde{\mathbf{M}}, v\|0)+\mathbf{A}_1\cdot \mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\widetilde{\mathbf{M}}, v\|1)\Big) & \text{if } |v| < \tau. \end{cases}$$

We treat $v \in \{0,1\}^\tau$ as a big-endian integer in $\mathbf{M}_v$.

Figure 7 illustrates the HVC labelling procedure. Starting at the leaves, each matrix $\mathbf{M}_i \in \mathcal{R}_{q'}^{\rho\times\nu}$ is vectorized as $\mathbf{v}_i$ and decomposed into a small-norm representative $\mathbf{u}_i = \mathrm{dec}_{q'}(\mathbf{v}_i)$ to resolve the domain mismatch between $\mathcal{R}_{q'}$ and $\mathcal{R}_q$. Leaf labels are then computed as $\mathbf{c}_i = \mathrm{dec}_q(\mathbf{B}\mathbf{u}_i) \in \mathcal{R}^{\omega\kappa}$ using the Ajtai hash matrix $\mathbf{B}$. Internal nodes are computed recursively: for child labels $\mathbf{c}_{\tilde{i}||0}, \mathbf{c}_{\tilde{i}||1}$, the parent label is $\mathbf{c}_{\tilde{i}} = \mathrm{dec}_q(\mathbf{A}_0\mathbf{c}_{\tilde{i}||0} + \mathbf{A}_1\mathbf{c}_{\tilde{i}||1})$, where $\mathbf{A}_0, \mathbf{A}_1$ are public matrices. This consistent use of decomposition ensures that Ajtai hash inputs always remain small-norm representatives, maintaining concrete efficiency throughout the tree.

Given the definition of Labeled Full Binary Tree, we are now ready to define our HVC construction.

Definition B.7. *Let $d, q, q', \alpha_w, N, \eta, \tau, \rho, \nu, \omega, \beta_{\mathrm{agg}} \in \mathbb{N}^+$ such that d is a power of 2 and q, q' are primes. We define the homomorphic vector commitment $\Pi_{\mathrm{HVC}} = (\mathsf{Setup}, \mathsf{Com}, \mathsf{Open}, \mathsf{iVrfy}, \mathsf{sVrfy}, \mathsf{wVrfy})$ for domain $\mathcal{S}_{\mathrm{dom}} = \mathcal{R}_{q'}^{\rho\times\nu}$, and vectors of length 2^τ by the algorithms given in Figure 4. Its commitments and openings are from $\mathcal{S}_{\mathrm{com}} = \mathcal{R}_q^{\omega}$ and $\mathcal{S}_{\mathrm{open}} = \mathcal{R}^{\tau\kappa}\times\mathcal{R}^{\tau\kappa}\times\mathcal{R}^{\rho\nu\kappa'}$, where $\kappa = \left\lceil \log_{2\eta+1} q \right\rceil$ and $\kappa' = \left\lceil \log_{2\eta+1} q' \right\rceil$.*

Similar to [8], the use of the encoding algorithm (Figure 6) implies that aggregated openings are not $\mathcal{R}$-linear in the standard sense. To recover homomorphic behavior, we define suitable operations under which our HVC is homomorphic.

Definition B.8 ([8]). *Let $d, \eta, q, q', \omega, \rho, \nu \in \mathbb{N}^+$ with d a power of 2, q, q' primes. Let $\kappa = \lceil \log_{2\eta+1} q\rceil$ and $\kappa' = \lceil \log_{2\eta+1} q'\rceil$. Consider the HVC construction Π_{HVC} from Figure 4. Let $\mathcal{S}_{\mathrm{open,un}} \subseteq (\mathcal{R}^{\omega\kappa})^{2\tau}\times(\mathcal{R}^{\kappa'})^{\rho\nu}$ be the unencoded opening space. Let $\mathsf{pp} := (\mathbf{B}, \mathbf{A}_0, \mathbf{A}_1) \leftarrow \mathsf{Setup}(1^\lambda)$. We call an unencoded opening $d = (\mathbf{p}_1, \ldots, \mathbf{p}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$ linearly verifying for time slot $t \in [2^\tau]$ iff the following conditions hold*

$$\mathbf{B}\cdot\mathbf{u} = \mathrm{proj}_q(\mathbf{p}_\tau)$$
$$\mathrm{proj}_q(\mathbf{p}_{j-1}) = \mathbf{A}_{\tilde{t}_j}\cdot\mathbf{p}_j + \mathbf{A}_{\tilde{t}_j\oplus 1}\cdot\mathbf{s}_j \quad \text{for all } 2 \le j \le \tau,$$

where $\tilde{t} = \mathrm{bin}_\tau(t-1)$. We define $\mathcal{S}^t_{\mathrm{open,un}} \subset \mathcal{S}_{\mathrm{open,un}}$ to be the subset of all linearly verifying unencoded openings for t.

Definition B.9 (Encoding/decoding openings at time slot t, [8]). *Fix $\mathsf{pp} = (\mathbf{B}, \mathbf{A}_0, \mathbf{A}_1)$ and $t \in [2^\tau]$ with $\tilde{t} = \mathrm{bin}_\tau(t-1)$. Let $d = (\mathbf{p}_1, \ldots, \mathbf{p}_\tau,\ \mathbf{s}_1, \ldots, \mathbf{s}_\tau,\ \mathbf{u}) \in \mathcal{S}^t_{\mathrm{open,un}}$. Define $\mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d)$ by encoding only the path nodes:*

$$\mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d) := (\overline{\mathbf{p}}_1, \ldots, \overline{\mathbf{p}}_\tau,\ \mathbf{s}_1, \ldots, \mathbf{s}_\tau,\ \mathbf{u}),$$

with

$$\overline{\mathbf{p}}_j := \mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{p}_j).$$

Define $\mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d})$ for $\bar{d} = (\overline{\mathbf{p}}_1, \ldots, \overline{\mathbf{p}}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$ by running the same decoding recursion as in Vrfy of Figure 4 but without checks:

$$\mathsf{hint}_\tau := \mathbf{B}\cdot\mathbf{u} \in \mathcal{R}_q^{\omega},$$

and for $j = \{\tau, \tau-1, \ldots, 1\}$:

$$\mathbf{p}_j := \mathsf{Decode}^{\mathcal{B}}_{\eta,q}(\overline{\mathbf{p}}_j, \mathsf{hint}_j), \qquad \mathsf{hint}_{j-1} := \mathbf{A}_{\tilde{t}_j}\cdot\mathbf{p}_j + \mathbf{A}_{\tilde{t}_j\oplus 1}\cdot\mathbf{s}_j \in \mathcal{R}_q^{\omega}.$$

Output $d = (\mathbf{p}_1, \ldots, \mathbf{p}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$.

In the following, we provide a lemma that is derived from item 4 of Thoerem 25 in [8].

Lemma B.2. *Let $d = (\mathbf{p}_1, \ldots, \mathbf{p}_\tau,\ \mathbf{s}_1, \ldots, \mathbf{s}_\tau,\ \mathbf{u}) \in \mathcal{S}^t_{\mathrm{open,un}}$, and $(\overline{\mathbf{p}}_1, \ldots, \overline{\mathbf{p}}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u}) := \bar{d} := \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d)$. Let $c \in \mathcal{S}_{\mathrm{com}}$ be any (possibly maliciously generated) comment and let Vrfy be as in Figure 4. If $\mathsf{Vrfy}(\mathsf{pp}, c, t, d, \beta) \neq \perp$ for some β, then*

$$\|\overline{\mathbf{p}}_i\|_{\max} < \frac{\beta}{2\eta} + \frac{1}{2} \text{ for all } i.$$

Definition B.10 (Module operations on encoded openings). *Fix pp, t. Let $\mathcal{S}^t_{\mathrm{open}}$ be the set of* encoded *openings $\bar{d}$ of the form $\bar{d} = (\overline{\mathbf{p}}_1, \ldots, \overline{\mathbf{p}}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$. Define for $w \in \mathcal{R}$ and $\bar{d}, \bar{d}' \in \mathcal{S}^t_{\mathrm{open}}$:*

$$w \odot \bar{d} := \mathsf{Encode}_{\mathrm{op}}\big(\mathsf{pp}, t,\ w\cdot \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d})\big)$$
$$\bar{d} \oplus \bar{d}' := \mathsf{Encode}_{\mathrm{op}}\big(\mathsf{pp}, t,\ \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d}) + \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d}')\big),$$

where the + and scalar multiplication on the right-hand side are taken componentwise on the unencoded tuple $(\mathbf{p}_1, \ldots, \mathbf{p}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$.

Lemma B.3 (Correctness of $\mathsf{Decode}_{\mathrm{op}} \circ \mathsf{Encode}_{\mathrm{op}}$). *For any pp, t and any unencoded opening $d = (\mathbf{p}_1, \ldots, \mathbf{p}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u})$, let $\bar{d} := \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d)$. Then*

$$\mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d}) = d.$$

Proof. By definition, $\mathsf{Decode}_{\mathrm{op}}$ computes $\mathsf{hint}_\tau = \mathbf{B}\cdot\mathbf{u}$ and then iterates downward. Assume inductively that at some level j we have $\mathsf{hint}_j = \mathrm{proj}_q(\mathbf{p}_j)$. Since $\overline{\mathbf{p}}_j = \mathsf{Encode}^{\mathcal{B}}_{\eta,q}(\mathbf{p}_j)$, Lemma B.1 (Item 2) gives $\mathsf{Decode}^{\mathcal{B}}_{\eta,q}(\overline{\mathbf{p}}_j, \mathsf{hint}_j) = \mathbf{p}_j$. Then $\mathsf{Decode}_{\mathrm{op}}$ sets

$$\mathsf{hint}_{j-1} = \mathbf{A}_{\tilde{t}_j}\cdot\mathbf{p}_j + \mathbf{A}_{\tilde{t}_j\oplus 1}\cdot\mathbf{s}_j = \mathrm{proj}_q(\mathbf{p}_{j-1}),$$

where the last equality is exactly the projection identity induced by the labeling recursion. The base case $j = \tau$ holds because $\mathrm{proj}_q(\mathbf{p}_\tau) = \mathbf{B}\cdot\mathbf{u}$ for honest leaves. Thus $\mathsf{Decode}_{\mathrm{op}}$ recovers every $\mathbf{p}_j$ and outputs d. □

We are now ready to show our constructed HVC satisfies all the Definitions in subsection Section B.1.

Theorem B.1. *Let $d, q, q', \alpha_w, N, \eta, \tau, \rho, \nu, \omega, \beta_{\mathrm{agg}} \in \mathbb{N}^+$ such that d is a power of 2 and q, q' are primes. Let $\epsilon_{\mathrm{hom}} \in (0,1)$. Let*

$$\beta_{\mathrm{agg}} \ \ge\ \eta\sqrt{2\alpha_w N\Big(\ln\frac{2d}{\epsilon_{\mathrm{hom}}} + \ln\big(N\cdot(2\tau\omega\kappa + \rho\nu\kappa' + 2\tau\omega)\big)\Big)}.$$

If $\mathsf{MSIS}^{\infty}_{q,\omega,2\omega\kappa,4\beta_{\mathrm{agg}}}$ and $\mathsf{MSIS}^{\infty}_{q,\omega,\rho\nu\kappa',4\beta_{\mathrm{agg}}}$ problems are hard, then the HVC construction in Figure 4 is an individually correct, $(N, \mathcal{T}_{\alpha_w}, \epsilon_{\mathrm{hom}})$-probabilistically homomorphic, robustly homomorphic and position binding for domain $\mathcal{R}_{q'}^{\rho\times\nu}$ and vector length 2^τ.

Proof. The proof of the theorem follows from Lemma B.4, Lemma B.5, Lemma B.6, and Lemma B.7 proven below. □

Lemma B.4 (Individual Correctness of HVC). *Let $d, q, q', \rho, \nu, \omega, \eta, \tau \in \mathbb{N}^+$ such that d is a power of 2 and q, q' are primes. Let $\mathcal{R}_q, \mathcal{R}_{q'}$ be the polynomial rings $\mathbb{Z}_q[X]/(X^d+1)$ and $\mathbb{Z}_{q'}[X]/(X^d+1)$, respectively. Then the HVC construction in Figure 4 is an individually correct HVC for domain $\mathcal{R}_{q'}^{\rho\times\nu}$ and vector length 2^τ as per Definition B.2.*

Proof. Let $\widetilde{\mathbf{M}} \in (\mathcal{R}_{q'}^{\rho\times\nu})^{2^\tau}$, $c = \mathsf{Com}(\mathsf{pp}, \widetilde{\mathbf{M}})$, $t \in [2^\tau]$, and$(\overline{\mathbf{p}}_1, \ldots, \overline{\mathbf{p}}_\tau, \mathbf{s}_1, \ldots, \mathbf{s}_\tau, \mathbf{u}) = \mathsf{Open}(\mathsf{pp}, c, \widetilde{\mathbf{M}}, t)$. Let $\tilde{t} = \mathsf{bin}_\tau(t-1)$ be as in the definition of Open. To disambiguate, we temporarily denote the equally named values $\mathbf{p}_j$ appearing in both Open and Vrfy by $\mathbf{p}_j^{\mathsf{Open}}$ and $\mathbf{p}_j^{\mathsf{Vrfy}}$ for $j \in [\tau]$. For notational convenience in the proof, we abuse $\mathbf{p}_0^{\mathsf{Open}}$ to denote the computation of $\mathbf{p}_0$ in Com. We first observe that for all $j \in [\tau]$ it holds in $\mathcal{R}_q^\omega$ that

$$\mathsf{proj}_q(\mathbf{p}_{j-1}^{\mathsf{Open}}) \tag{29}$$
$$=\mathsf{proj}_q\big(\mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\widetilde{\mathbf{M}}, \tilde{t}_{<j})\big) \tag{30}$$
$$=\mathsf{proj}_q\Big(\mathsf{dec}_q\big(\mathbf{A}_0 \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|0) + \mathbf{A}_1 \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|1)\big)\Big) \tag{31}$$
$$=\mathbf{A}_0 \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|0) + \mathbf{A}_1 \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|1) \tag{32}$$
$$=\mathbf{A}_{\tilde{t}_j} \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|\tilde{t}_j) + \mathbf{A}_{\tilde{t}_j\oplus 1} \cdot \mathsf{label}(\widetilde{\mathbf{M}}, \tilde{t}_{<j}\|(\tilde{t}_j \oplus 1)) \tag{33}$$
$$=\mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^{\mathsf{Open}} + \mathbf{A}_{\tilde{t}_{j\oplus 1}} \cdot \mathbf{s}_j. \tag{34}$$

The first equality comes from the definition of Com and Open. The second equlity comes from label, $|\tilde{t}_{<j}| < \tau$. The third equlity comes from Item 3 of Proposition B.3. The final equality comes from the definition of Open. Observe that for $j = 1$ this yields

$$c = \mathsf{proj}_q(\mathbf{p}_0^{\mathsf{Open}}) = \mathbf{A}_{\tilde{t}_1} \cdot \mathbf{p}_1^{\mathsf{Open}} + \mathbf{A}_{\tilde{t}_1\oplus 1} \cdot \mathbf{s}_1 \quad \text{in } \mathcal{R}_q^\omega.$$

Further it holds that

$$\mathsf{proj}_q(\mathbf{p}_\tau^{\mathsf{Open}}) = \mathsf{proj}_q\big(\mathsf{label}_{\mathbf{B},\mathbf{A}_0,\mathbf{A}_1}(\mathbf{M}, \tilde{t})\big) \tag{35}$$
$$= \mathsf{proj}_q\Big(\mathsf{dec}_q\big(\mathbf{B} \cdot \mathsf{dec}_{q'}(\mathsf{vec}(\mathbf{M}_t))\big)\Big) \tag{36}$$
$$= \mathbf{B} \cdot \mathsf{dec}_{q'}(\mathsf{vec}(\mathbf{M}_t)) \tag{37}$$
$$= \mathbf{B} \cdot \mathbf{u}. \tag{38}$$

where the first and last equalities hold by definition of Open. The second equality holds true by Def. of label, leaf case and the second last equality holds by Item 3 of Proposition B.3. We argue that, on an honestly generated opening, the verifier decodes the same vectors $\mathbf{p}_j$ that were used by the opener, i.e., $\mathbf{p}_j^{\mathsf{Vrfy}} = \mathbf{p}_j^{\mathsf{Open}}$ for all $j \in [\tau]$. To this end, we show the following invariant for the hints computed by Vrfy:

$$(\star_j) \qquad \mathsf{hint}_j = \mathsf{proj}_q(\mathbf{p}_j^{\mathsf{Open}}) \in \mathcal{R}_q^\omega \quad \text{for all } j \in \{0, 1, \ldots, \tau\}.$$

We prove $(\star_j)$ by backward induction on j:

- *Base case ($j = \tau$).* In Vrfy the verifier sets $\mathsf{hint}_\tau = \mathbf{B} \cdot \mathbf{u}$. By Equation (35), we have $\mathsf{proj}_q(\mathbf{p}_\tau^{\mathsf{Open}}) = \mathbf{B} \cdot \mathbf{u}$, hence $(\star_\tau)$ holds.
- *Inductive step.* Fix $j \in \{1, \ldots, \tau\}$, we assume $(\star_j)$, i.e., $\mathsf{hint}_j = \mathsf{proj}_q(\mathbf{p}_j^{\mathsf{Open}})$, and prove $(\star_{j-1})$, i.e., $\mathsf{hint}_{j-1} = \mathsf{proj}_q(\mathbf{p}_{j-1}^{\mathsf{Open}})$.

In Vrfy, we have

$$\mathbf{p}_j^{\mathsf{Vrfy}} = \mathsf{Decode}_{\eta,q}^{\mathcal{B}}(\overline{\mathbf{p}}_j, \mathsf{hint}_j) \tag{39}$$
$$= \mathsf{Decode}_{\eta,q}^{\mathcal{B}}\big(\mathsf{Encode}_{\eta,q}^{\mathcal{B}}(\mathbf{p}_j^{\mathsf{Open}}), \mathsf{proj}_q(\mathbf{p}_j^{\mathsf{Open}})\big) \tag{40}$$
$$= \mathbf{p}_j^{\mathsf{Open}} \tag{41}$$

where the first equality is satisfied by Def. of Vrfy, the second equality is true by Def. of Open and hypothesis and last equality comes from item 2 of Lemma B.1. We obtain

$$\begin{aligned}\mathsf{hint}_{j-1} &= \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^{\mathsf{Vrfy}} + \mathbf{A}_{\tilde{t}_j\oplus 1} \cdot \mathbf{s}_j \\ &= \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^{\mathsf{Open}} + \mathbf{A}_{\tilde{t}_j\oplus 1} \cdot \mathbf{s}_j \\ &= \mathsf{proj}_q(\mathbf{p}_{j-1}^{\mathsf{Open}}),\end{aligned}$$

where the first equality is satisfied by Def. of Vrfy and the second and third equalities are held by Equation (39) and Equation (29), respectively. This exactly proves $(\star_{j-1})$. This completes the backward induction and establishes $(\star_j)$ for all j. In particular, $(\star_0)$ implies that on honestly generated openings the verifier obtains

$$\mathsf{hint}_0 = \mathsf{proj}_q(\mathbf{p}_0^{\mathsf{Open}}) = c,$$

and hence the check $c = \mathsf{hint}_0$ succeeds.

It only remains to check that the norm bounds are not violated for iVrfy, i.e. for $\beta = \eta$. For every $j \in [\tau]$, the vectors $\mathbf{p}_j^{\mathsf{Open}}$ and $\mathbf{s}_j$ are outputs of label and thus (depending on the node depth) are obtained by applying dec_q to an element of $\mathcal{R}_q^\omega$. Similarly, $\mathbf{u} = \mathsf{dec}_{q'}(\mathsf{vec}(\mathbf{M}_t))$ is obtained by applying $\mathsf{dec}_{q'}$ to an element of $\mathcal{R}_{q'}^{\rho\nu}$. By design of dec_q and $\mathsf{dec}_{q'}$ (base $2\eta+1$), all resulting coefficients lie in $\{-\eta, \ldots, \eta\}$, and therefore $\|\mathbf{p}_j^{\mathsf{Vrfy}}\|_{\max} \le \eta$, $\|\mathbf{s}_j\|_{\max} \le \eta$, and $\|\mathbf{u}\|_{\max} \le \eta$. Moreover, applying $\mathsf{proj}_{\eta,\kappa}$ to $\mathbf{p}_j^{\mathsf{Vrfy}}$ or $\mathbf{s}_j$ yields the centered representative of the decomposed element, so

$$\big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{p}_j^{\mathsf{Vrfy}})\big\|_{\max}, \ \big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{s}_j)\big\|_{\max} \le \frac{q-1}{2}.$$

Thus all checks in $\mathsf{iVrfy}(\mathsf{pp}, c, t, d) = \mathsf{Vrfy}(\mathsf{pp}, c, t, d, \eta)$ pass, and the algorithm outputs

$$\mathsf{vec}^{-1}\big(\mathsf{proj}_{q'}(\mathbf{u})\big) = \mathsf{vec}^{-1}\big(\mathsf{proj}_{q'}(\mathsf{dec}_{q'}(\mathsf{vec}(\mathbf{M}_t)))\big) = \mathbf{M}_t.$$

This proves individual correctness. □

Lemma B.5 (Probabilistic Homomorphism). *Let $d, q, q', \alpha_w, N, \eta, \tau, \rho, \nu, \omega, \beta_{\mathrm{agg}} \in \mathbb{N}^+$ and $\epsilon_{\mathrm{hom}} \in_{\mathbb{R}} (0,1)$ such that d is a power of 2 and q, q' are primes. Let $\mathcal{R}_q, \mathcal{R}_{q'}$ be the polynomial rings $\mathbb{Z}_q[X]/(X^d+1)$ and $\mathbb{Z}_{q'}[X]/(X^d+1)$, respectively. Set $\kappa = \lceil \log_{2\eta+1}(q) \rceil$ and $\kappa' = \lceil \log_{2\eta+1}(q') \rceil$. If*

$$\beta_{\mathrm{agg}} \ge \eta\sqrt{2\alpha_w N\Big(\ln\frac{2d}{\epsilon_{\mathrm{hom}}} + \ln\big(N \cdot (2\tau\omega\kappa + \rho\nu\kappa' + 2\tau\omega)\big)\Big)}, \tag{42}$$

then the HVC construction in Figure 4 is $(N, \mathcal{T}_{\alpha_w}, \epsilon_{\mathrm{hom}})$-probabilistically homomorphic for domain $\mathcal{R}_{q'}^{\rho\times\nu}$ and vector length 2^τ as per Definition Definition B.3.

Proof. Let $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$ and fix $t \in [2^\tau]$ with $\tilde{t} = \mathsf{bin}_\tau(t-1)$. Let $(c^{(i)}, d^{(i)})$ for $i \in [N]$ be such that

$$\mathsf{iVrfy}(\mathsf{pp}, c^{(i)}, t, d^{(i)}) = \mathbf{M}_t^{(i)} \neq \perp,$$

where we parse each decommitment as

$$d^{(i)} = (\overline{\mathbf{p}}_1^i, \dots, \overline{\mathbf{p}}_\tau^i,\ \mathbf{s}_1^i, \dots, \mathbf{s}_\tau^i,\ \mathbf{u}^i).$$

Let $w^{(1)}, \dots, w^{(N)} \leftarrow \mathcal{T}_{\alpha_w}$ be arbitrary weights, and define the aggregated pair

$$c^* := \sum_{i=1}^N w^{(i)} \cdot c^{(i)} \in \mathcal{R}_q^\omega, \qquad d^* := \bigoplus_{i=1}^N \big(w^{(i)} \odot d^{(i)}\big) \in \mathcal{S}_{\mathrm{open}}^t,$$

where $\odot, \oplus$ are as in Definition B.10. Write

$$d^* = (\overline{\mathbf{p}}_1^*, \dots, \overline{\mathbf{p}}_\tau^*,\ \mathbf{s}_1^*, \dots, \mathbf{s}_\tau^*,\ \mathbf{u}^*).$$

First, we check that the linear relations in Vrfy hold deterministically for the aggregated opening. For each $i \in [N]$, let $\bar{d}^{(i)} := d^{(i)}$ be the encoded opening output by Open, and define the corresponding unencoded opening by decoding:

$$d_{\mathrm{un}}^{(i)} := \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d}^{(i)}) = (\mathbf{p}_1^i, \dots, \mathbf{p}_\tau^i, \mathbf{s}_1^i, \dots, \mathbf{s}_\tau^i, \mathbf{u}^i).$$

Since $\mathsf{iVrfy}(\mathsf{pp}, c^{(i)}, t, \bar{d}^{(i)}) \neq \perp$, the verifier's decoding recursion is satisfied and produces hints $\mathsf{hint}_j^i \in \mathcal{R}_q^\omega$ such that $\mathsf{hint}_0^i = c^{(i)}$ and, for all $j \in [\tau]$,

$$\mathsf{hint}_\tau^i = \mathbf{B} \cdot \mathbf{u}^i, \qquad \mathsf{hint}_{j-1}^i = \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^i + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^i.$$

Now define the aggregated unencoded opening component-wise as $d_{\mathrm{un}}^* := \sum_{i=1}^N w^{(i)} \cdot d_{\mathrm{un}}^{(i)}$ i.e.,

$$\mathbf{p}_j^* = \sum_i w^{(i)} \mathbf{p}_j^i,\ \mathbf{s}_j^* = \sum_i w^{(i)} \mathbf{s}_j^i,\ \mathbf{u}^* = \sum_i w^{(i)} \mathbf{u}^i),$$

and define the aggregated *encoded* opening as

$$\bar{d}^* := \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d_{\mathrm{un}}^*) = \bigoplus_{i=1}^N (w^{(i)} \odot \bar{d}^{(i)}) \in \mathcal{S}_{\mathrm{open}}^t.$$

Let $c^* := \sum_{i=1}^N w^{(i)} \cdot c^{(i)} \in \mathcal{R}_q^\omega$, and define aggregated hints $\mathsf{hint}_j^* := \sum_{i=1}^N w^{(i)} \cdot \mathsf{hint}_j^i$. Then for every $j \in [\tau]$, by linearity in $\mathcal{R}_q^\omega$ we obtain

$$\mathsf{hint}_{j-1}^* = \sum_{i=1}^N w^{(i)} \cdot \mathsf{hint}_{j-1}^i = \mathbf{A}_{\tilde{t}_j} \cdot \Big(\sum_{i=1}^N w^{(i)} \mathbf{p}_j^i\Big) + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \Big(\sum_{i=1}^N w^{(i)} \mathbf{s}_j^i\Big)$$
$$= \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^* + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^*.$$

In particular, $\mathsf{hint}_0^* = \sum_i w^{(i)} c^{(i)} = c^*$. Moreover, Lemma B.3 implies $\mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, \bar{d}^*) = d_{\mathrm{un}}^*$. Hence, when Vrfy runs its decoding loop on input $\bar{d}^*$, it reconstructs $\mathbf{p}_j^{\mathsf{Vrfy}} = \mathbf{p}_j^*$ for all $j \in [\tau]$ (and similarly $\mathbf{s}_j^{\mathsf{Vrfy}} = \mathbf{s}_j^*$ and $\mathbf{u}^{\mathsf{Vrfy}} = \mathbf{u}^*$), and therefore all linear (mod-q) equalities checked by Vrfy hold deterministically for $(c^*, \bar{d}^*)$.

Next, we bound the failure probability of the norm checks. Writing out the conditions (cf. $\mathsf{sVrfy}(\mathsf{pp}, \cdot, \cdot, \cdot) = \mathsf{Vrfy}(\mathsf{pp}, \cdot, \cdot, \cdot, \beta_{\mathrm{agg}})$), we need to bound the probability p at the top of the next page.

Observe that this is an $\|\cdot\|_{\max}$-bound for a total of

$$N_{\mathrm{bounds}} = N \cdot \big(2\tau\omega\kappa + \rho\nu\kappa' + 2\tau\omega\big)$$

many ring elements. For each of these N_{bounds} ring elements, we apply Lemma A.3 with growth factor $\zeta = \beta_{\mathrm{agg}}/\eta$. Taking an N_{bounds}-fold union bound gives

$$P \le N_{\mathrm{bounds}} \cdot 2d \cdot \exp\Big(-\frac{\beta_{\mathrm{agg}}^2}{2\eta^2\alpha_w N}\Big).$$

By (42), we have

$$\frac{\beta_{\mathrm{agg}}^2}{2\eta^2\alpha_w N} \ge \ln\Big(\frac{2d}{\epsilon_{\mathrm{hom}}} \cdot N \cdot (2\tau\omega\kappa + \rho\nu\kappa' + 2\tau\omega)\Big) = \ln\Big(\frac{2d}{\epsilon_{\mathrm{hom}}} \cdot N_{\mathrm{bounds}}\Big),$$

and hence $P \le \epsilon_{\mathrm{hom}}$. Finally, we prove the correctness of the aggregated output. Therefore, with probability at least $1 - \epsilon_{\mathrm{hom}}$ over the choice of $w^{(1)}, \dots, w^{(N)} \leftarrow \mathcal{T}_{\alpha_w}$, the strong verification algorithm outputs

$$\mathsf{proj}_{q'}(\mathbf{u}^*) = \mathsf{proj}_{q'}\Big(\sum_{i=1}^N w^{(i)} \cdot \mathbf{u}^i\Big) = \sum_{i=1}^N w^{(i)} \cdot \mathsf{proj}_{q'}(\mathbf{u}^i)$$
$$= \sum_{i=1}^N w^{(i)} \cdot \mathsf{iVrfy}(\mathsf{pp}, c^{(i)}, t, d^{(i)}) = \sum_{i=1}^N w^{(i)} \cdot \mathbf{M}_t^{(i)},$$

where the first equality is due to linearity of $\mathsf{proj}_{q'}$ as required. This proves $(N, \mathcal{T}_{\alpha_w}, \epsilon_{\mathrm{hom}})$-probabilistic homomorphism. □

Lemma B.6 (Robust Homomorphism). *Let $d, q, q', \alpha_w, N, \eta, \tau, \rho, \nu, \omega, \beta_{\mathrm{agg}} \in \mathbb{N}^+$ such that d is a power of 2 and q, q' are primes. Let $\mathcal{R}_q, \mathcal{R}_{q'}$ be the polynomial rings $\mathbb{Z}_q[X]/(X^d+1)$ and $\mathbb{Z}_{q'}[X]/(X^d+1)$, respectively. Then the HVC in Figure 4 is robustly homomorphic as per Definition Definition B.3.*

Proof. Let $c^{(1)}, c^{(2)} \in \mathcal{R}_q^\omega$ and $d^{(1)}, d^{(2)} \in \mathcal{S}_{\mathrm{open}}^t$ be arbitrary, and let $t \in [2^\tau]$ with $\tilde{t} := \mathsf{bin}_\tau(t-1)$, such that

$$\mathsf{sVrfy}(\mathsf{pp}, c^{(1)}, t, d^{(1)}) = \mathbf{M}_1 \text{ and } \mathsf{sVrfy}(\mathsf{pp}, c^{(2)}, t, d^{(2)}) = \mathbf{M}_2 \quad (43)$$

with $\mathbf{M}_1, \mathbf{M}_2 \neq \perp$. We need to show that

$$\mathsf{wVrfy}(\mathsf{pp}, c^{(1)} - c^{(2)}, t, d^{(1)} \ominus d^{(2)}) = \mathbf{M}_1 - \mathbf{M}_2,$$

where $\ominus$ is the subtraction in $\mathcal{S}_{\mathrm{open}}^t$ opposed to $\oplus$. For $i \in \{1, 2\}$, parse $d^{(i)} = (\overline{\mathbf{p}}_1^i, \dots, \overline{\mathbf{p}}_\tau^i,\ \mathbf{s}_1^i, \dots, \mathbf{s}_\tau^i,\ \mathbf{u}^i)$. Let

$$d_{\mathrm{un}}^{(i)} := \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, d^{(i)}) = (\mathbf{p}_1^i, \dots, \mathbf{p}_\tau^i, \mathbf{s}_1^i, \dots, \mathbf{s}_\tau^i, \mathbf{u}^i).$$

Then we have

$$\begin{aligned} & d^{(1)} \ominus d^{(2)} \\ =& \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, d^{(1)}) - \mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, d^{(2)})) \\ =& \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d_{\mathrm{un}}^{(1)} - d_{\mathrm{un}}^{(2)}) \\ =& \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, (\mathbf{p}_1^1 - \mathbf{p}_1^2, \dots, \mathbf{p}_\tau^1 - \mathbf{p}_\tau^2, \mathbf{s}_1^1 - \mathbf{s}_1^2, \dots, \mathbf{s}_\tau^1 - \mathbf{s}_\tau^2, \mathbf{u}^1 - \mathbf{u}^2)) \end{aligned}$$

Since $\mathsf{sVrfy}(\mathsf{pp}, c^{(i)}, t, d^{(i)}) \neq \perp$, the verifier's decoding recursion is satisfied and produces hints $\mathsf{hint}_j^i \in \mathcal{R}_q^\omega$ such that:

(1) $\mathsf{hint}_\tau^i = \mathbf{B} \cdot \mathbf{u}^i$,
(2) for all $j \in [\tau]$,

$$\begin{aligned} \mathsf{hint}_{j-1}^i &= \mathbf{A}_{\tilde{t}_j} \cdot \mathsf{Decode}_{\eta,q}^{\mathcal{B}}\big(\overline{\mathbf{p}}_j, \mathsf{hint}_j\big) + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^i \\ &= \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^i + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^i \end{aligned} \quad (44)$$

(3) $\mathsf{hint}_0^i = c^i$

Now define

$$d_{\mathrm{un}}^* := d_{\mathrm{un}}^{(1)} - d_{\mathrm{un}}^{(2)} \quad (\text{i.e., } \mathbf{p}_j^* = \mathbf{p}_j^1 - \mathbf{p}_j^2,\ \mathbf{s}_j^* = \mathbf{s}_j^1 - \mathbf{s}_j^2,\ \mathbf{u}^* = \mathbf{u}^1 - \mathbf{u}^2),$$

and

$$d^* := \mathsf{Encode}_{\mathrm{op}}(\mathsf{pp}, t, d_{\mathrm{un}}^*) = d^{(1)} \ominus d^{(2)} \in \mathcal{S}_{\mathrm{open}}^t.$$

$$P := \Pr\Big[\, w^{(1)}, \ldots, w^{(N)} \leftarrow \mathcal{T}_{\alpha_w} : \exists j \in [\tau] \text{ s.t. } \Big\|\sum_{i=1}^{N} w^{(i)} \cdot \mathbf{p}_j^i\Big\|_{\max} > \beta_{\mathrm{agg}} \;\vee\; \Big\|\sum_{i=1}^{N} w^{(i)} \cdot \mathbf{s}_j^i\Big\|_{\max} > \beta_{\mathrm{agg}}$$
$$\vee\; \Big\|\sum_{i=1}^{N} w^{(i)} \cdot \mathbf{u}^i\Big\|_{\max} > \beta_{\mathrm{agg}} \;\vee\; \Big\|\sum_{i=1}^{N} w^{(i)} \cdot \mathsf{proj}_{\eta,\kappa}(\mathbf{p}_j^i)\Big\|_{\max} > \frac{q\beta_{\mathrm{agg}}}{2\eta} \;\vee\; \Big\|\sum_{i=1}^{N} w^{(i)} \cdot \mathsf{proj}_{\eta,\kappa}(\mathbf{s}_j^i)\Big\|_{\max} > \frac{q\beta_{\mathrm{agg}}}{2\eta}\,\Big].$$

Let $c^* := c^{(1)} - c^{(2)} \in \mathcal{R}_q^{\omega}$, and define hints $\mathsf{hint}_j^* := \mathsf{hint}_j^1 - \mathsf{hint}_j^2$. Then for every $j \in [\tau]$, by linearity in $\mathcal{R}_q^{\omega}$ we obtain

$$\begin{aligned}
\mathsf{hint}_{j-1}^* &= \mathsf{hint}_j^1 - \mathsf{hint}_j^2 \\
&= \Big(\mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^1 + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^1\Big) - \Big(\mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^2 + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^2\Big) \\
&= \mathbf{A}_{\tilde{t}_j}(\mathbf{p}_j^1 - \mathbf{p}_j^2) + \mathbf{A}_{\tilde{t}_j \oplus 1}(\mathbf{s}_j^1 - \mathbf{s}_j^2) \\
&= \mathbf{A}_{\tilde{t}_j} \cdot \mathbf{p}_j^* + \mathbf{A}_{\tilde{t}_j \oplus 1} \cdot \mathbf{s}_j^*,
\end{aligned}$$

where the second equality is due to Equation (44). In particular, $\mathsf{hint}_0^* = c^{(1)} - c^{(2)}$.

Moreover, Lemma B.3 implies $\mathsf{Decode}_{\mathrm{op}}(\mathsf{pp}, t, d^*) = d^*_{\mathrm{un}}$. Hence, when Vrfy runs its decoding loop on input d^*, it reconstructs $\mathbf{p}_j^{\mathsf{Vrfy}} = \mathbf{p}_j^*$ for all $j \in [\tau]$ (and similarly $\mathbf{s}_j^{\mathsf{Vrfy}} = \mathbf{s}_j^*$ and $\mathbf{u}^{\mathsf{Vrfy}} = \mathbf{u}^*$), and therefore all linear (mod-q) equalities checked by Vrfy hold deterministically for (c^*, d^*). Because sVrfy accepts, for each $i \in \{1,2\}$ and every $j \in [\tau]$ we have:

$$\|\mathbf{p}_j^i\|_{\max} \le \beta_{\mathrm{agg}}, \quad \|\mathbf{s}_j^i\|_{\max} \le \beta_{\mathrm{agg}}, \quad \text{and} \quad \|\mathbf{u}^i\|_{\max} \le \beta_{\mathrm{agg}}, \quad (45)$$

as well as

$$\big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{p}_j^i)\big\|_{\max} \le \frac{q\beta_{\mathrm{agg}}}{2\eta} \text{ and } \big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{s}_j^i)\big\|_{\max} \le \frac{q\beta_{\mathrm{agg}}}{2\eta}. \quad (46)$$

From (45) and (46), we obtain

$$\|\mathbf{p}_j^1 - \mathbf{p}_j^2\|_{\max} \le 2\beta_{\mathrm{agg}}, \quad \|\mathbf{s}_j^1 - \mathbf{s}_j^2\|_{\max} \le 2\beta_{\mathrm{agg}}, \quad \|\mathbf{u}^1 - \mathbf{u}^2\|_{\max} \le 2\beta_{\mathrm{agg}}.$$

and

$$\big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{p}_j^1 - \mathbf{p}_j^2)\big\|_{\max} \le \frac{q\beta_{\mathrm{agg}}}{\eta} \quad \text{and} \quad \big\|\mathsf{proj}_{\eta,\kappa}(\mathbf{s}_j^1 - \mathbf{s}_j^2)\big\|_{\max} \le \frac{q\beta_{\mathrm{agg}}}{\eta}.$$

Thus all checks in wVrfy go through for input $(\mathsf{pp}, c^{(1)} - c^{(2)}, t, d^{(1)} \ominus d^{(2)})$, hence $\mathsf{wVrfy}(\mathsf{pp}, c^{(1)} - c^{(2)}, t, d^{(1)} \ominus d^{(2)})$ equals

$$\begin{aligned}
&= \mathsf{vec}^{-1}(\mathsf{proj}_{q'}(\mathbf{u}^*)) \\
&= \mathsf{vec}^{-1}(\mathsf{proj}_{q'}(\mathbf{u}^{(1)})) - \mathsf{vec}^{-1}(\mathsf{proj}_{q'}(\mathbf{u}^{(2)})) \\
&= \mathbf{M}_1 - \mathbf{M}_2,
\end{aligned}$$

as required. □

Lemma B.7 (Position Binding). *Let $d, q, q', \alpha_w, N, \eta, \tau, \rho, \nu, \omega, \beta_{\mathrm{agg}} \in \mathbb{N}^+$ and $\epsilon_{\mathrm{hom}} \in_{\mathbb{R}} (0,1)$ such that d is a power of 2 and q, q' are primes. Let $\mathcal{R}_q, \mathcal{R}_{q'}$ be the polynomial rings $\mathbb{Z}_q[X]/(X^d + 1)$ and $\mathbb{Z}_{q'}[X]/(X^d + 1)$, respectively. If $\mathsf{MSIS}^{\infty}_{q,\,\omega,\,2\omega\kappa,\,4\beta_{\mathrm{agg}}}$ and $\mathsf{MSIS}^{\infty}_{q,\,\omega,\,\rho\nu\kappa',\,4\beta_{\mathrm{agg}}}$ are hard, then the construction in Figure 4 is position binding of Definition Definition B.5.*

Proof. Let $\mathcal{A}$ be an arbitrary PPT adversary against position binding. By the law of total probability it holds that

$$\Pr\left[\begin{array}{c} \mathsf{pp} \leftarrow \mathsf{Setup}(1^{\lambda}) \\ (c, t, d^{(1)}, d^{(2)}) \leftarrow \mathcal{A}(\mathsf{pp}) \\ \mathbf{M}_1 \leftarrow \mathsf{wVrfy}(\mathsf{pp}, c, t, d^{(1)}) \\ \mathbf{M}_2 \leftarrow \mathsf{wVrfy}(\mathsf{pp}, c, t, d^{(2)}) \end{array} : \mathbf{M}_1 \neq \mathbf{M}_2 \wedge \perp \notin \{\mathbf{M}_1, \mathbf{M}_2\}\right]$$

$$= \Pr\big[\mathsf{Bad} \wedge \mathsf{proj}_q(\mathbf{p}_\tau^1) = \mathsf{proj}_q(\mathbf{p}_\tau^2)\big] + \Pr\big[\mathsf{Bad} \wedge \mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)\big],$$

where Bad denotes the event $\mathbf{M}_1 \neq \mathbf{M}_2$ and $\perp \notin \{\mathbf{M}_1, \mathbf{M}_2\}$, and where (for $b \in \{1,2\}$) we parse

$$d^b = (\overline{\mathbf{p}}_1^b, \ldots, \overline{\mathbf{p}}_\tau^b,\ \mathbf{s}_1^b, \ldots, \mathbf{s}_\tau^b,\ \mathbf{u}^b)$$

and let $\mathbf{p}_\tau^b$ denote the decoded value appearing in $\mathsf{wVrfy}(\mathsf{pp}, c, t, d^b)$. We now bound the two probabilities separately.

Case 1: $\mathsf{proj}_q(\mathbf{p}_\tau^1) = \mathsf{proj}_q(\mathbf{p}_\tau^2)$. Using the definition of wVrfy (in particular, its output is $\mathsf{inv\text{-}vec}(\mathsf{proj}_{q'}(\mathbf{u}^b))$ on acceptance), we have

$$\begin{aligned}
&\Pr\big[\mathsf{Bad} \;\wedge\; \mathsf{proj}_q(\mathbf{p}_\tau^1) = \mathsf{proj}_q(\mathbf{p}_\tau^2)\big] \\
\le &\Pr\Big[\mathsf{proj}_{q'}(\mathbf{u}^1) \neq \mathsf{proj}_{q'}(\mathbf{u}^2) \wedge \mathbf{B} \cdot \mathbf{u}^1 = \mathbf{B} \cdot \mathbf{u}^2 \wedge \|\mathbf{u}^1\|, \|\mathbf{u}^2\| \le 2\beta_{\mathrm{agg}}\Big] \\
\le &\Pr\Big[\mathbf{u}^1 \neq \mathbf{u}^2 \wedge \mathbf{B} \cdot (\mathbf{u}^1 - \mathbf{u}^2) = \mathbf{0} \bmod q \wedge \|\mathbf{u}^1 - \mathbf{u}^2\| \le 4\beta_{\mathrm{agg}}\Big] \\
= &\Pr\Big[(\mathbf{u}^1 - \mathbf{u}^2) \in \mathcal{B}_{4\beta_{\mathrm{agg}}} \setminus \{\mathbf{0}\} \wedge \mathbf{B} \cdot (\mathbf{u}^1 - \mathbf{u}^2) = \mathbf{0} \bmod q\Big] \\
\le &\mathsf{negl}(\lambda),
\end{aligned}$$

where $\mathcal{B}_B$ denotes the ball of radius B under the norm used by wVrfy, and the last inequality follows from hardness of $\mathsf{MSIS}^{\infty}_{q,\omega,\rho\nu\kappa',4\beta_{\mathrm{agg}}}$ together with the fact that all algorithms are PPT.

Case 2: $\mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)$. We now analyze

$$\Pr\big[\mathsf{Bad} \;\wedge\; \mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)\big].$$

We construct a PPT algorithm $\overline{\mathcal{A}}$ that solves $\mathsf{MSIS}^{\infty}_{q,2\omega\kappa,4\beta_{\mathrm{agg}}}$ as follows. On input a uniformly random matrix

$$\mathbf{A} \;\in\; \mathcal{R}_q^{\omega \times 2\omega\kappa}$$

the algorithm $\overline{\mathcal{A}}$ parses $\mathbf{A}$ as a horizontal concatenation $\mathbf{A} = [\mathbf{A}_0 \mid \mathbf{A}_1]$ with $\mathbf{A}_0, \mathbf{A}_1 \in \mathcal{R}_q^{\omega \times \omega\kappa}$, samples $\mathbf{B} \leftarrow \mathcal{R}_q^{\omega \times \rho\nu\kappa'}$, sets $\mathsf{pp} := (\mathbf{B}, \mathbf{A}_0, \mathbf{A}_1)$, and runs

$$(c, t, d^{(1)}, d^{(2)}) \leftarrow \mathcal{A}(\mathsf{pp}).$$

For $b \in \{1,2\}$, let $\mathbf{M}_b := \mathsf{wVrfy}(\mathsf{pp}, c, t, d^{(b)})$. If $\mathbf{M}_1 = \mathbf{M}_2$, or $\perp \in \{\mathbf{M}_1, \mathbf{M}_2\}$, or $\mathsf{proj}_q(\mathbf{p}_\tau^1) = \mathsf{proj}_q(\mathbf{p}_\tau^2)$, then $\overline{\mathcal{A}}$ aborts. Otherwise, parse each

$$d^b = (\overline{\mathbf{p}}_1^b, \ldots, \overline{\mathbf{p}}_\tau^b,\ \mathbf{s}_1^b, \ldots, \mathbf{s}_\tau^b,\ \mathbf{u}^b)$$

and let $\mathbf{p}_j^b$ be the decoded path nodes computed inside wVrfy (so that $\mathbf{p}_0^b := \mathsf{dec}_q(c)$ and $\tilde{t} := \mathsf{bin}_\tau(t)$). Let $j^* \in [\tau]$ be the *largest* index such that

$$\mathsf{proj}_q(\mathbf{p}_{j^*}^1) \neq \mathsf{proj}_q(\mathbf{p}_{j^*}^2),$$

which exists since $\mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)$ by assumption, and satisfies $j^* \geq 1$ while $\mathsf{proj}_q(\mathbf{p}_{j^*-1}^1) = \mathsf{proj}_q(\mathbf{p}_{j^*-1}^2)$ by maximality.

Define the vector $\mathbf{z} \in \mathcal{R}^{2\omega\kappa}$ by

$$\mathbf{z} := \begin{cases} (\mathbf{p}_{j^*}^1 - \mathbf{p}_{j^*}^2,\ \mathbf{s}_{j^*}^1 - \mathbf{s}_{j^*}^2) & \text{if } \tilde{t}_{j^*} = 0, \\ (\mathbf{s}_{j^*}^1 - \mathbf{s}_{j^*}^2,\ \mathbf{p}_{j^*}^1 - \mathbf{p}_{j^*}^2) & \text{if } \tilde{t}_{j^*} = 1. \end{cases}$$

Since $\mathsf{proj}_q(\mathbf{p}_{j^*-1}^1) = \mathsf{proj}_q(\mathbf{p}_{j^*-1}^2)$ and both openings pass wVrfy, the linear consistency check at level j^* gives

$$\mathsf{proj}_q(\mathbf{p}_{j^*-1}^b) = \mathbf{A}_{\tilde{t}_{j^*}} \cdot \mathbf{p}_{j^*}^b + \mathbf{A}_{\tilde{t}_{j^*} \oplus 1} \cdot \mathbf{s}_{j^*}^b \qquad (b \in \{0,1\}).$$

Subtracting the two equalities and using linearity of proj_q yields

$$\mathbf{A}_{\tilde{t}_{j^*}} \cdot (\mathbf{p}_{j^*}^1 - \mathbf{p}_{j^*}^2) + \mathbf{A}_{\tilde{t}_{j^*} \oplus 1} \cdot (\mathbf{s}_{j^*}^1 - \mathbf{s}_{j^*}^2) = \mathbf{0} \bmod q.$$

By construction of $\mathbf{z}$ (swapping the two blocks depending on $\tilde{t}_{j^*}$) this is exactly

$$\mathbf{A} \cdot \mathbf{z} = \mathbf{0} \bmod q.$$

By definition of wVrfy, both openings satisfy $\|\mathbf{p}_{j^*}^b\|_{\max} \leq 2\beta_{\mathrm{agg}}$ and $\|\mathbf{s}_{j^*}^b\|_{\max} \leq 2\beta_{\mathrm{agg}}$ for $b \in \{0,1\}$. Hence,

$$\|\mathbf{z}\|_{\max} \leq \max\{\|\mathbf{p}_{j^*}^1\|_{\max}, \|\mathbf{s}_{j^*}^1\|_{\max}\} + \max\{\|\mathbf{p}_{j^*}^2\|_{\max}, \|\mathbf{s}_{j^*}^2\|_{\max}\} \leq 4\beta_{\mathrm{agg}}.$$

By choice of j^* we have $\mathsf{proj}_q(\mathbf{p}_{j^*}^1) \neq \mathsf{proj}_q(\mathbf{p}_{j^*}^2)$, and thus $\mathbf{p}_{j^*}^1 \neq \mathbf{p}_{j^*}^2$, implying $\mathbf{z} \neq \mathbf{0}$.

Therefore, whenever $\mathcal{A}$ succeeds in event Bad $\wedge\ \mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)$, the reduction $\overline{\mathcal{A}}$ outputs a nonzero $\mathbf{z}$ with $\|\mathbf{z}\|_{\max} \leq 4\beta_{\mathrm{agg}}$ and $\mathbf{A} \cdot \mathbf{z} = \mathbf{0} \bmod q$, solving $\mathsf{MSIS}^\infty_{q,\omega,2\omega\kappa,4\beta_{\mathrm{agg}}}$. Hence,

$$\Pr\big[\mathsf{Bad}\ \wedge\ \mathsf{proj}_q(\mathbf{p}_\tau^1) \neq \mathsf{proj}_q(\mathbf{p}_\tau^2)\big]\ \leq\ \mathsf{negl}(\lambda).$$

Combining the two cases, we conclude that

$$\Pr\big[\mathsf{Bad}\big]\ \leq\ \mathsf{negl}(\lambda),$$

as required. □

C Details of Lemur Multi-Signature

C.1 Formal Definitions

We begin by formally defining synchronized multi-signatures, following the modular framework of [8].

Definition C.1 (Synchronized N-wise Multi-Signatures). *A synchronized N-wise multi-signature scheme for 2^τ time periods is defined by six PPT algorithms* $\Pi_{\mathsf{MSIG}} = (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{Aggregate}, \mathsf{iVrfy}, \mathsf{aVrfy})$.

- $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$*: The setup algorithm takes as input a security parameter* $\lambda \in \mathbb{N}^+$ *and outputs a set of public parameters* pp.
- $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KGen}(\mathsf{pp})$*: The key generation algorithm takes as input a set of public parameters* pp *and outputs a secret key* sk *and the corresponding public key* pk.
- $\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{sk}, t, \mu)$*: The signing algorithm takes as input a set of public parameters* pp*, a secret key* sk*, a time period* $t \in [2^\tau]$*, and a message* μ*, and outputs a signature* σ.
- $\sigma_{\mathrm{agg}} \leftarrow \mathsf{Aggregate}(\mathsf{pp}, P, t, \mu, S)$*: The aggregation algorithm takes as input a set of public parameters* pp*, a list* P *of public keys, a time period* $t \in [2^\tau]$*, a message* μ*, and a list* S *of signatures such that* $|P| = |S| = N$*, and outputs an aggregated signature* σ_{agg} *or an error symbol* $\perp$.
- $b \leftarrow \mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}, t, \mu, \sigma)$*: The deterministic individual verification algorithm takes as input a set of public parameters* pp*, a public key* pk*, a time period* $t \in [2^\tau]$*, a message* μ*, and a signature* σ*, and outputs a bit* b *indicating acceptance/rejection.*
- $b \leftarrow \mathsf{aVrfy}(\mathsf{pp}, P, t, \mu, \sigma_{\mathrm{agg}})$*: The deterministic aggregated verification algorithm a set of public parameters* pp*, a list* P *of public keys, a time period* $t \in [2^\tau]$*, a message* μ*, and a signature* σ_{agg}*, and outputs a bit* b *indicating acceptance/rejection.*

We require that a synchronized N-wise multi-signature satisfies the following three properties: probabilistic individual correctness, probabilistic aggregation correctness, and existential unforgeability.

Definition C.2 (Probabilistic Individual Correctness). *Let* $\Pi_{\mathsf{MSIG}} = (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{Aggregate}, \mathsf{iVrfy}, \mathsf{aVrfy})$ *be a synchronized N-wise multi-signature scheme for 2^τ time periods. Let* $\epsilon_{\mathrm{cor}} \in [0,1)$. *We say that* Π_{MSIG} *is* ϵ_{cor}*-individually correct, if for all security parameters* $\lambda \in \mathbb{N}^+$*, public parameters* $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$*, key pairs* $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KGen}(\mathsf{pp})$*, time periods* $t \in [2^\tau]$*, and messages* $\mu \in \{0,1\}^*$*, for signatures* $\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{sk}, t, \mu)$ *it holds that*

$$\Pr[\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}, t, \mu, \sigma) = 1] \geq 1 - \epsilon_{\mathrm{cor}}$$

Definition C.3 (Probabilistic Aggregation Correctness). *Let* $\Pi_{\mathsf{MSIG}} = (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{Aggregate}, \mathsf{iVrfy}, \mathsf{aVrfy})$ *be a synchronized N-wise multi-signature scheme for 2^τ time periods. We say that* Π_{MSIG} *has* correct aggregations in the presence of rogue keys and signatures *if for all security parameters* $\lambda \in \mathbb{N}^+$*, public parameters* $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$*, time periods* $t \in [2^\tau]$*, messages* $\mu \in \{0,1\}^*$*, lists of public keys* $P = (\mathsf{pk}^{(1)}, \ldots, \mathsf{pk}^{(N)})$ *and signatures* $S = (\sigma^{(1)}, \ldots, \sigma^{(N)})$ *such that for all* $i \in [N]$ *it holds that* $\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}^{(i)}, t, \mu, \sigma^{(i)}) = 1$*, we have*

$$\Pr\Big[\sigma_{\mathrm{agg}} \leftarrow \mathsf{Aggregate}(\mathsf{pp}, P, t, \mu, S) :\ \mathsf{aVrfy}(\mathsf{pp}, P, t, \mu, \sigma_{\mathrm{agg}}) = 1\Big]$$

is greater than or equal to $1 - \mathsf{negl}(\lambda)$.

Definition C.4 (Existential Unforgeability (EUF)). *Let* $\Pi_{\mathsf{MSIG}} = (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{Aggregate}, \mathsf{iVrfy}, \mathsf{aVrfy})$ *be a synchronized N-wise multi-signature scheme for 2^τ time periods. Consider the game* $\mathsf{EUF}_{\Pi_{\mathsf{MSIG}},\mathcal{A}}(\lambda)$ *in Figure 8. We say that* Π_{MSIG} *is* existentially unforgeable (EUF) *if for all security parameters* λ *and all PPT adversaries* $\mathcal{A}$ *it holds that*

$$\Pr\left[\mathsf{EUF}_{\Pi_{\mathsf{MSIG}},\mathcal{A}}(\lambda) = 1\right]\ \leq\ \mathsf{negl}(\lambda)\,.$$

C.2 Lemur Construction

In this subsection, we provide a formal description of the Lemur construction (See Figure 5). At a high level, the scheme is built by composing a key-homomorphic one-time signature (KOTS) and a homomorphic vector commitment (HVC), following the blueprint from Squirrel [9] and Chipmunk [8].

Definition C.5 (The Lemur Scheme). *Let* $d, q, q', \eta, \tau, N, \rho, \nu, \alpha_w, m, n, k, \ell, \alpha_H, \in \mathbb{N}^+$ *such that* d *is a power of 2, and* q *and* q' *are primes. Let* Π_{KOTS} *be a key-homomorphic one-time signature scheme defined in Figure 3 and* Π_{HVC} *be a homomorphic vector commitment scheme defined in Figure 4. We define the synchronized multi-signature scheme Lemur* $= (\mathsf{Setup}, \mathsf{KGen}, \mathsf{Sign}, \mathsf{Aggregate}, \mathsf{iVrfy}, \mathsf{aVrfy})$ *as specified in Figure 5, with* $\rho = k$ *and* $\nu = n$.

Game $\mathsf{EUF}_{\Pi_{\mathsf{MSIG}},\mathcal{A}}(\lambda)$:
(1) $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$.
(2) $(\mathsf{sk}^*, \mathsf{pk}^*) \leftarrow \mathsf{KGen}(\mathsf{pp})$.
(3) Initialize $Q \leftarrow \emptyset$ (set of queried time-message pairs).
(4) $(P^*, t^*, \mu^*, \sigma^*_{\mathsf{agg}}) \leftarrow \mathcal{A}^{\mathsf{Sign}(\mathsf{pp},\mathsf{sk}^*,\cdot,\cdot)}(\mathsf{pp}, \mathsf{pk}^*)$.
(5) Output 1 iff:
  (a) $\mathsf{aVrfy}(\mathsf{pp}, P^*, t^*, \mu^*, \sigma^*_{\mathsf{agg}}) = 1$,
  (b) $\mathsf{pk}^* \in P^*$, *// Target pk is included in signer set*
  (c) $(t^*, \mu^*) \notin Q$, *// Forgery is on a non-queried message/time*
  (d) $|Q_t| \leq 1,\ \forall t \in [2^\tau]$. *// Q_t denotes the set of signing queries made at time slot t*

Oracle $\mathsf{Sign}(\mathsf{pp}, \mathsf{sk}^*, t, \mu)$:
(1) If $Q_t \neq \emptyset$, return $\perp$. *// One-time constraint*
(2) $Q \leftarrow Q \cup \{(t, \mu)\}$.
(3) Return $\sigma \leftarrow \mathsf{Sign}(\mathsf{pp}, \mathsf{sk}^*, t, \mu)$.

Figure 8: EUF security game for multi-signatures.

Theorem C.1. *Let $\lambda, d, q', N, n, k, \gamma, \tau \in \mathbb{N}^+$, $\epsilon_{\mathsf{cor}} \in (0, 1)$, and $\epsilon_{\mathsf{hom}} \in (0, \frac{1}{2})$ with d being a power of 2, q' being prime, and $\gamma \geq \lambda/\log_2\left(\frac{1}{2\epsilon_{\mathsf{hom}}}\right)$. Let $\mathcal{R}_{q'}$ be the polynomial ring $\mathbb{Z}_{q'}[X]/(X^d + 1)$. Let $W \subseteq \mathcal{R}$ be a set such that $|W| > 2^\lambda$ and let $W' := \{w^0 - w^1 \mid w^0, w^1 \in W\}$. Let $H : \{0, 1\}^* \to W^N$ be a random oracle. Let Π_{KOTS} be a key homomorphic one-time signature scheme with public keys in $\mathcal{R}_{q'}^{k\times n}$ as defined in Figure 3 and let Π_{HVC} be a homomorphic vector commitment for domain $\mathcal{R}_{q'}^{k\times n}$ as defined in Figure 4.*

If Π_{KOTS} is ϵ_{cor}-individually correct, $(N, W, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, robustly homomorphic, and W'-multi-user existentially unforgeable under rerandomized keys and Π_{HVC} is individually correct, $(N, W, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, robustly homomorphic, and position binding, then the Lemur construction from Figure 5 is an unforgeable synchronized N-wise multi-signature scheme that is ϵ_{cor}-individually correct and has correct aggregations in the presence of rogue keys and signatures.

Proof. We prove the theorem via Lemma C.1, Lemma C.2, and Lemma C.3 below. □

Lemma C.1. *Let $\lambda, d, q', n, k, \gamma, N, \tau \in \mathbb{N}^+$ with d being a power of 2, q' being prime. Let $R_{q'}$ be the polynomial ring $\mathbb{Z}_{q'}[X]/(X^d+1)$. Let Π_{KOTS} be a key-homomorphic one-time signature scheme with public keys in $R_{q'}^{k\times n}$, and let Π_{HVC} be a homomorphic vector commitment for domain $R_{q'}^{k\times n}$.*

If Π_{KOTS} is ϵ_{cor}-individually correct and Π_{HVC} is individually correct, then the Lemur construction from Figure 5 is ϵ_{cor}-individually correct.

Proof. Let $\mathsf{pp} = (\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{pp}_{\mathsf{HVC}}) \leftarrow \mathsf{Setup}(1^\lambda)$, $(\mathsf{sk}, \mathsf{pk}) = ((\mathsf{OSK}, \mathsf{OPK}), c) \leftarrow \mathsf{KGen}(\mathsf{pp})$, $t \in [2^\tau]$, $\mu \in \{0, 1\}^*$, and $\sigma = (\sigma', d) = \mathsf{Sign}(\mathsf{pp}, \mathsf{sk}, t, \mu)$, where:

$$\sigma' \leftarrow \Pi_{\mathsf{KOTS}}.\mathsf{Sign}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{osk}^{(t)}, \mu)$$
$$d \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{Open}(\mathsf{pp}_{\mathsf{HVC}}, c, \mathsf{OPK}, t)$$

Now, consider the execution of $\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}, t, \mu, \sigma)$. The algorithm proceeds as follows:

- HVC Verification. It computes $\mathsf{opk} \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c, t, d)$. By the individual correctness of the Π_{HVC} scheme, for any c that is a valid commitment to the vector $\mathsf{OPK} = (\mathsf{opk}^{(1)}, \ldots, \mathsf{opk}^{(2^\tau)})$ and any valid opening d for index t, the verification algorithm $\Pi_{\mathsf{HVC}}.\mathsf{sVrfy}$ must return the original committed value at that position. Therefore, we have:
$$\mathsf{opk} = \mathsf{opk}^{(t)}$$
Since $t \leq 2^\tau$ and $\mathsf{opk} = \mathsf{opk}^{(t)} \neq \perp$, the algorithm does not return 0 in line 3.
- KOTS Verification. The algorithm finally returns the result of $\Pi_{\mathsf{KOTS}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}, \mu, \sigma')$. Substituting $\mathsf{opk} = \mathsf{opk}^{(t)}$ from the previous step, this is equivalent to:
$$\Pi_{\mathsf{KOTS}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}^{(t)}, \mu, \sigma')$$
By the ϵ_{cor}-individual correctness of the Π_{KOTS} scheme, since σ' was generated using $\Pi_{\mathsf{KOTS}}.\mathsf{Sign}$ with the secret key $\mathsf{osk}^{(t)}$ corresponding to the public key $\mathsf{opk}^{(t)}$, this verification algorithm return 1 with probability $\geq 1 - \epsilon_{\mathsf{cor}}$.

So with probability $\geq 1 - \epsilon_{\mathsf{cor}}$, the Lemur individual verification algorithm iVrfy returns 1. This concludes the proof. □

Lemma C.2. *Let $\lambda, d, q', n, k, \gamma, N, \tau \in \mathbb{N}^+$ with d being a power of 2, q' being prime. Let $\epsilon_{\mathsf{hom}} \in_{\mathbb{R}} (0, \frac{1}{2})$ such that $\gamma \geq \lambda/\log_2\left(\frac{1}{2\epsilon_{\mathsf{hom}}}\right)$. Let $R_{q'}$ be the polynomial ring $\mathbb{Z}_{q'}[X]/(X^d + 1)$. Let $W \subseteq \mathcal{R}$ be a set such that $|W| > 2^\lambda$ and let $W' := \{w^0 - w^1 \mid w^0, w^1 \in W\}$. Let $H : \{0, 1\}^* \to W^N$ be a random oracle. Let Π_{KOTS} be a key-homomorphic one-time signature scheme with public keys in $R_{q'}^{k\times n}$, and let Π_{HVC} be a homomorphic vector commitment for domain $R_{q'}^{k\times n}$.*

If both Π_{KOTS} and Π_{HVC} are $(N, W, \epsilon_{\mathsf{hom}})$-probabilistically homomorphic, then the Lemur construction from Figure 5 has correct aggregations in the presence of rogue keys and signatures.

Proof. Let $\mathsf{pp} = (\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{pp}_{\mathsf{HVC}})$, and let $(\mathsf{pk}^{(i)}, \sigma^{(i)})$ be pairs such that $\mathsf{iVrfy}(\mathsf{pp}, \mathsf{pk}^{(i)}, t, \mu, \sigma^{(i)}) = 1$. By the definition of iVrfy in Figure 5, this implies that for each $i \in [N]$, there exists a decoded one-time public key $\mathsf{opk}^{(i)}$ such that:

$$\mathsf{opk}^{(i)} = \Pi_{\mathsf{HVC}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c^{(i)}, t, d^{(i)})$$
$$\Pi_{\mathsf{KOTS}}.\mathsf{iVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}^{(i)}, \mu, \sigma'^{(i)}) = 1,$$

where $\mathsf{pk}^{(i)} = c^{(i)}$ and $\sigma^{(i)} = (\sigma'^{(i)}, d^{(i)})$.The aggregation algorithm Aggregate performs at most χ attempts to find a salt j such that the aggregated signature (σ', d, j) is accepted by aVrfy. In each iteration j, the algorithm samples weights $(w^{(1)}, \ldots, w^{(N)}) \leftarrow H(t, \mu, P, j)$ and computes:

$$\sigma' = \sum_{i=1}^{N} w^{(i)}\sigma'^{(i)}, \quad d = \sum_{i=1}^{N} w^{(i)} d^{(i)}, \quad c = \sum_{i=1}^{N} w^{(i)} c^{(i)}$$

According to aVrfy, the signature is accepted if:

- **HVC Homomorphism.** $\Pi_{\mathsf{HVC}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c, t, d) = \mathsf{opk}$ for some $\mathsf{opk} \neq \perp$. By the $(N, W, \epsilon_{\mathsf{hom}})$-probabilistic homomorphism of Π_{HVC}, since $d^{(i)}$ are valid openings for $c^{(i)}$ at position t with values $\mathsf{opk}^{(i)}$, then $d = \sum w^{(i)} d^{(i)}$ is a valid opening for $c = \sum w^{(i)} c^{(i)}$ at position t with value $\mathsf{opk} = \sum w^{(i)}\mathsf{opk}^{(i)}$, except with probability ϵ_{hom}.
- **KOTS Homomorphism.** $\Pi_{\mathsf{KOTS}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}, \mu, \sigma') = 1$. Given that $\mathsf{opk} = \sum w^{(i)}\mathsf{opk}^{(i)}$ from the step above, the $(N, W, \epsilon_{\mathsf{hom}})$-probabilistic homomorphism of Π_{KOTS} ensures that the linear

combination of signatures $\sigma' = \sum w^{(i)}\sigma'^{(i)}$ is a valid signature for message μ under the aggregated key opk, except with probability ϵ_{hom}.

By a union bound, the probability that a single iteration j fails to satisfy both conditions is at most $2\epsilon_{\mathsf{hom}}$. Since the algorithm makes χ independent attempts (using different salts j for the random oracle), the probability that all χ attempts fail is at most $(2\epsilon_{\mathsf{hom}})^{\chi}$. Since $(2\epsilon_{\mathsf{hom}})^{\chi} \le 2^{-\lambda}$, which is negligible in the security parameter. Thus, the aggregation algorithm succeeds with probability $1 - \mathsf{negl}(\lambda)$. □

Lemma C.3. *Let $\lambda, d, q', n, k, \gamma, N, \tau \in \mathbb{N}^+$ with d being a power of 2, q' being prime. Let $R_{q'}$ be the polynomial ring $\mathbb{Z}_{q'}[X]/(X^d+1)$. Let $W \subseteq \mathcal{R}$ be a set such that $|W| > 2^{\lambda}$ and let $W' := \{w^0 - w^1 \mid w^0, w^1 \in W\}$. Let $H : \{0,1\}^* \to W^N$ be a random oracle. Let Π_{KOTS} be a key-homomorphic one-time signature scheme with public keys in $R_{q'}^{k\times n}$, and let Π_{HVC} be a homomorphic vector commitment for domain $R_{q'}^{k\times n}$.*

If both Π_{KOTS} is W'-multi-user existentially unforgeable under rerandomized keys and Π_{HVC} is position binding, then the Lemur construction from Figure 5 is existentially unforgeable. Specifically, for any PPT adversary $\mathcal{A}$ against Lemur, there exist PPT adversaries $\mathcal{B}_1$ and $\mathcal{B}_2$ such that:

$$\mathsf{Adv}^{\mathsf{euf}}_{\mathsf{Lemur},\mathcal{A}}(\lambda) \le \mathsf{Adv}^{\mathsf{binding}}_{\Pi_{\mathsf{HVC}},\mathcal{B}_1}(\lambda) + \mathsf{Adv}^{\mathsf{MU\text{-}EUF\text{-}RK}}_{\Pi_{\mathsf{KOTS}},\mathcal{B}_2}(\lambda)$$

Proof. The proof follows the same idea as in [8]. For completeness, we still draft the proof idea in here. Suppose an adversary $\mathcal{A}$ outputs a valid forgery $(P^*, t^*, \mu^*, \sigma^*_{\mathsf{agg}})$ where $\sigma^*_{\mathsf{agg}} = (\sigma', d, j)$. By the winning conditions of the EUF game, the target honest key $\mathsf{pk}^* = c^*$ must be in P^*, and the pair (t^*, μ^*) must not have been queried to the signing oracle. Let $(w^{(1)}, \ldots, w^{(N)}) \leftarrow H(t^*, \mu^*, P^*, j)$ be the weights derived from the random oracle. We define $\mathsf{opk} \leftarrow \Pi_{\mathsf{HVC}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{HVC}}, c, t^*, d)$, where $c = \sum w^{(i)}c^{(i)}$. For the forgery to be valid, aVrfy must return 1, implying $\mathsf{opk} \neq \perp$ and $\Pi_{\mathsf{KOTS}}.\mathsf{sVrfy}(\mathsf{pp}_{\mathsf{KOTS}}, \mathsf{opk}, \mu^*, \sigma') = 1$. We distinguish between two cases based on the behavior of the adversary:

Case 1: Breaking HVC Binding. In this case, the aggregated public key opk extracted from the forgery is different from the intended linear combination of one-time public keys. Let $\mathsf{opk}^{(i,t^*)}$ be the t^*-th one-time public key of the i-th signer. If $\mathsf{opk} \neq \sum_{i=1}^{N} w^{(i)}\mathsf{opk}^{(i,t^*)}$, then the opening d is a valid opening for commitment c at position t^* to a value other than the sum of the committed values. Since the HVC is homomorphic, $\sum w^{(i)}d^{(i)}$ is a valid opening for $\sum w^{(i)}\mathsf{opk}^{(i,t^*)}$. Providing a different valid opening d for the same commitment c at the same position t^* directly violates the position binding property of Π_{HVC}. We can construct $\mathcal{B}_1$ to output these two different openings to win the binding game.

Case 2: Breaking KOTS Security. If Case 1 does not occur, then $\mathsf{opk} = \sum_{i=1}^{N} w^{(i)}\mathsf{opk}^{(i,t^*)}$. This means the adversary has produced a valid signature σ' for the message μ^* under the aggregated key opk, which is a random linear combination of the one-time keys. Since (t^*, μ^*) was never queried, the adversary has never seen a signature for μ^* at time t^* for the honest signer. The W'-multi-user existential unforgeability of Π_{KOTS} ensures that it is hard to forge a signature even under a linear combination of keys (rerandomized keys). Specifically, $\mathcal{B}_2$ can simulate the Lemur environment by embedding its own MU-EUF-RK challenge keys into the HVC commitments. A successful forgery σ' by $\mathcal{A}$ then serves as a valid forgery against the KOTS challenge.

Since $\mathcal{A}$ must succeed in one of these two cases to produce a valid forgery, the advantage of $\mathcal{A}$ is bounded by the sum of the advantages of $\mathcal{B}_1$ and $\mathcal{B}_2$. □

Contribution	**Size in Bits**
Public Parameters	(can be generated from a seed instead of the expanded sizes below)
HVC: Ajtai's hash	$d(\omega kn\kappa' + 2\omega^2\kappa)\lceil\log q\rceil$
KOTS	$d(m-n)n\lceil\log q'\rceil$
Public Key	
HVC commitment	$d\omega\lceil\log q\rceil$
Secret Key	
2^{τ} KOTS keys	may regenerate on the fly
Individual Signature	
KOTS signature	$d\ell m\lceil\log(2\beta_z+1)\rceil$
HVC opening	$(\tau\omega\kappa + kn\kappa')d\lceil\log(2\eta+1)\rceil$
Aggregate Signature	
agg. KOTS signature	$d\ell m\lceil\log(2\beta_\sigma+1)\rceil$
agg. HVC opening	$(\tau\omega\kappa + kn\kappa')d\lceil\log(2\beta_{\mathsf{agg}}+1)\rceil + \tau\omega\kappa d\lceil\log(2\beta_{\mathsf{encode}}+1)\rceil$
index j of attempt	$\lceil\log\gamma\rceil$

Table 6: Sizes of Lemur components without any entropic encoding.

We also provide a summary of theoretical size of Lemur components in Table 6.

D Additional Lemmas and Proofs

Proof of Lemma 2.6. The algorithm $\mathcal{B}$ runs as follows on input the basis $\mathbf{B}_0 = (\mathbf{b}_{0,1}, \ldots, \mathbf{b}_{0,r})$ for a rank r lattice $\Lambda_0 := \mathbf{B}_0 \cdot \mathbb{Z}^r \subseteq \mathbb{R}^n$, a vector $\mathbf{c}_0 \in \mathbb{R}^n$ and a positive definite symmetric matrix $\Sigma_0 \in \mathbb{R}^{n\times n}$:

(1) Let $\mathrm{S}_0 := \sqrt{\Sigma_0} \in \mathbb{R}^{n\times n}$ (i.e. $\mathrm{S}_0\mathrm{S}_0^T = \Sigma_0$).
(2) Let lattice $\Lambda := \mathrm{S}_0^{-1}\cdot\Lambda_0$ with $\mathbb{Z}$-basis $\mathbf{B} := \mathrm{S}_0^{-1}\cdot\mathbf{B}_0 \in \mathbb{R}^{n\times r}$, and vector $\mathbf{c} := \mathrm{S}_0^{-1}\cdot\mathbf{c}_0 \in \mathbb{R}^n$.
(3) Let $V := \mathrm{span}_{\mathbb{R}}(\mathbf{B})$ denote the rank r vector space over $\mathbb{R}$ spanned by the columns of $\mathbf{B}$, and compute an orthonormal column $\mathbb{R}$-basis $\mathbf{U} \in \mathbb{R}^{n\times r}$ for V (e.g. $\mathbf{U}$ can be computed as the normalized Gram-Schmidt Orthogonalization of $\mathbf{B}$).
(4) Let $\mathbf{c}_{\parallel} := \mathbf{U}\mathbf{U}^T\mathbf{c} \in V$ denote the projection of $\mathbf{c}$ onto V, and let $\mathbf{c}_{\perp} := \mathbf{c} - \mathbf{c}_{\parallel} \in V^{\perp}$ denote the projection of $\mathbf{c}$ onto the orthogonal complement $V^{\perp} := \{\mathbf{v}_{\perp} \in \mathbb{R}^n : \mathbf{v}_{\perp}^T \cdot V = \mathbf{0}\}$ of V.
(5) Let $\mathbf{B}' := \mathbf{U}^T\cdot\mathbf{B} \in \mathbb{R}^{r\times r}$ denote a basis for the full rank lattice $\Lambda' := \mathbf{U}^T\cdot\Lambda \subseteq \mathbb{R}^r$, let $\mathbf{t} := \mathbf{U}^T\cdot\mathbf{c} \in \mathbb{R}^r$, and $s := 1$.
(6) Call algorithm $\mathcal{A}$ on input $(\mathbf{B}', \mathbf{t}, s)$ to sample $\mathbf{y} \leftarrow \mathcal{D}_{\Lambda'+\mathbf{t},1}$.
(7) Let $\mathbf{x} := \mathbf{U}\cdot\mathbf{y} + \mathbf{c}_{\perp}$ (remark: $\mathbf{x}$ is distributed as $\mathcal{D}_{\Lambda+\mathbf{c},1}$).
(8) Let $\mathbf{x}_0 := \mathrm{S}_0\cdot\mathbf{x}$ (remark: $\mathbf{x}_0$ is distributed as $\mathcal{D}_{\Lambda_0+\mathbf{c}_0,\sqrt{\Sigma_0}}$).
(9) Return $\mathbf{x}_0$.

To analyze the output distribution of $\mathbf{x}_0$, we first observe that in the call to algorithm $\mathcal{A}$ in line (6), the required parameter condition $s \ge \eta(r)\cdot\max_i\|\mathbf{b}'_i\|_2$ is satisfied. Indeed, the assumed condition (3) and the setting $s = 1$ implies that $s \ge \eta(r)\cdot\max_{i\in[r]}\|\mathrm{S}_0^{-1}\mathbf{b}_{0,i}\|_2 \ge \eta(r)\cdot\max_{i\in[r]}\|\mathbf{U}^T\cdot\mathrm{S}_0^{-1}\mathbf{b}_{0,i}\|_2 = \eta(r)\cdot\max_i\|\mathbf{b}'_i\|_2$, as required, where in the inequality we used the fact that the mapping $\mathbf{v} \mapsto \mathbf{U}^T\mathbf{v}$ is a projection, and hence cannot increase the norm. It follows that $\mathbf{y}$ is distributed as $\mathcal{D}_{\Lambda'+\mathbf{t},1}$.

Next, we show that $\mathbf{x}$ computed in line (7) is distributed as $\mathcal{D}_{\Lambda+\mathbf{c},1}$. Indeed, since $\mathbf{y} \in \Lambda' + \mathbf{t}$, $\Lambda' = \mathbf{U}^T\cdot\Lambda$ and $\mathbf{t} := \mathbf{U}^T\cdot\mathbf{c}$, we have

$\mathbf{y} = \mathbf{U}^T \cdot \mathbf{v} + \mathbf{U}^T \cdot \mathbf{c}$ for some $\mathbf{v} \in \Lambda$. Hence $\mathbf{x} := \mathbf{U} \cdot \mathbf{y} + \mathbf{c}_\perp = \mathbf{U}\mathbf{U}^T \cdot \mathbf{v} + \mathbf{U}\mathbf{U}^T \cdot \mathbf{c} + \mathbf{c}_\perp$. Since $\mathbf{v} \in V$, the projection $\mathbf{U}\mathbf{U}^T$ onto V acts as the identity on $\mathbf{v}$ (indeed, we have $\mathbf{v} = \mathbf{U}\mathbf{w}$ for some $\mathbf{w}$ and hence $\mathbf{U}\mathbf{U}^T\mathbf{v} = \mathbf{U}\mathbf{U}^T\mathbf{U}\mathbf{w} = \mathbf{U}\mathbf{w} = \mathbf{v}$ since $\mathbf{U}^T\mathbf{U} = \mathbf{I}$ due to the orthonormal columns of $\mathbf{U}$). Furthermore, $\mathbf{U}\mathbf{U}^T \cdot \mathbf{c} = \mathbf{c}_\parallel$. So we get $\mathbf{x} = \mathbf{v} + \mathbf{c}_\parallel + \mathbf{c}_\perp = \mathbf{v} + \mathbf{c}$ and hence the support of $\mathbf{x}$ is $\Lambda + \mathbf{c}$. Furthermore, for each $\bar{\mathbf{x}} = \mathbf{v} + \mathbf{c}_\parallel + \mathbf{c}_\perp \in \Lambda + \mathbf{c}$ with $\mathbf{v} \in \Lambda$, we have

$$
\begin{aligned}
\Pr[\mathbf{x} = \bar{\mathbf{x}}] &= \Pr[\mathbf{U}\mathbf{y} + \mathbf{c}_\perp = \mathbf{v} + \mathbf{c}_\parallel + \mathbf{c}_\perp] && (47)\\
&= \Pr[\mathbf{U}\mathbf{y} = \mathbf{v} + \mathbf{c}_\parallel] && (48)\\
&= \Pr[\mathbf{y} = \mathbf{U}^T \cdot (\mathbf{v} + \mathbf{c}_\parallel)] && (49)\\
&= \mathcal{D}_{\Lambda' + \mathbf{t}, 1}\left(\mathbf{U}^T \cdot (\mathbf{v} + \mathbf{c}_\parallel)\right) && (50)\\
&= \alpha \cdot \exp\left(-\pi \|\mathbf{U}^T \cdot (\mathbf{v} + \mathbf{c}_\parallel)\|^2\right) \quad \text{(normaliz. factor } \alpha)\\
&= \alpha \cdot \exp\left(-\pi \|\mathbf{v} + \mathbf{c}_\parallel\|^2\right) \quad (\mathbf{U}^T \text{ preserves norm on V})\\
&= \alpha \exp(\pi \|\mathbf{c}_\perp\|^2) \cdot \exp\left(-\pi \|\mathbf{v} + \mathbf{c}_\parallel\|^2\right) \cdot \exp(-\pi \|\mathbf{c}_\perp\|^2) && (51)\\
&= \alpha \exp(\pi \|\mathbf{c}_\perp\|^2) \cdot \exp\left(-\pi \|\mathbf{v} + \mathbf{c}_\parallel + \mathbf{c}_\perp\|^2\right) && (52)\\
&= \mathcal{D}_{\Lambda + \mathbf{c}, 1}(\bar{\mathbf{x}}), && (53)
\end{aligned}
$$

as required, where in the second last equality above we used the orthogonality of $\mathbf{c}_\parallel$ and $\mathbf{c}_\perp$, and in the last equality, the fact that $\alpha \exp(\pi \|\mathbf{c}_\perp\|^2)$ is a normalization factor independent of $\mathbf{v}$. Finally, we show that $\mathbf{x}_0$ computed in step (8) of the above algorithm is distributed as $\mathcal{D}_{\Lambda_0 + \mathbf{c}_0, \sqrt{\Sigma_0}}$. Indeed, since $\mathbf{x} = \mathbf{v} + \mathbf{c} \in \Lambda + \mathbf{c}$ for some $\mathbf{v} \in \Lambda$, we have $\mathbf{x}_0 := \mathrm{S}_0 \cdot (\mathbf{v} + \mathbf{c}) \in \mathrm{S}_0 \cdot (\Lambda + \mathbf{c}) = \Lambda_0 + \mathbf{c}_0$, as required. Furthermore, for each $\bar{\mathbf{x}}_0 = \bar{\mathbf{v}}_0 + \mathbf{c}_0 \in \Lambda_0 + \mathbf{c}_0$ with $\mathbf{v}_0 \in \Lambda_0$, we have

$$
\begin{aligned}
\Pr[\mathbf{x}_0 = \bar{\mathbf{x}}_0] &= \Pr[\mathrm{S}_0 \cdot \mathbf{x} = \bar{\mathbf{v}}_0 + \mathbf{c}_0] && (54)\\
&= \Pr[\mathbf{x} = \mathrm{S}_0^{-1} \cdot (\bar{\mathbf{v}}_0 + \mathbf{c}_0)] && (55)\\
&= \mathcal{D}_{\Lambda + \mathbf{c}, 1}\left(\mathrm{S}_0^{-1} \cdot \bar{\mathbf{x}_0}\right) && (56)\\
&= \alpha' \cdot \exp\left(-\pi \bar{\mathbf{x}}_0^T \cdot (\mathrm{S}_0 \mathrm{S}_0^{-T}) \cdot \bar{\mathbf{x}_0}\right) \quad \text{(normaliz. factor } \alpha')\\
&= \mathcal{D}_{\Lambda_0 + \mathbf{c}_0, \sqrt{\Sigma_0}}(\bar{\mathbf{x}_0}), && (57)
\end{aligned}
$$

as required. The time bound follows from the fact that all the linear algebra operations run in polynomial time. ◻

Lemma D.1. *Let λ, q', d, m, n, k, ℓ, α_H, $\beta_z \in \mathbb{N}^+$ where d is a power of 2 and $k > \ell$. Let $\alpha, \epsilon_{\mathrm{cor}} \in \mathbb{R}$ and $\varepsilon_1 \in (0, \frac{1}{2})$, such that $\alpha \geq \sqrt{2}\eta_{\varepsilon_1}(\mathbb{Z})$, $\beta_z \geq 6\alpha\sqrt{1 + (k-\ell)\alpha_H}$, and $\epsilon_{\mathrm{cor}} \geq \ell m d \cdot \left(2^{-162} + 2(k-\ell)\alpha_H \varepsilon_1\right)$. Let $\mathsf{H} : \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell \times (k-\ell)}$ be a hash function. Then Π_{KOTS} in Figure 3 is ϵ_{cor}-individually correct.*

Proof. Let $\mathrm{pp} = \mathbf{A} \leftarrow \mathsf{Setup}(1^\lambda)$ and $(\mathrm{osk}, \mathrm{opk}) = (\mathbf{S}, \mathbf{T}) \leftarrow \mathsf{KGen}(\mathrm{pp})$, so that $\mathbf{T} = \mathbf{S}\mathbf{A}$. For any $\mu \in \mathcal{M}$. The signer computes $\mathbf{H}' = \mathsf{H}(\mu)$, forms $\mathbf{H} = [\mathbf{I}_\ell \mid \mathbf{H}']$, and outputs $\sigma = \mathbf{Z} := \mathbf{H}\mathbf{S}$. The algebraic check always passes since $\mathbf{Z}\mathbf{A} = (\mathbf{H}\mathbf{S})\mathbf{A} = \mathbf{H}\mathbf{T} \pmod{q'}$.

We show $\|\mathbf{Z}\|_{\max} \leq \beta_z$ except with probability at most ϵ_{cor}. Fix indices $(r, c, i) \in [\ell] \times [m] \times [\![d]\!]$. Since $\mathbf{Z} = \mathbf{H}\mathbf{S}$, the i-th coefficient of $z_{r,c}$ expands as $z_{r,c}^{(i)} = s_{r,c}^{(i)} + \sum_{j=1}^{k-\ell} (h'_{r,j} s_{\ell+j,c})^{(i)}$. Because $\|h'_{r,j}\|_1 = \alpha_H$ and $\|h'_{r,j}\|_{\max} = 1$, the term $(h'_{r,j} s_{\ell+j,c})^{(i)}$ is a signed sum of exactly α_H coefficients of $s_{\ell+j,c}$. Since coefficients of S are sampled independently from $\mathcal{D}_{\mathbb{Z},\alpha}$, $z_{r,c}^{(i)}$ is distributed identically to the sum of $1 + (k-\ell)\alpha_H$ independent samples from $\mathcal{D}_{\mathbb{Z},\alpha}$.

Since $\alpha \geq \sqrt{2}\eta_{\varepsilon_1}(\mathbb{Z})$, Lemma A.4 bounds the statistical distance between $z_{r,c}^{(i)}$ and $\mathcal{D}_{\mathbb{Z},\alpha\sqrt{1+(k-\ell)\alpha_H}}$ by $2(k-\ell)\alpha_H\varepsilon_1$. Furthermore, because $\beta_z \geq 6\alpha\sqrt{1 + (k-\ell)\alpha_H}$, Lemma A.5 ensures a sample from this Gaussian exceeds β_z with probability at most 2^{-162}. Combining these, $\Pr[|z_{r,c}^{(i)}| > \beta_z] \leq 2^{-162} + 2(k-\ell)\alpha_H\varepsilon_1$. Taking a union bound over all $\ell m d$ coefficients in $\mathbf{Z}$ yields $\Pr[\|\mathbf{Z}\|_{\max} > \beta_z] \leq \ell m d\left(2^{-162} + 2(k-\ell)\alpha_H\varepsilon_1\right) \leq \epsilon_{\mathrm{cor}}$, as required. ◻

Lemma D.2. *Let $\lambda, q', d, m, n, k, \ell, \beta_z, \beta_\sigma, \alpha_H, \alpha_w, N \in \mathbb{N}^+$, where d is a power of 2 and $k > \ell$. Let β_z be as in Lemma D.1. Let $\epsilon_{\mathrm{hom}} \in (0, 1)$ such that $\beta_\sigma \geq \beta_z\sqrt{2\alpha_w N \ln(\frac{2\ell m d}{\epsilon_{\mathrm{hom}}})}$. Let $\mathsf{H}\colon \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell \times (k-\ell)}$ be a hash function, and $W = \mathcal{T}_{\alpha_w} \subseteq \mathcal{R}$. Then Π_{KOTS} in Figure 3 is $(N, \mathcal{T}_{\alpha_w}, \epsilon_{\mathrm{hom}})$-probabilistically homomorphic.*

Proof. Let $\mathrm{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, $\mu \in \mathcal{M}$, and let tuples $(\mathrm{opk}^i, \sigma^i) = (\mathbf{T}^i, \mathbf{Z}^i)$ for $i \in [N]$ satisfy $\mathsf{iVrfy}(\mathrm{pp}, \mathrm{opk}^i, \mu, \sigma^i) = 1$. Let $\mathbf{H} = [\mathbf{I}_\ell \mid \mathsf{H}(\mu)]$. Sampling $w^1, \ldots, w^N \leftarrow W$, we define $\mathbf{T}^* := \sum_{i=1}^N w^i \mathbf{T}^i$ and $\mathbf{Z}^* := \sum_{i=1}^N w^i \mathbf{Z}^i$. Since individual correctness ensures $\mathbf{Z}^i\mathbf{A} = \mathbf{H}\mathbf{T}^i \pmod{q'}$, linearity gives $\mathbf{Z}^*\mathbf{A} = \sum_{i=1}^N w^i(\mathbf{Z}^i\mathbf{A}) = \sum_{i=1}^N w^i(\mathbf{H}\mathbf{T}^i) = \mathbf{H}\mathbf{T}^* \pmod{q'}$. Thus, the algebraic check in sVrfy always holds.

It remains to bound the probability that $\|\mathbf{Z}^*\|_{\max} > \beta_\sigma$. Let $\mathbf{Z}^i = (z_{r,c}^i)$ for $r \in [\ell], c \in [m]$. Since iVrfy passed, $\|z_{r,c}^i\|_{\max} \leq \beta_z$ for all i, r, c. For any fixed entry (r, c), applying Lemma A.3 with $x_i := z_{r,c}^i$ and $\zeta = \beta_\sigma/\beta_z$ yields $\Pr_{w^i \leftarrow W}\left[\|\sum_{i=1}^N w^i z_{r,c}^i\|_{\max} > \beta_\sigma\right] \leq 2d\exp(-\frac{\beta_\sigma^2}{2\alpha_w N \beta_z^2})$. Taking a union bound over all ℓm entries, the probability that $\|\mathbf{Z}^*\|_{\max} > \beta_\sigma$ is at most $2\ell m d \exp(-\frac{\beta_\sigma^2}{2\alpha_w N \beta_z^2})$. By our assumed lower bound on β_σ, this exponent simplifies, bounding the overall failure probability by ϵ_{hom}. ◻

Lemma D.3. *Let $\lambda, q', d, m, n, k, \ell, \beta_\sigma, \alpha_H, \alpha_w, N \in \mathbb{N}^+$, where d a power of 2, $k > \ell$, and q' is prime. Let $\mathsf{H} : \mathcal{M} \to (\mathcal{T}_{\alpha_H})^{\ell \times (k-\ell)}$ be a hash function. Then Π_{KOTS} in Figure 3 is robustly homomorphic.*

Proof. Fix $\lambda \in \mathbb{N}$, $\mathrm{pp} = \mathbf{A} \leftarrow \mathsf{Setup}(1^\lambda)$, $\mu \in \mathcal{M}$, and let pairs $(\mathrm{opk}^b, \sigma^b) = (\mathbf{T}^b, \mathbf{Z}^b)$ for $b \in \{0, 1\}$ satisfy $\mathsf{sVrfy}(\mathrm{pp}, \mathrm{opk}^b, \mu, \sigma^b) = 1$. Let $\mathbf{H} = [\mathbf{I}_\ell \mid \mathsf{H}(\mu)]$. By the definition of sVrfy, we have $\mathbf{Z}^b\mathbf{A} = \mathbf{H}\mathbf{T}^b \pmod{q'}$ and $\|\mathbf{Z}^b\|_{\max} \leq \beta_\sigma$ for $b \in \{0, 1\}$. By linearity, $(\mathbf{Z}^0 - \mathbf{Z}^1)\mathbf{A} = \mathbf{Z}^0\mathbf{A} - \mathbf{Z}^1\mathbf{A} = \mathbf{H}\mathbf{T}^0 - \mathbf{H}\mathbf{T}^1 = \mathbf{H}(\mathbf{T}^0 - \mathbf{T}^1) \pmod{q'}$. Furthermore, the triangle inequality ensures $\|\mathbf{Z}^0 - \mathbf{Z}^1\|_{\max} \leq \|\mathbf{Z}^0\|_{\max} + \|\mathbf{Z}^1\|_{\max} \leq 2\beta_\sigma$. Thus, $\mathsf{wVrfy}(\mathrm{pp}, \mathrm{opk}^0 - \mathrm{opk}^1, \mu, \sigma^0 - \sigma^1) = 1$ holds, proving robust homomorphism. ◻